FOUNDATIONS

SECTION UIP 1

FOUNDATION COSTS

Foundation costs include all labor, materials and equipment, plus overhead and profit for the installing contractor.

All foundations are normally below the level of the lowest floor. If the building has a basement, the basement floor is the lowest floor. Whether there is a basement or not, the cost of foundations should be included in all buildings. These costs, converted to a Segregated Cost basis, are in Sections SEG 1 through SEG 6 for the various types of construction and building occupancy and they include an allowance for foundation excavation and backfill. Basement excavation must be priced separately.

In Section UIP 1, foundation costs are directly related to size and composition, but not to occupancy. Specific foundation costs can be developed using this section. Foundation costs may normally comprise excavation or trenching, concrete footing, foundation wall, waterproofing and drain tile.

Foundation and frame costs are dependent on the total weight that must be supported. The foundation cost for a one story with basement building is approximately the same as for a two story building without a basement. In using the Segregated Costs, the total area including basement, which is supported, must be used in order to develop a reasonable cost.

Buildings with heavy floor loads, such as warehouses, must have larger foundations than lighter structures. Also buildings with greater wall heights or heavier walls must have heavier foundations even though they have comparable floor areas.

Buildings with slab floors constructed on the ground usually have lighter foundations. In light construction where heavily insulated shallow foundations are employed, these buildings may have very little foundation; however, a cost of 40% to 60% of the normal foundation cost should be included to allow for perimeter footings and preparatory work not included in the slab cost. In these areas, ordinary foundation costs may be 40% lower than in the conventional deep foundations. Where the slab and footings are poured simultaneously (monolithically poured) and form one unit, the costs will be lower.

HILLSIDE CONSTRUCTION

Buildings located on hillsides have added costs because of the more extensive underground footings, the higher foundation walls required on the low side, the difficulty of construction on steep slopes, added excavation and the additional cost of transporting materials to the site. In addition, there is usually a great deal of site preparation, which must be estimated separately.

The costs of hillside foundations can run up to more than five times as much for a similar building on flat land. Generally, reinforced concrete foundation costs will run 150% to 200% more than the costs for normal footings and walls listed on Page 2. Beam and column supports will cost 25% to 75% more than those shown on Pages 7 and 8. Hillside screen walls (finished one side) can be built-up from Section UIP 5.

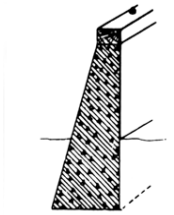
SUBTERRANEAN CONSTRUCTION

Earth sheltered residences and similar buildings usually require additional waterproofing; this includes materials similar to those used on roofs and exterior walls of standard structures above ground. These costs can be found on Page 3 of this section.

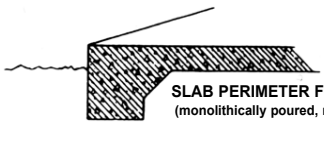
WOOD FOUNDATIONS

The mineral content of soils, in certain areas, makes treated wood a viable and desirable alternative to concrete and concrete block for underground foundation walls (including full basements). Costs for these types of walls are on Page 3 of this section.

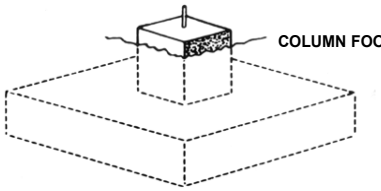
Mudsills are a substandard construction practice of resting a wood frame directly on the earth. This is not to be confused with current terminology, where the term refers to the first wooden framing member that is in direct contact with a foundation stem wall or slab, to which it is attached by anchor bolts or power pins. In this case as well as where the frame rests on light wood blocking, there is little or no foundation cost and it is often included in the frame cost. In some areas a so called concrete mudsill is used, which is merely a thin layer of concrete spread under the frame to remove it from contact with the earth and give a level bearing surface. This cost is usually negligible.

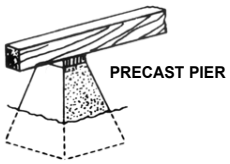


FOUNDATION WALLS

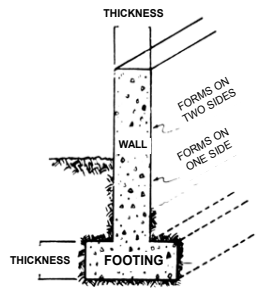


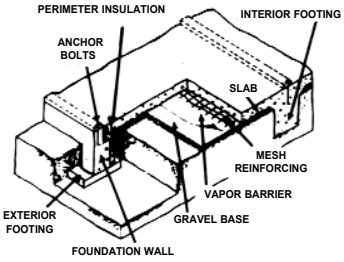
CONCRETE SLAB AND FOUNDATION

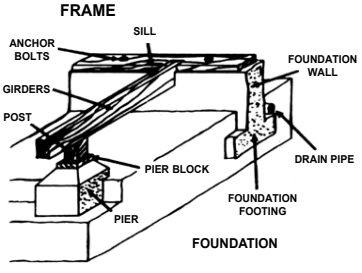


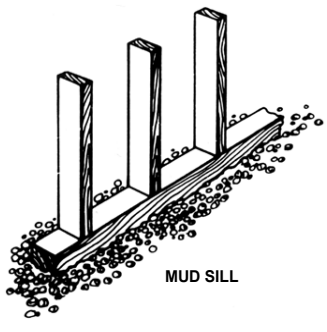


WOOD FLOOR CONSTRUCTION
(with interior frame supports)









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FOUNDATIONS

SECTION UIP 1

BULK CONCRETE FOR FOUNDATIONS

(in place)

	LOW	AVERAGE	GOOD
Concrete, plain (cubic foot)	$ 10.05	$ 11.75	$ 13.40
Concrete, reinforced (cubic foot)	12.85	15.15	18.00
Concrete, reinforced deep caissons (cubic foot)	23.50	31.00	40.25
Forms, surface area (square foot)	5.61	7.71	10.50
Concrete, piers, precast residential (each)	15.15	32.75	70.50

MISCELLANEOUS SITE PREPARATION

SITE CLEARING . . . Cost per acre. Does not include demolition.

	COST RANGE
Level with light grass	$ 895.00 – $ 1660.00
Rolling with medium brush and small trees	1660.00 – 3575.00
Steep with heavy growth and trees	3900.00 – 7650.00

General lot grading, cut and fill, etc., is listed in the Segregated Cost sections.

SOIL STABILIZATION . . . Cost per square yard.

	COST RANGE
Lime stabilization	$ 7.71 – $ 15.00
Gabions filled with stone	117.00 – 353.00
Riprap	113.00 – 160.00
Geotextile fabric	0.88 – 6.49
Synthetic matting	14.20 – 24.35
hexagonal matrix	17.10 – 28.00
rigid with integral drain	20.30 – 31.75

BULK EXCAVATION . . . Cost per cubic yard, includes allowance for one mile haul, disposal and return. For each additional mile add $ 3.31 to $ 5.45

	COST RANGE
Soft earth	$ 6.41 – $ 8.92
Medium earth	6.99 – 9.81
Hard earth	7.71 – 10.55
Clay or adobe	11.75 – 16.45
Loose rock, hardpan or sand (requiring shoring)	13.40 – 21.90

Excavation by hand costs $ 50.00 to $ 105.00 per cubic yard.

UTILITY TUNNELS

Costs of tunnels carrying utilities between buildings will vary greatly depending on the depth of the tunnel and the soil type, but the following costs are smoothed averages of reinforced concrete lined tunnels per cubic foot of tunnel, including lighting and drainage.

Wall Thickness	Light Soil	Medium Soil	Hardpan	Rock
3” – 5”	$ 29.50	$ 32.25	$ 35.50	$ 39.00
5” – 7”	34.25	38.50	42.25	46.75
7” – 10”	40.00	44.75	50.50	56.00

TRENCHING

Cost per linear foot, includes chain trencher and operator.

Width x Depth	Soft Earth	Medium Earth	Hard Earth
6” x 12”	$ 1.11	$ 1.27	$ 1.35
6” x 24”	1.14	1.30	1.40
6” x 48”	1.86	2.22	2.38
12” x 12”	1.35	1.57	1.70
12” x 24”	1.40	1.64	1.79
12” x 48”	2.36	2.75	2.95

TRENCHING

Cost per linear foot, including backhoe, operator, backfill and compaction. Bucket sizes vary and costs may vary plus or minus 50%.

Width x Depth	Soft Earth	Medium Earth	Hard Earth
24” x 2’	$ 2.12	$ 2.41	$ 3.63
24” x 4’	4.27	4.78	7.28
36” x 3’	4.80	5.32	8.08
36” x 6’	9.66	10.70	16.20
36” x 8’	12.80	14.20	21.55
48” x 3’	6.38	7.12	10.75
48” x 4’	8.60	9.45	14.40
48” x 6’	12.40	14.20	21.85
48” x 8’	16.85	18.95	29.25
72” x 4’	13.10	14.65	22.70
72” x 8’	25.25	28.25	42.75

EARTH WORK

(Cost per cubic yard)

	Soft Earth	Medium Earth	Hard Earth
Cut and fill and compact	$ 4.48	$ 8.92	$ 15.00
dock height fill	13.00	15.80	20.30
Backfill and compaction, Unconfined	20.30	24.70	31.75
Confined area	34.00	42.50	53.00
Handwork	99.50	153.00	238.00

Haul and Fill	1 mile	3 miles	5 miles	8 mile
5-CY truck	$ 6.99	$ 16.55	$ 23.15	$ 31.75
10-CY truck	5.85	10.55	16.55	21.20

CONCRETE CONTINUOUS (STRIP) FOOTINGS

Cost per linear foot, includes forms and reinforcing as noted.

SIZE	LEVEL GRADE		SLOPING GRADE	
Depth or Thickness/ Width	Plain	Reinforced	Plain	Reinforced
6”x12”	$ 15.20	$ 19.80	$ 17.35	$ 22.65
6”x14”	17.10	21.75	19.60	24.70
8”x12”	19.30	23.15	21.40	26.25
8”x18”	22.25	28.00	24.70	31.50
8”x24”	24.45	30.00	28.00	34.75
12”x18”	24.70	30.25	29.50	35.25
12”x24”	31.50	37.75	35.75	43.50
12”x36”	40.25	48.25	45.50	55.00
12”x48”	46.00	56.00	52.00	62.50
16”x48”	61.50	71.50	68.00	79.50
16”x60”	71.00	84.50	78.00	91.50
24”x72”	112.00	127.00	121.00	139.00
24”x96”	139.00	160.00	147.00	169.00

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FOUNDATIONS

SECTION UIP 1

CONCRETE COLUMN (SPREAD) FOOTINGS

Cost each, includes forms and reinforcing as noted.

Square x Depth	Plain	Rein-forced	Square x Depth	Plain	Rein-forced
3' x 12"	$ 252.00	$ 291.00	8' x 24"	$ 1430.00	$ 1770.00
4' x 12"	344.00	406.00	12' x 36"	4075.00	5000.00
4' x 16"	445.00	515.00	16' x 36"	6750.00	8700.00
6' x 16"	780.00	915.00	20' x 48"	12900.00	16600.00

CONCRETE GRADE BEAMS

Cost per linear foot, includes forming and reinforcing.

SIZE Width x Depth	COST	SIZE Width x Depth	COST
12" x 16"	$ 40.25	14" x 28"	$ 63.00
12" x 18"	43.25	16" x 30"	73.00
12" x 20"	48.25	16" x 32"	74.00
12" x 24"	55.50	20" x 52"	115.00
14" x 26"	60.50	24" x 52"	127.00

CONCRETE FOUNDATION WALLS*

Cost per square foot of wall area, includes forming and reinforcing as noted. For walls formed on one side only, reduce costs by 15% to 25%.

Thickness	Plain	Reinforced
6"	$ 18.35	$ 20.30
8"	19.95	22.40
10"	20.75	24.35
12"	22.40	25.75
16"	25.25	28.50

For stay-in-place (EPS) forming, add $ 0.74 to $ 2.68

***NOTE:** For hillside retaining walls, costs can be 100% to 300% higher.

CONCRETE BLOCK FOUNDATION WALLS*

Cost per square foot of wall area, includes grouting and reinforcing as noted.

	PLAIN			REINFORCED	
Thickness	Above Ground	Below Grade	Thickness	Above Ground	Below Grade
8"	$ 14.10	$ 12.50	8"	$ 15.30	$ 13.80
12"	16.70	15.15	12"	19.25	18.45

*NOTE: For hillside retaining walls, costs can be 100% to 300% higher.

WOOD FOUNDATION WALLS

Cost per square foot of wall area, includes treated wood, studs, sheathing and waterproofing. Add for insulation and interior finish from Section UIP 5.

SIZE	COST RANGE	SIZE	COST RANGE
2" x 4" – 12" OC	$ 8.36 – $ 10.10	2" x 8" – 12" OC	$ 10.30 – $ 15.15
2" x 4" – 16" OC	7.64 – 9.15	2" x 8" – 16" OC	8.90 – 13.75
2" x 6" – 12" OC	10.05 – 11.95	2" x 10" – 12" OC	10.45 – 17.55
2" x 6" – 16" OC	8.66 – 9.56	2" x 10" – 12" OC	10.10 – 17.15

WOOD MUDSILLS

Cost per linear foot, treated wood.

SIZE	COST	SIZE	COST
4" x 4"	$ 3.28	6" x 8"	$ 10.60
4" x 6"	4.65	8" x 8"	13.75
6" x 6"	6.83	10" x 10"	21.45

SLAB ON GRADE

Cost per square foot of slab area, includes forming and reinforcing as noted. For post-tension reinforcing, add 10%.

Thickness	Plain	Reinforced
3"	$ 3.99 – $ 4.24	$ 4.48 – $ 5.03
4"	4.24 – 4.87	4.94 – 5.92
6"	5.45 – 6.17	6.17 – 7.23
8"	6.49 – 7.38	7.71 – 8.76
10"	7.87 – 8.92	8.99 – 10.50
12"	8.99 – 10.05	10.05 – 11.75

PERIMETER FOOTING. . . Monolithically poured with slab, cost per linear foot, including forms.

Depth Below Slab	Cost	Depth Below Slab	Cost
8"	$ 18.35	20"	$ 52.00
12"	27.25	24"	69.00
16"	39.00	30"	91.50

ADJUSTMENTS. . . To slab on grade; cost per square foot of slab area.

Thickness		.002"	.006"	.010"
Polyethylene vapor barrier		$ 0.30	$ 0.36	$ 0.42
Depth		**4"**	**6"**	**12"**
Base:	Gravel	$ 0.56	$ 0.84	$ 1.56
	Sand.	0.92	1.28	2.44
	Crushed Stone . .	1.14	1.57	2.87

WATERPROOFING . . . Cost per square foot.	COST RANGE
Asphalt, bituminous, built–up, 1 ply	$ 1.54 – $ 2.03
2 ply	2.36 – 2.93
3 ply	3.31 – 4.15
For protector board, add.	1.14 – 1.54
Asphalt or pitch, one coat	0.98 – 1.54
Two coats	1.14 – 1.86
Asphalt with fibers, 1/16" thick	1.54 – 2.10
1/8" thick	2.03 – 2.93
Asphalt–coated board and mastic, 1/4" thick . . .	2.03 – 3.00
1/2" thick	3.16 – 4.70
Bentonite clay, 3/8" thick	2.93 – 4.24
Bituthene, 1/8" thick	3.99 – 5.69
Cement parging, two coats, 1/2" thick.	3.49 – 5.45
Elastomeric, Neoprene, 1/16"	5.45 – 7.87
Gypsum board, 1/2" thick thermal barrier or backer board	1.38 – 2.03
Polyethylene sheet, .006" thick	0.27 – 0.35
.010" thick	0.30 – 0.40
Silicone spray, one coat	0.58 – 0.68
Two coats	0.98 – 1.17

PERIMETER INSULATION . . . Cost per square foot.	
Cork, asphalt impregnated 1/2" thick	$ 2.36 – $ 2.68
1"	2.93 – 3.49
Polystyrene beadboard, 1" thick	1.54 – 2.10
1½"	1.86 – 2.51
2"	2.36 – 2.93
3"	3.42 – 4.15
SITE TREATMENT, TERMITE . . . Cost per sq. ft.	0.52 – 0.92

DRAIN PIPE . . . Cost per linear foot, includes bedding.

Diameter	4"	6"
Cement fiber/ metal.	$ 10.55 –	$ 14.10
City	11.95 –	17.50
Concrete	9.09 –	12.50
Plastic	9.57 –	11.75

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PILING

SECTION UIP 1

EXPLANATION

Pilings and special foundations must be priced as an addition to other foundation costs. The costs given below are averages of total costs in place, exclusive of architect's fees. Setting up and dismantling a job depends mainly on the size of equipment needed and the difficulty of access to the job. For a small residential job or boat dock where light piling is used and a mobile rig can run up to the work, setup and dismantling may amount to only a few hundred dollars. While a major dam or seawall project far from a shipping point and requiring piecemeal movement of heavy equipment, the cost might run twice the average as shown. When placement is in wet conditions, riverfront, marshland, over water, etc., the costs should be increased 25% to 50%.

Where light residential piles are used as common practice, installed costs have been as low as $5.38 per linear foot for one hundred 8" tapered wood piles, and as high as $ 17.55 per linear foot for composition wood pole and concrete lined pipe over water, and up to $38.00 per linear foot for fifteen 12" precast concrete piles, plus $11900.00 for setup costs in wetlands.

For pile tests, add $186.00 to $ 510.00 per ton of load per test pile.

EXAMPLE

A piling job requires 100 treated wood pilings, each 50' long by 14" diameter.

Cost of setup and dismantling	$ 25,600
100 piles 50' long = 5,000 linear feet @ $48.50	242,000
Total job cost	$267,000

SHEET PILING

TYPE	COST RANGE
Sheeting, bulkheads, left in place, per square foot	
concrete, poured in place	$ 29.75 – $39.75
Metal, aluminum panels	31.00 – 35.50
Steel, 27# average	43.00 – 60.50
Wood, untreated	17.55 – 20.35
Vinyl	17.90 – 19.95
Treated	20.35 – 23.20

Seawalls, cost per linear foot where typically installed, 10' – 14' depth for small residential jobs. For large commercial projects, costs may be 50% lower.

Treated wood, 8" – 12", including tiebacks	$454.00 – $615.00
Concrete, precast, 5" – 6" including Ties or piling	775.00 – 1270.00
Masonry block, decorative, 1' solid, Including 1' of bedding	870.00 – 1220.00
Rubble stone, 3', including 1' of bedding	1050.00 – 1370.00

PILING COSTS

TYPE OF PILING		COST PER LINEAR FOOT PILE	DRIVING	TOTAL	SETUP COST
Untreated Wood	10"	$ 10.25	$ 13.50	$ 23.85	$18000.00
	12"	15.45	14.60	30.25	22400.00
	14"	22.30	15.70	38.00	25600.00
	16"	30.25	16.85	47.50	29600.00
Treated Wood	10"	18.10	13.50	31.75	18000.00
	12"	24.95	14.60	39.75	22400.00
	14"	33.00	15.70	48.50	25600.00
	16"	42.00	16.85	58.50	29600.00
Steel. "H"	8" x 8"	39.75	15.05	55.00	20000.00
	10" x 10"	55.00	17.40	72.00	23100.00
	12" x 12"	71.50	19.80	91.50	27300.00
	14" x 14"	89.00	21.65	110.00	30500.00
Concrete, precast	10"	29.75	18.35	48.25	24900.00
	12"	38.75	21.65	60.00	29300.00
	14"	48.25	24.75	73.00	33000.00
	16"	58.00	27.75	86.00	37400.00
	18"	68.50	30.50	100.00	42000.00
	24"	105.00	39.00	143.00	54750.00
Steel pipe, concrete filled	8"	43.75	15.75	59.50	17100.00
	10"	55.00	17.95	72.50	20500.00
	12"	65.00	19.80	84.50	25600.00
	16"	86.00	23.05	110.00	34600.00
	18"	105.00	26.00	130.00	42000.00
	20"	119.00	31.00	152.00	47700.00
	24"	143.00	34.50	177.00	54750.00
	30"	169.00	41.50	208.00	63500.00
Concrete, in drilled holes	12"	-----	-----	48.25	-----
	16"	-----	-----	57.00	-----
	24"	-----	-----	86.00	-----
	36"	-----	-----	155.00	-----
	48"	-----	-----	276.00	-----

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FRAMES

SECTION UIP 1

EXPLANATIONS

DEFINITION

Frames are independent structures which hold up the floors and roof of a building. In many cases they will also support or brace the walls. Frame costs are listed in the Segregated Cost sections.

Bearing walls and partitions are priced under Walls and Interior Construction, respectively. In addition, some buildings with bearing walls have floor supports for wood floors, which are priced separately under Frames in the Segregated Cost sections.

Pilasters and bond beams for bearing walls are priced under Walls as an additional wall cost.

Roof trusses are priced under Trusses and Girders in the Segregated Cost sections. When a complete frame cost is used, generally the trusses or girders would be included as the horizontal members of the frame. In certain cases, a low frame cost (Rank 1) could be used to account for just the vertical portions, and then the appropriate truss or space frame cost could be priced separately.

BEARING WALLS

A bearing wall is constructed to support the weight of the floors or roof above. In addition, in many buildings the interior walls will aid in supporting the roof and floors. In the case of a bearing wall, the costs given under Walls in the Segregated Cost sections are all inclusive, with an average amount of wall materials, windows, doors, plaster, paint, etc., commensurate with the classification. Since this cost is complete, no further charge is made for supporting structure except for pilasters and bond beams, which are listed under Concrete or Masonry Walls, or floor supports, which are listed separately under Frames in the Segregated Cost sections.

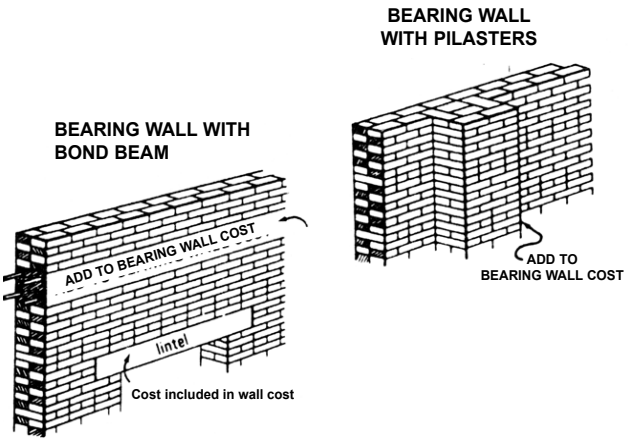
If bearing walls are used in part, they will substitute to that extent for the frame, and the frame cost is reduced accordingly, as described to the right.

PILASTERS

Brick and concrete walls are often strengthened with added thicknesses forming columns at intervals. Often the roof trusses will bear on these columns or pilasters, which are priced as an additive to the bearing wall cost in the Segregated Cost sections.

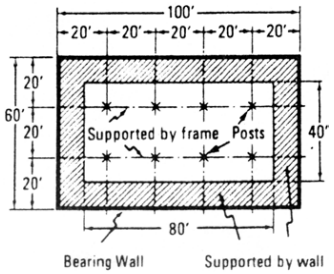
BOND BEAMS

Masonry walls also may be strengthened by horizontal concrete beams, bonded to the bricks or concrete blocks, which distribute vertical loads. These costs are also found as an additive to the bearing wall costs.



PARTIAL BEARING WALLS

When a building has a combination of frame and bearing walls, it is more accurate to first price the bearing walls with pilasters and bond beams and then price the frame on the basis of the area supported by the frame, or to price the columns and girders separately. In the drawing below it is assumed that each wall supports the roof halfway to the posts or columns as shown by the shaded area. Therefore, the balance is supported by the frame. For this example, assume a frame cost from the Segregated Cost section of $5.25 per square foot.



Area supported by frame = 40' x 80' = 3,200 sq. ft.
Total area supported by walls and frame = 60' x 100' = 6,000 sq. ft.
Percentage of total supported by frame = 53% (3,200 ÷ 6,000).
.53 x $5.25 = $2.78 per square foot of total area for the frame cost including pipe columns or posts and girders.

The table below indicates approximate percentages which may be used in lieu of this computation.

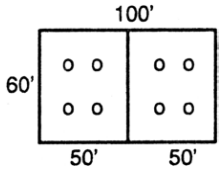
PARTIALLY FRAMED BUILDINGS

If a building is supported partially by frame and partially by bearing walls, and the exact proportion is difficult to determine in the manner shown above, then the following table may be useful. The distances listed are the smallest dimension between bearing walls.

SHORTEST DISTANCE	PERCENTAGE OF FRAME COST	SHORTEST DISTANCE	PERCENTAGE OF FRAME COST
30'	30%	75'	60%
40'	40%	90'	70%
50'	50%	120' and over	75%

EXAMPLE

This building is 60' x 100' without interior bearing walls; therefore, the shortest dimension is 60'. If there were an interior bearing wall like that shown by the broken line, the shortest dimension would be 50'.



From table above: 50' span, use 50% of full frame cost.
75' span, use 60% of full frame cost.
By interpolation 60' span, use 54% of full frame cost.

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FRAMES

SECTION UIP 1

EXPLANATIONS

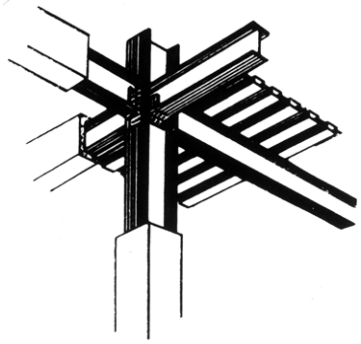
CLASS A AND B FRAMES

Class A and B frames are of fireproof construction, steel or concrete. In these buildings, especially in composite construction, there must be some arbitrary breakdown between roof, floors, walls and frame. This has been done by making the frame costs residual to the costs of other elements of the building shell, or in other words, the frame costs include all supporting structure costs which are not charged to the wall, roof or floors.

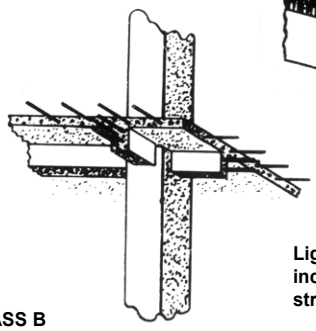
The walls of Class A and B buildings may rest on the frame or may be hung on it, and usually bear no more than their own weight. These are known as curtain walls since they merely enclose space.

CLASS C AND D FRAMES

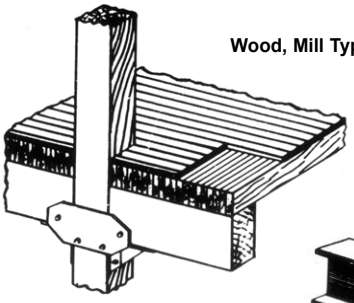
Class C and D buildings may also be of framed construction where an independent frame supports the floors and roof, and if all walls and partitions were removed, the skeleton would still stand insofar as vertical loads are concerned. This frame may consist of wood or steel posts and girders, wood poles, Glulam arches, etc. The vert cal members can be encased in fire resistive coverings or enclosed within the walls. It is often called an open frame since it lacks the complete fire resistance of a Class A or B frame. Costs are listed under Frames in the Segregated Cost sections and further descriptions and pictures may be found in the introduction.



CLASS A
Fireproofed Steel

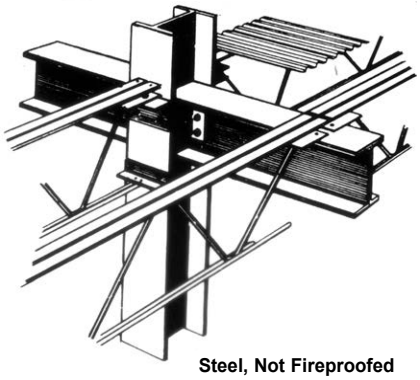


CLASS B
Reinforced Concrete



Wood, Mill Type

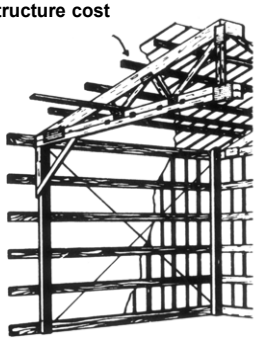
Light purlin supports included in roof structure cost



Steel, Not Fireproofed

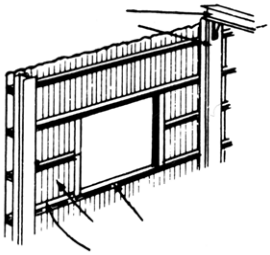
CLASS S FRAMES

Class S frames are typically open steel skeleton members consisting of post and beam and/or rigid bent configurations. The vertical members can be enclosed within the walls but the frame is basically non fire resistive. Costs are listed under Frames in the Segregated Cost sections. Costs for light pre-engineered frame can also be found in Section UIP 11.

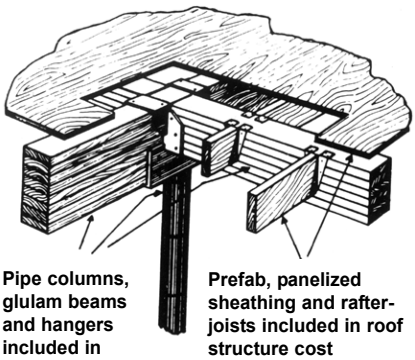


CLASS D Pole Frame (posts and prefab. trussed rafters)

Columns, beams and ties included in frame cost

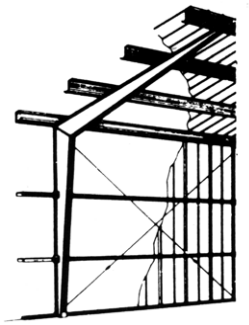


Wall cover, girts, and windows, included in wall cost



Pipe columns, glulam beams and hangers included in frame cost

Prefab, panelized sheathing and rafter-joists included in roof structure cost



Post and beam frame

Rigid tapered plate frame

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FRAME COSTS

SECTION UIP 1

WOOD BEAMS AND COLUMNS

Average cost, in place, per linear foot, including bolts, gussets and miscellaneous ties. For other sizes, use the equivalent cross section. For finished and wrapped or treated structural lumber add 10% to 20%.

SIZE	TIMBER	BUILT UP	SIZE	TIMBER
4" x 4"	$ 6.19 – $ 7.05	$ 6.19	8" x 24"	$ 77.50 – $ 88.00
4" x 8"	11.15 – 14.15	11.30	10" x 10"	37.75 – 43.25
4" x 12"	16.35 – 19.20	16.80	10" x 20"	68.00 – 79.50
6" x 6"	12.85 – 16.15	13.65	10" x 30"	105.00 – 113.00
6" x 10"	21.10 – 25.50	21.95	12" x 12"	53.50 – 61.50
6" x 14"	29.25 – 34.50	27.50	12" x 24"	97.50 – 108.00
6" x 20"	42.00 – 48.75	37.00	12" x 36"	152.00 – 168.00
8" x 8"	22.75 – 29.50	26.00	14" x 14"	75.50 – 85.50
8" x 12"	37.75 – 43.75	33.75	14" x 28"	146.00 – 161.00
8" x 16"	51.00 – 59.00	41.50	16" x 32"	214.00 – 249.00

For posts and girders supporting the floor structure of the first floor, use costs above, deducting 15%.

WOOD GLULAM BEAMS

Average cost, in place, per linear foot, including erection, hangers and miscellaneous connectors. For other sizes, use the equivalent cross section. For finished and wrapped lumber add 10%.

SIZE	COST	SIZE	COST	SIZE	COST	SIZE	COST
3" x 9"	$ 24.10	7" x 15"	$ 54.50	9" x 21"	$ 117.00	11" x 21"	$ 149.00
3" x 12"	28.75	7" x 18"	64.00	9" x 21"	133.00	11" x 24"	168.00
3" x 15"	33.25	7" x 21"	74.50	9" x 24"	149.00	11" x 27"	191.00
5" x 12"	37.25	7" x 24"	84.00	9" x 27"	168.00	11" x 30"	206.00
5" x 15"	43.75	7" x 27"	94.50	9" x 30"	184.00	11" x 33"	219.00
5" x 18"	50.50	7" x 30"	104.00	9" x 33"	198.00	11" x 36"	242.00
5" x 21"	55.50	7" x 33"	112.00	9" x 36"	214.00	11" x 42"	280.00
5" x 24"	60.50	9" x 18"	102.00	9" x 42"	227.00	11" x 45"	303.00

WOOD POLE FRAMES

Average cost, in place, per linear foot above ground for pressure-treated columns. Add wood trussed rafters from Section UIP 7. For untreated lumber, deduct 10%.

4" x 4" posts	$ 6.50 – $ 8.90	6" x 6" posts	$ 11.25 – $ 13.40
4" x 6" posts	8.03 – 10.45	6" x 8" posts	14.90 – 17.35
8" x 8" posts	18.20 – 20.85	12" to 14" poles	29.50 – 44.50

WOOD SILLS

Average cost, in place, per linear foot of bolted, pressure-treated or redwood sills. For sill sealer add $ 0.56 to $ 0.80 per linear foot.

2" x 4"	$ 3.04 – $ 3.77	2" x 6"	$ 4.33 – $ 5.38
2" x 8"	6.11 – 6.50		

REINFORCED CONCRETE COLUMNS

Average cost per linear foot. The amount of reinforcement can vary considerably based on the weight supported, and costs may vary plus or minus 15%.

CIRCULAR COLUMNS DIA.	COST RANGE	ADD FOR CAPS (each)	SQUARE COLUMNS DIA.	COST RANGE
12"	$ 49.50 – $ 66.50	$ 426.00	12"	$ 68.50 – $ 85.00
14"	62.50 – 85.00	464.00	14"	82.00 – 105.00
16"	74.50 – 103.00	505.00	16"	97.50 – 127.00
18"	88.00 – 125.00	520.00	18"	113.00 – 149.00
20"	103.00 – 146.00	570.00	20"	130.00 – 170.00
24"	133.00 – 193.00	650.00	24"	163.00 – 224.00
28"	163.00 – 239.00	770.00	28"	201.00 – 280.00
32"	201.00 – 302.00	880.00	32"	239.00 – 333.00
36"	239.00 – 356.00	1000.00	36"	271.00 – 396.00

For ornamental nonbearing columns, see Section UIP 6.

REINFORCED CONCRETE BEAMS

Cost per linear foot, includes erection and ties. For other sizes, use equivalent cross section. Add 10% for stressing.

SIZE	COST RANGE	SIZE	COST RANGE
4" x 6"	$ 21.40 – $ 31.75	10" x 18"	$ 71.50 – $ 91.00
6" x 8"	35.25 – 49.00	12" x 16"	77.00 – 94.00
8" x 10"	49.00 – 70.00	12" x 20"	83.50 – 105.00
10" x 12"	63.50 – 74.50	12" x 24"	92.50 – 114.00

PILASTERS AND BOND BEAMS

Average cost in place per linear foot.

BLOCK (BOND BEAMS)	CONCRETE	MASONRY (PILASTERS)
$ 17.65 – $ 24.60	$ 35.50 – $ 57.50	$ 45.00 – $ 68.00

PIPE AND TUBE COLUMNS

Typical cost range, per linear foot, including ancillary items and/or welding.

SIZE	ROUND COST RANGE	SIZE	SQUARE COST RANGE
3"	$ 41.00 – $ 49.50	3"	$ 40.25 – $ 50.00
4"	53.00 – 64.50	4"	50.50 – 66.50
6"	76.00 – 94.00	6"	75.50 – 97.50
8"	102.00 – 125.00	8"	99.00 – 128.00
10"	125.00 – 154.00	10"	122.00 – 163.00
12"	149.00 – 185.00	12"	147.00 – 193.00

Add $ 0.48 to $ 0.70 each inch of diameter per linear foot for concrete fill.

Adjustable jack columns for residential basements cost $ 88.00 to $ 154.00 each.

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FRAME COSTS

SECTION UIP 1

PRE-ENGINEERED STEEL FRAMES

Costs are for industrial type integral roof and wall column truss frames including ties, connections and/or welding. Design loads vary and costs may vary by plus or minus 25%.

TYPE	COST – EACH TRUSS					
Span	20	30	40	50	60	80
Open-web tapered truss	1880.00	2370.00	3175.00	3925.00	4825.00	7700.00
Post and web truss	2050.00	2525.00	3250.00	4150.00	5300.00	8050.00
Post and beam	2120.00	2700.00	3525.00	4400.00	5550.00	8650.00
Tapered plate	3000.00	3725.00	4750.00	6000.00	7550.00	12000.00

Add 4% for high profile (4:12 roof slope) buildings.

Add or deduct 4% for each foot of deviation from 14' base eave height.

STEEL

(Wide Flange "H" and "I" Beams)

Average cost per linear foot. Steel weights and types vary considerably and costs may vary from a plus 50% to minus 25%. Beams are designated by web length, where the flange width at each end increases the beam weight and cost as the flange gets larger.

Example: W8 cost range

Low, 15# per linear foot "I" beam, 8" web, 4" flange

High, 40# per linear foot "H" beam, 8" web, 8" flange

SIZE (web length)	COST RANGE	SIZE (web length)	COST RANGE
3"	$ 41.00 – $42.75	18"	$125.00 – $201.00
4"	48.75 – 57.50	21"	146.00 – 208.00
5"	57.00 – 65.50	24"	163.00 – 224.00
6"	63.50 – 74.50	27"	177.00 – 239.00
8"	68.00 – 97.50	30"	201.00 – 349.00
10"	74.50 – 125.00	33"	215.00 – 363.00
12"	85.00 – 163.00	36"	232.00 – 426.00
14"	100.00 – 177.00		

For trusses, see Section UIP 4, Roofs.

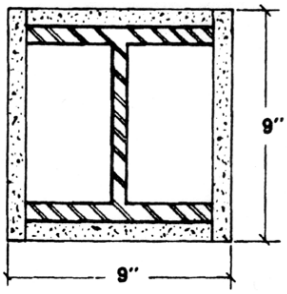
FIREPROOFING

The following are average costs per square foot of covered or enclosed area for fireproofing of structural steel.

TYPE OF ENCASEMENT	FIRE RATING			
	1 HR	2 HR	3 HR	4 HR
Sprayed fiber, beams	$ 2.89	$ 4.18	$ 4.75	$ 5.42
columns	3.78	5.66	7.44	9.26
Drywall	10.55	14.25	17.20	19.40
Lath and plaster	15.95	20.20	23.00	25.50
Concrete	17.60	21.90	24.70	27.00
Masonry units	-----	17.30	22.15	26.25

Sprayed fiber on underside of decking costs $ 2.37 to $ 3.03 per square foot.

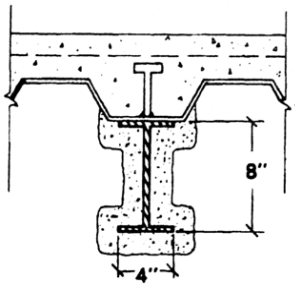
To convert the amount of covered area to a cost per linear foot, add up the total number of covered sides and divide by 12 inches.



A Wide-Flange "H" Column (W8 x 40#) enclosed by drywall boxed all around with a 2-hour fire rating.

$$\frac{9 + 9 + 9 + 9}{12} =$$

3.00 sq. ft. x $14.25 = $42.75/LF



A Wide-Flange "I" Beam (W8 x 15#) covered by sprayed fiber all around except for one flange side with a 3-hour fire rating

$$\frac{8 + 4 + 4 + 8 + 4}{12} =$$

2.33 sq. ft. x $4.75 = $11.07/LF

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INTERIOR CONSTRUCTION

SECTION UIP 2

INTERIOR PARTITIONS

Costs are averages for wall construction only, per square foot of wall area of nonbearing partitions. Additions should be made for doors from the Interior Door table without deducting for the areas of the openings. See Section UIP 5 for basic walls, wall covers, insulation and finishes not shown here. For shear walls or double layers of gypsum or sound board, add costs from the Sheathing category found on Page 4 of Section UIP 5.

MASONRY

TYPE	UNFINISHED	PLASTER ONE SIDE	PLASTER TWO SIDES
Clay tile, 4”	$13.10 – $15.05	$16.30 – $20.20	$ 19.35 – $25.00
6”	15.25 – 17.90	18.20 – 22.90	21.35 – 28.00
8”	17.15 – 19.10	20.20 – 24.10	23.15 – 29.25
Concrete block,			
2" solid	8.83 – 11.10	11.95 – 16.00	14.95 – 21.05
4” solid	10.05 – 12.50	13.40 – 17.85	16.45 – 22.70
6” solid	11.95 – 14.20	15.05 – 19.10	18.05 – 24.15
4” hollow	9.80 – 12.10	12.85 – 17.15	15.90 – 22.10
6” hollow	11.25 – 13.10	14.30 – 18.20	17.15 – 23.20
8” hollow	12.35 – 14.60	15.55 – 19.65	18.35 – 24.75

NOTE: For gypsum board, deduct $ 0.53 to $1.13 For furring, add $ 1.44 to $ 3.20

For glazed block or tile, add $8.52 to $11.80 for each glazed side.

FRAME

TYPE	2½" – 3" WALL	4" WALL	6" WALL
Steel studs,			
Plaster one side	$ 6.26 – $12.30	$ 6.44 – $12.50	$ 6.98 – $13.10
Two sides	9.80 – 19.60	9.94 – 20.05	10.80 – 20.50
Solid wall	11.95 – 15.55	----- – -----	----- – -----
Gypsum board			
One side	5.07 – 7.71	5.16 – 8.03	5.58 – 8.57
Two sides	6.90 – 10.05	7.00 – 10.60	7.60 – 11.55
Solid wall	11.30 – 14.15	----- – -----	----- – -----
NOTE: Add insulation to solid wall for 4” shaft wall cost.			
Wood studs,			
Plaster one side	5.07 – 7.56	5.29 – 8.19	5.75 – 8.99
Two sides	8.43 – 12.65	8.83 – 13.40	9.58 – 14.30
Gypsum board			
One side	3.37 – 5.07	3.70 – 5.47	4.01 – 6.44
Two sides	5.47 – 7.22	5.95 – 8.03	6.47 – 8.91

NOTE: For staggered studs on 2” x 6” plate, add $1.81 per square foot to 4” wall.

For lead lined lath, add $14.40 to $ 35.75 per square foot.

For gypsum green board (wet walls), add $0.25 per square foot per side.

MISCELLANEOUS

TYPE	COST RANGE
Glass block, white or aqua	$ 54.50 – $ 75.00
Color and intaglio	75.00 – 102.00
Folding partitions, wood and plastic	23.65 – 36.00
Acoustical	57.00 – 121.00
Add for electrical operation	32.25 – 35.00
Movable office partitions, hardboard/plastic	14.90 – 22.45
Softwood/plywood	17.70 – 29.75
Hardwood	27.00 – 42.00
Metal partitions, single thickness	21.35 – 33.25
2” insulated	23.65 – 39.75
3” insulated	27.00 – 41.00
add for glazing	1.93 – 3.70
Clear glass, full height	38.75 – 100.00
Frameless, structural glazing	302.00 – 390.00
Metal channels, 2¼” gypsum	12.05 – 17.70
Cement fiber panels	19.25 – 35.00
Woven wire, including doors	10.45 – 20.50
X–ray viewing, clear–opaque	
Lead–plastic panels	273.00 – 442.00

For hospital privacy curtains, see Section UIP 12, Page 3.

TOILET PARTITIONS

Average cost of each unit completely installed, including door. For privacy screens use the low side of the price range, deducting 10% to 20%. For stall shower with dressing cabinet, use high side of the price range plus 50% to 75%.

TYPE	COST RANGE
Laminated plastic	$ 965.00 – $1860.00
High-density solid plastic	1370.00 – 2600.00
Marble or Corian	1610.00 – 2825.00
Metal, baked enamel	1010.00 – 1240.00
Stainless steel or porcelainized	1410.00 – 2900.00
Wood	605.00 – 1280.00

For grab bars, add $97.00 to $ 236.00 each.

INTERIOR DOORS

Average cost in place per square foot of door opening, including hardware.

Accordion	$ 18.85 – $ 36.00
Bypass folding	11.50 – 23.45
Cafe (bar) doors	23.30 – 31.25
Flush doors, hollow core:	
Hardboard	12.85 – 18.10
Metal	35.00 – 50.50
Plastic–clad	34.00 – 44.25
Wood	20.50 – 31.00
add for solid core	4.42 – 6.19
add for raised panel door	1.33 – 9.24
add for decorative pattern or carved	39.00 – 60.00
add for metal frame	3.12 – 6.19
add for lead–lined frame	44.25 – 59.00
add for lead–lined door or glass	78.00 – 88.50
add for glass areas	14.05 – 26.50
add for Underwriters’ label	11.50 – 23.80
add for security locksets (restrooms, hotel/motel)	5.79 – 16.45
card entry, excluding monitoring system	16.45 – 22.90
French doors, wood	26.50 – 34.00
Louver doors, wood	19.25 – 30.25
Mirror wardrobe panels	20.35 – 35.75
Add for automatic door closers, each	102.00 – 331.00

For panic hardware, see Section UIP 6, Page 6; smoke-activated closers, Section UIP 3, Page 12.

INTERIOR STAIRWAYS

Apply cost to each full flight of stairs. The following costs include stair framing, finish and railings. For staircases that have a large number of the same railings and balustrades along an open landing or balcony, increase cost by 20% to 30%. For partial flights use proportionate costs. For every foot of floor height above 10’, add 5%. Custom curved or floating stairways are constructed of hardwood, hardwood with some tile or stone or with marble, with costs ascending in that order. The width of the staircase, landings if any, overall height, along with materials used, workmanship, and complexity of design (split staircase), must be considered when determining quality.

Disappearing attic stairs, slide-up pivot	
type, manual	$ 805.00 – $ 2900.00
Motorized	15200.00 – 19300.00
Folding type, wood	1010.00 – 2240.00
Aluminum	3950.00 – 5100.00
Telescoping access ladder	4200.00 – 5100.00
Spiral stairs, aluminum	5100.00 – 9200.00
Steel	4500.00 – 9700.00
Wood	7050.00 – 10600.00
Wood, residential, with balustrade,	
Softwood	1860.00 – 5900.00
Hardwood	2575.00 – 9800.00
Spiral or curved	6950.00 – 21800.00
Add or deduct 10% – 15% for each ft. deviation from 42" base width.	
Custom stairways, spiral	12200.00 – 30300.00
Curved	22800.00 – 75750.00
Floating	23900.00 – 107000.00

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INTERIOR CONSTRUCTION

SECTION UIP 2

FLOOR COVERING

(Cost per square foot)

TYPE	COST RANGE
Access (computer) floor, raised	$ 24.55 – $ 48.50
add for ramps, each	1240.00 – 2020.00
Railing, per linear foot	84.00 – 133.00
Full office floors	16.50 – 30.50
Asphalt tile	2.89 – 5.38
Bamboo, laminated plank	12.85 – 28.25
Block, creosoted wood, in asphalt	9.24 – 17.70
Brick, acidproof, industrial	20.10 – 52.50
Brick, common in mortar	11.10 – 22.25
Brick, pavers in concrete	13.40 – 27.25
Carpet, light grade	2.02 – 5.16
Medium grade	4.66 – 7.87
Heavy grade (includes carpet tiles)	6.67 – 14.40
Custom (special order)	12.45 – 62.00
Indoor/outdoor	2.56 – 7.32
add for pad	0.71 – 2.02
Concrete, color	1.16 – 3.21
Staining	5.16 – 7.07
Concrete, foamed, 1–5/8”	1.52 – 2.80
Concrete hardener and sealer	0.97 – 3.30
Heavy duty	2.24 – 5.86
Aggregate topping, 1/2” – 1”	6.10 – 18.25
Cork	7.32 – 14.30
Diato, magnesite, etc.	9.24 – 17.70
Epoxy terrazzo, 3/8” to 1/2” tile or poured	12.45 – 27.00
Epoxy, urethane, neoprene, 1/32” – 1/16”	4.83 – 12.05
1/8” – 3/8”	8.43 – 18.85
add for colored chips or glitter	2.24 – 5.07
Felt underlay	0.41 – 0.78
Flagstone, random local stone, in concrete	17.70 – 35.00
Granite	35.25 – 98.50
Gratings, plastic	12.85 – 63.00
Steel or aluminum	14.05 – 61.00
Stainless steel	60.00 – 206.00
Linoleum	4.66 – 9.80
Marble	32.25 – 103.00
Cast tile	19.25 – 36.50
Melamine laminated tiles or sheet	3.85 – 13.00
Deluxe	9.87 – 18.85
Plastic tile, interlocking	9.62 – 14.70
Rubber fabric tile	12.45 – 24.90
Rubber tile or sheet	4.91 – 16.60
Slate, grouted	20.10 – 39.50
Terrazzo, exclusive of base slab	13.65 – 31.75
Tile, 1” to 1–1/2”	25.75 – 48.25
Tile, ceramic or quarry	12.85 – 33.75
Custom tile	31.00 – 67.50
Vinyl composition tile or sheet	2.49 – 5.29
Premium, designer grade	4.49 – 7.07
Vinyl sheet	3.77 – 13.55
Vinyl tile	4.02 – 14.20
Premium, designer grade	12.35 – 23.55

Add 15% for conductive flooring.

For exterior decking, see Section UIP 13. Synthetic sports surfaces, see Section UIP 14.

WOOD FLOORING

(Cost per square foot)

Cost per square foot, exclusive of floor structure and subfloor, including wood base at walls, sanding and finish. Vapor barrier or building paper underlayment not included. Add $1.05 to $ 2.87 per square foot for sleepers. Custom patterns and/or special species may run two to three times costs listed. For custom finishes, add $0.72 to $ 0.97 for each additional layer of urethane.

TYPE	COMMON	SELECT	CLEAR
Hardwood			
Strip, 1/2"	$11.55 – $13.00	$15.05 – $17.95	$ 22.15 – $24.40
3/4" – 13/16"	11.50 – 13.10	15.55 – 18.65	24.00 – 26.50
Plank, imitation			
3/8" – 1/2"	13.40 – 14.95	16.65 – 20.55	23.65 – 28.75
3/4" – 13/16"	14.40 – 16.45	17.55 – 23.45	24.40 – 33.25
Parquet,			
3/4" – 13/16"	15.55 – 17.40	19.10 – 26.00	26.50 – 39.00
3/16" – 1/2" in mastic	10.60 – 12.30	14.45 – 19.75	22.70 – 32.50
3/4" – 13/16" in mastic	12.60 – 14.60	19.60 – 23.30	25.00 – 37.25
Softwood			
Strip, 3/4" – 13/16"	$ 9.06 – $10.50	$ 11.10 – $ 13.40	$ 15.15 – $17.15
Gymnasium Flooring			
Hardwood flooring on subfloor and sleepers, per square foot			$ 15.85 – $ 26.00
Portable floors on 2 x 4s and 2 x 6s, per square foot			16.95 – 28.25

For synthetic and other special sports surfaces, see Section UIP 14.

MOLDING AND TRIM

(Cost per square foot)

FLOOR BASE	
Ceramic or quarry tile	$ 13.40 – $ 19.65
Epoxy or diato topset	9.55 – 14.05
Hardwood	2.98 – 6.26
Ornate	7.71 – 12.60
Plastic	2.80 – 5.16
Softwood	2.74 – 5.47
Terrazzo	15.90 – 22.90
Vinyl or rubber topset	2.56 – 4.35
CEILING TRIM	
Hardwood	$ 3.77 – $ 15.15
Ornate	17.90 – 60.00
Metal	8.91 – 21.05
Plaster, cast	23.65 – 60.00
Plastic	3.54 – 9.62
Softwood	3.37 – 12.10
CHAIR AND WALL RAILS	
Hardwood	$ 4.27 – $ 14.60
Metal	30.00 – 62.00
Stainless steel	90.00 – 155.00
Plaster, reinforced	20.90 – 54.00
Plastic	3.54 – 8.91
Softwood	3.37 – 11.50
MISCELLANEOUS	
Ceiling medallions, wood or cast plaster, each, small	$ 265.00 – $ 442.00
Large, over 2' diameter	565.00 – 1890.00
Wall niches, cast plaster, each	1240.00 – 2390.00

For mantels, see Section UIP 9.

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INTERIOR CONSTRUCTION

SECTION UIP 2

CEILING COSTS

(Cost per square foot)

Costs of ceilings, with the exception of suspended or dropped ceilings, do not include framing, as flat plaster on lath is usually su ported by the bottom of the floor joists or roof structure. Painting or other finish typical of the quality and type of ceiling is included. For wallpaper, see Section UIP 5.

For finished buildings and residences, ceiling joists are considered in the roof structure costs. In plain buildings such as industrials and warehouses, no such allowance is made, and where a ceiling exists under a roof, the cost of ceiling joists must be added, as in all instances where special ceiling framing is needed.

A ceiling structure hung by wires or supports is a suspended ceiling, and the cost of the suspension system must be added, except as noted.

To convert costs to floor area for high-pitched ceilings, use the roof slope multipliers from Section UIP 4.

TYPE	COST RANGE
Acoustical ceilings, tile or panels:	
Metal panels, including pads and suspension system	$ 9.24 – $ 21.75
Mineral fiber, fiberglass, tiles (1' x 1')	2.80 – 7.00
panels (2' x 2' or 4')	1.53 – 5.95
Mirror–faced panels	24.90 – 42.00
Organic fiber, wood or cane	2.49 – 4.73
Embossed metal	6.10 – 22.05
Fiberboard panels	2.02 – 3.70
Gypsum board, taped and painted	2.56 – 4.35
Spray–on, texture	2.49 – 3.85
Hardboard paneling, 1/4", decorative	3.54 – 8.12
Paint only, on bottom of roof or floor structure	0.81 – 2.42
Plaster on lath: acoustical	5.07 – 9.16
Spray–on (thincoat w/texture) on lath	3.62 – 6.34
Standard, add 20% for Keene's	4.49 – 8.12
add for metal lath	0.71 – 1.33
Plaster on masonry soffit: acoustical	4.27 – 7.87
Spray–on (thincoat w/texture)	2.89 – 5.16
Standard, add 24% for Keene's	3.62 – 6.76
Plaster panels, reinforced, decorative	18.10 – 29.75
Plastic panels, including suspension system but excluding lighting (included in electrical cost)	7.79 – 18.35
Plywood paneling, softwood, 1/4" 3/8"	3.85 – 7.87
Hardwood, 1/4" 3/8"	5.54 – 12.45
Coffered or vaulted panel with molding	12.45 – 39.50
Custom woods, site built	32.25 – 65.00
Polylaminated panels	4.02 – 6.67
Insulated panels	6.03 – 9.94
Wood, boards or T&G, softwood, 1/2"	3.77 – 8.57
Hardwood, 1/2"	6.44 – 14.20
Hardwood, decorative carved or coffered	11.30 – 46.75
Add for wood furring	1.15 – 2.74
Add for metal furring	1.93 – 4.49
Add for ceramic tile finish	15.60 – 43.50
Custom tile	34.00 – 75.00
Add for ceiling structure which is not part of the roof or floor structure	1.93 – 4.49
Add for suspended ceiling, metal	1.77 – 4.35
Add for suspended ceiling, fiberglass	2.56 – 5.95
Add for suspended ceiling, hardwood decorative	4.58 – 13.40
Add for suspended ceiling, seismic supports	0.27 – 0.81

For ceiling insulation or exposed plank or deck, see detailed costs in Section UIP 4.

For art glass, see Section UIP 6.

INTERIOR DECORATING

The following costs are for the highly specialized type of ornamental finishes associated with auditoriums, public buildings and some high-value theaters. These figures are generally included in the costs in the Calculator and Segregated Cost Methods. In some cases, however, it may be desirable to make separate estimates of ornamented interiors. If a figure on the ornamentation is desired, the method suggested is to multiply the total area of all decorated surfaces by the costs given below.

The assessor who uses these costs must ascertain the proper classification. As a guide, small or neighborhood cinemas will usually fit into the plain category, the medium–size theater will be in the average category and the expensive major theater will normally be in the expensive classification.

A study of the individual structures will enable the assessor to place the type of decoration in the cost range. Two or more of the classifications listed may be found in one building. Thus, a theater auditorium may be extensively decorated and the foyer relatively plain. Both the extent of the decoration and the relative cost of the medium employed should be considered.

PLAIN DECORATING . . . This consists largely of flat or plain color work, some border or cove stenciling, or a minimum of plaster staff or other ornamental medium.

$ 6.26 to $10.10 per square foot

AVERAGE DECORATING . . . This includes basic flat work and considerable ornamental treatment of openings, columns, beams and arch facings; also a limited amount of staff, terra cotta, marble or tile, such as for niches, plaques or crestings.

$16.05 to $ 45.75 per square foot

EXTENSIVE DECORATING . . . This ranges from the upper limit of the average class to the finest type wherein the surfaces are largely covered with intricate designs in expensive media.

$ 57.00 to $118.00 per square foot

Detailed costs for stone and marble veneers can be found in Section UIP 6.

Hand-painted murals by famous artists must be valued by a qualified art appraiser, but those by commercial or billboard artists usually cost $106.00 to $162.00 per square foot. Simple hand-painted stenciling will cost $ 32.75 to $ 66.00 per square foot.

Photo murals in black and white or sepia, pasted on the wall, will cost $ 25.75 to $46.00 per square foot. In color, the cost is $ 63.50 to $ 141.00 .

Staff, stone or wood ornamentation will usually take the same depreciation as the building, but items of artistic merit may depreciate little, and ornamentation which is subject to wear may depreciate at a high rate. Works designated as having historical or artistic merit must be valued by fine arts specialists.

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INTERIOR CONSTRUCTION

SECTION UIP 2

CABINETS

(Cost per linear foot)

TYPE	LOW	AVG.	GOOD	HIGH COST
Base cabinets				
paint or print, laminates	$ 124.00	$ 192.00	$ 295.00	$ 810.00
natural, lacquer finish	162.00	250.00	376.00	960.00
Resins, baked enamel	1050.00	1240.00	1490.00	1800.00
Metal	169.00	250.00	368.00	860.00

For island cabinetry, add 10% to 20%; peninsula (doors both sides), add 55% to 65%.
For lazy-susan, add 10%.
For matching stove/sink fronts only, deduct 60%.

Wall cabinets, 12" deep by 30" high				
Paint or print, laminates	$ 106.00	$ 162.00	$ 243.00	$ 545.00
Natural, lacquer finish	124.00	177.00	287.00	700.00
Resins, baked enamel	805.00	915.00	1070.00	1240.00
Metal	124.00	192.00	280.00	605.00

For peninsula (double sided), add 60% to 75%.
For each 6" variation from base, add or deduct 10%.
For undercabinet canopies with recessed lighting, add $27.25 to $ 56.00 per linear foot.

Full height, utility (broom, water heater, wardrobe)				
Paint or print, laminates	$ 177.00	$ 287.00	$ 471.00	$ 1240.00
Natural, lacquer finish	236.00	368.00	605.00	1550.00
Resins, baked enamel	1650.00	2020.00	2420.00	2925.00
Metal	243.00	361.00	550.00	1360.00

For storage, oven or linen, add 30% to 35%. High cost, add 5% to 10%.
For garage storage or work benches, deduct 10% to 25%.

Pullman or vanity bases				
Paint or print, laminates	$103.00	$ 162.00	$250.00	$ 620.00
Natural, lacquer finish.	124.00	192.00	295.00	810.00
Resins, baked enamel	735.00	900.00	1120.00	1320.00
Built–in desks				
Paint or print, laminates	147.00	206.00	280.00	595.00
Natural, lacquer finish.	162.00	243.00	346.00	735.00
Resins, baked enamel	1090.00	1150.00	1230.00	1320.00
Open shelves				
8” paint or print, laminates	6.19	13.65	27.25	125.00
Natural, lacquer finish.	7.07	14.45	32.25	155.00
Metal	12.05	20.50	40.25	137.00
12” paint or print,				
Laminates	7.07	14.90	32.25	155.00
Natural, lacquer finish	8.43	17.30	38.00	184.00
Metal	12.85	23.65	45.25	162.00

COUNTER OR SINK TOPS

(Cost per linear foot. Do not deduct sink or other built-in space.)

TYPE	COST RANGE
Laminated plastic	$ 22.45 – $ 177.00
Solid plastic	
(Avonite, Corian, Nevamar, etc.)	140.00 – 273.00
add for integral sink	75.00 – 184.00
Marble, cultured with integral sink	116.00 – 177.00
Marble or granite	177.00 – 471.00
Metal, stainless steel	144.00 – 280.00
Tile	66.50 – 250.00
Custom hand painted	144.00 – 368.00
Wood	69.00 – 192.00

RESTROOM ACCESSORIES

(Cost each except as noted)

Baby changing station, foldout	$ 464.00 – $ 765.00
Stainless steel	2210.00 – 3175.00
Child protection seat	133.00 – 258.00
Dispensers, air freshener	53.00 – 177.00
Facial tissue, surface	62.50 – 295.00
Custom residential or recessed	280.00 – 565.00
Paper cup	45.25 – 155.00
Paper towel, surface	81.00 – 980.00
Combination waste receptacle	331.00 – 1300.00
Recessed	177.00 – 1200.00
Combination waste receptacle	295.00 – 1640.00
Sanitary napkin	317.00 – 1050.00
Combination disposal unit	920.00 – 1640.00
Seat cover, surface	70.50 – 177.00
Recessed	133.00 – 565.00
Dual unit	580.00 – 580.00
Soap	62.50 – 184.00
Surface mounted	45.25 – 309.00
Recessed	162.00 – 354.00
Custom dishes	116.00 – 324.00
Three station w/ liquid storage system	1750.00 – 2420.00
Toilet paper	53.00 – 390.00
Recessed	62.50 – 184.00
Custom holders or dual unit	106.00 – 765.00
Grab bars	53.00 – 106.00
L shape	169.00 – 302.00
Swinging or custom	317.00 – 1080.00
Fold–up seat, handicapped	162.00 – 1280.00
Hand dryers	715.00 – 2525.00
Medicine cabinets, surface	106.00 – 650.00
Recessed	162.00 – 875.00
Mirrors, to 12 sq. ft., framed, per sq. ft.	53.00 – 116.00
Custom residential	162.00 – 457.00
Over 12 sq. ft., unframed	27.25 – 70.50
Robe hooks	17.70 – 98.50
Custom	81.00 – 317.00
Shelves, per linear foot	45.25 – 133.00
Custom residential	162.00 – 486.00
Toilet seats	53.00 – 258.00
Custom	177.00 – 860.00
Raised seat, handicapped	162.00 – 354.00
Towel bars	35.25 – 155.00
Custom	221.00 – 1080.00
Rings, custom	116.00 – 442.00
Urns	155.00 – 427.00
Waste receptacles	116.00 – 580.00
Custom residential	420.00 – 915.00
Sanitary napkin disposal	53.00 – 442.00

NOTE: Equipment and cabinet costs for some occupancies can be found in Section UIP 12.

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INTERIOR CONSTRUCTION

SECTION UIP 2

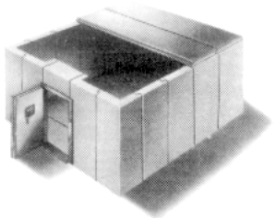
VAULTS

Costs per square foot of floor area for poured-in-place construction are based on an average eight foot ceiling height, exclusive of floors and doors, but including built in alarm systems, ventilators and interior finish. Add or deduct 2% for each foot of height variation.

Security vault	$ 246.00	– $ 393.00
Record storage	97.00	– 134.00

MODULAR VAULT SYSTEMS

Costs per square foot of floor area are based on an average eight foot ceiling height exclusive of floors and doors, but including built-in alarm systems, ventilators and interior finishes for solid precast panel construction. All modular vault systems shown are comprised of U.L. listed components. For other than ground level construction, add 30% to 40% for security vault systems comprised of laminated lightweight panels.



Security Vault Systems – Attack Resistance Rated Components		
Class M	$ 76.00	– $ 86.50
Class 1	86.50	– 118.00
Class 2	109.00	– 164.00
Class 3	186.00	– 209.00
Insulated/Record Storage Vault Systems – Fire Resistance Rated Components		
1 Hour	$ 63.50	– $ 83.50
2 Hour	77.50	– 86.50
4 Hour	85.00	– 103.00
6 Hour	101.00	– 118.00

VAULT DOORS

Costs for security vault doors include door delivered to premises and installed with time lock, day gate, sill and other hardware. Cost for insulated/record storage doors includes complete installation. Costs shown are for doors with clear opening of 78" high x 40" wide; reduce costs by 15% for doors 78" high x 32" wide. These vault doors are shown by U.L. listing. Older, unrated rectangular doors are shown in approximate equivalence to current U.L. listings. Circular doors are no longer made; the listed costs are normal at the time installed trended to date.

Security Vault Doors – Attack-Resistance Rating		
Class M (2" – 3" thick)	$ 12700.00	– $ 18500.00
Class 1 (3½" – 6" thick)	34100.00	– 49300.00
Class 2 (7" – 9" thick)	41000.00	– 64000.00
Class 3 (10" thick)	57250.00	– 92250.00
Insulated/Record Storage Vault Doors – Fire-Resistance Rating		
1 Hour	$ 5050.00	– $ 7350.00
2 Hour	5650.00	– 7900.00
4 Hour	6800.00	– 8500.00
6 Hour	7900.00	– 10700.00
Circular Vault Doors – Thickness		
8"	$ 200000.00	– $252000.00
10"	213000.00	– 270000.00
12"	228000.00	– 287000.00
14"	247000.00	– 305000.00
16"	262000.00	– 325000.00

SAFE DEPOSIT BOXES AND LOCKERS

Costs of modular steel units, 24" deep, completely installed, per box

Add 10% for custom-built units. Deduct 25% for aluminum units.

TYPE	COST RANGE	
3" x 5"	$ 133.00	– $ 171.00
5" x 5"	179.00	– 201.00
3" x 10"	186.00	– 209.00
5" x 10"	231.00	– 276.00
10" x 10"	356.00	– 483.00
5" x 16"	445.00	– 520.00
11" x 16"	715.00	– 1060.00
22" x 16"	1580.00	– 2010.00
44" x 16" (includes 4" base)	2900.00	– 3475.00

BANK FIXTURES

Costs for teller stations include teller cages, undercounter equipment, service counters, islands, alarm systems, and other equipment not listed individually on this page.

Cost per teller station	$ 13300.00	– $ 17700.00

Costs for hardened steel plates include complete installation with necessary accessories. Hardened steel plate, per square foot.

1/2"	$ 113.00	– $ 179.00
1"	179.00	– 283.00
2"	261.00	– 371.00

For security surveillance equipment, see Section UIP 3.

TRANSACTION EQUIPMENT

Night deposit chutes and boxes, each	$ 18300.00	– $ 34700.00
Drive-up/walk-up windows, each	15300.00	– 27400.00
Vision window only, per station	3250.00	– 3875.00
Pneumatic drive-up system, excluding canopy, window, and other site improvements.		
For multiple installations: deduct 20% for second unit; 2% for each unit thereafter.		
Cost per station	31600.00	– 56250.00
Customer TV monitor systems,		
Per station	93750.00	– 113000.00
add for remote cameras, each	3075.00	– 3875.00
Automated teller machines		
Lobby/retail units, each	13900.00	– 52750.00
Drive-up/walk-up through-wall units,		
Stand-alone island units	38300.00	– 83750.00

For ATM structures, see Section UIP 11.

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INTERIOR CONSTRUCTION

SECTION UIP 2

CLEAN ROOMS DEFINITIONS

Each clean room space is specially engineered based on how clean the room must be. Costs can vary greatly for each of the three basic cost ranges and should be used with caution. We would recommend that, wherever possible, actual contract costs be obtained.

Clean room space is subdivided into three types classifed by the number of particles of a certain size present in a cubic foot of air. The classifications used in this book are described below:

Type 1: Class 100,000 to 10,000. This cost range represents cleaned space that has no more than 100,000 to 10,000 particles of 0.5 microns or larger per cubic foot of air.

Type 2: Class 1,000 to 100. This cost range represents cleaned space that has no more than 1,000 particles of 0.5 microns or larger, to 100 particles of 0.3 microns or larger, per cubic foot of air.

Type 3: Class 100 to 10. This cost range represents cleaned space that has no more than 100 particles no larger than 0.3 microns, to 10 or fewer particles of 0.12 microns or smaller, per cubic foot of air.

MODULAR ROOM ENCLOSURE SYSTEMS

Cost each, based on an average 8 foot ceiling height exclusive of floors or roof deck, but including downflow filtered air flow, lighting and ceiling panel and wall modules appropriate for the two encl sure types.

PREFABRICATED HARDWALL ROOMS

ROOM SIZE	COST RANGE		
(SQ. FT.)	Type 1	Type 2	Type 3
50	$ 12900.00	$ 27700.00	$ 43800.00
100	20900.00	42100.00	65500.00
150	29600.00	57500.00	92750.00
200	37900.00	75250.00	120000.00
300	49900.00	98250.00	159000.00
400	65250.00	132000.00	224000.00

VINYL CURTAIN ROOMS

ROOM SIZE	COST RANGE		
(SQ. FT.)	Type 1	Type 2	Type 3
50	$ 7050.00	$ 10600.00	$ 31200.00
100	11300.00	15300.00	45500.00
150	16100.00	22200.00	66750.00
200	20300.00	28000.00	88000.00
300	27100.00	38700.00	115000.00
400	40200.00	59750.00	175000.00

For curtain tunnels and horizontal airflow, add 50% to 70%.

For pass-thru chambers, add $ 665.00 to $ 6200.00 .

Personnel air showers, 16 to 30 square feet, add $ 25200.00 to $ 40200.00 per unit.

CLEAN ROOM EQUIPMENT

	COST EACH	COST RANGE
Clean room storage cabinets, filtered	$ 8100.00	– $10600.00
Unfiltered	5900.00	– 7050.00
Containment weigh station	9450.00	– 12800.00
Work stations, laminated finish, horizontal laminar flows,		
3' to 5' long	8100.00	– 12800.00
6' to 8' long	12400.00	– 18900.00

For table mount units without base cabinet, deduct 10%. For downflow units, add 10%; vertical recirculation or negative flow, add 25%. For polypropylene units, add an additional 100%. For wet stations without hood (air flow), deduct 40%.

For general laboratory equipment, see Section UIP 12.

MISCELLANEOUS INTERIOR ITEMS

	UNIT	COST RANGE
Building directories, 6 to 12 sq. ft., per sq. ft.	$ 258.00	– $ 685.00
12 to 24 sq. ft., per sq. ft.	214.00	– 575.00
Mail chutes, per floor	4100.00	– 7450.00
Mail collection boxes, single, each	2850.00	– 3650.00
Double, each	4500.00	– 6400.00
Mail boxes, apartment type, 4 8 doors,		
Vertical, each door	62.50	– 169.00
Over 8 doors, each door	53.00	– 162.00
Horizontal, each door	70.50	– 106.00
Post office type, wall unit, each door	53.00	– 192.00
Free–standing, each door	12.45	– 35.25
Plaques, bronze, per square foot	457.00	– 840.00
Sound system, base cost	1230.00	– 2210.00
add per speaker, paging only	124.00	– 236.00
Music and voice	302.00	– 420.00
Telephone enclosures, wall hung	1300.00	– 2575.00
Full height	5250.00	– 7100.00
Handicapped, portable amplifiers	70.50	– 88.50
Telecommunicators	685.00	– 1140.00
Television antenna system, per outlet	192.00	– 302.00

For waste/linen chutes, see Section UIP 9.

KITCHEN UNITS

Average costs per linear foot of single-unit steel kitchens such as those found in motels, efficiency apartments, etc., containing sink, stove, oven and refrigerator. Add $106.00 per foot for snack bar. For microwave oven see, SEG 2.

	UNIT	COST RANGE
Base section only (72" to 30")	$ 1070.00	– $ 1360.00
Base section and upper cabinets	1460.00	– 1640.00

Lump Sum Method:

For Units having kitchens or built-in kitchen units, add the following:

Excellent (stove, refrigerator, sink and cabinet unit)	$ 7500.00
Good	5500.00
Average (cabinets and sink)	4100.00
Low Cost (kitchenette)	3050.00

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HEATING
SECTION UIP 3

EXPLANATION

The following costs are averages of typical installations, including cost of unit, miscellaneous materials and labor included in installation, cost of roughing in necessary utilities and vents, and a pro rata share of contractor's profit and overhead through a general contractor.

Heating and cooling costs per square foot of floor area are included in the Calculator and Segregated Cost Sections.

FLOOR AND WALL FURNACES

Costs are for gas fired units. Add $ 206.00 to $292.00 per thermostat. Add $261.00 to $330.00 for electric ignition. Add $ 206.00 to $ 306.00 for circulating fan. For direct vent thru-wall or freestanding units, add $161.00 to $398.00 .

RATED CAPACITY	FLOOR FURNACES		WALL FURNACES	
	SINGLE	DUAL	SINGLE	DUAL
25,000 B.T.U .	$ 1500.00	-----	$ 1320.00	$ 1470.00
35,000	1640.00	$ 1990.00	1400.00	1550.00
45,000	1780.00	2120.00	1520.00	1660.00
55,000	1870.00	2280.00	1640.00	1800.00
65,000	1930.00	2420.00	1800.00	1940.00
75,000	-----	2625.00	-----	-----

FORCED-AIR AND GRAVITY FURNACES

Average cost of installed gas fired units with electric ignition including thermostat. Gravity and upflow furnaces will typically be at the lower end of the cost range while downflow (counterflow), horizontal and electric furnaces will be at the higher end. For high-efficiency models use higher side of the range, adding an additional $ 505 for deluxe low-NOx emission high-efficiency condensing furnaces. Add $ 306 to $ 445 per outlet for ducts, registers and grills. Deduct $276 to $ 368 for standing pilot ignition systems. Add 10% for oil-fired or stoker-fired units plus cost for a storage tank below. For dual chamber woodburning units, add 100%; for outdoor wood-burning units, add 110% to 130%. See Section UIP 12 for chimneys.

RATED CAPACITY	COST RANGE	RATED CAPACITY	COST RANGE
65,000 BTU .	2675.00 – 3825.00	125,000 BTU . .	3350.00 – 4725.00
75,000	2775.00 – 3975.00	150,000	3675.00 – 5150.00
85,000	2925.00 – 4200.00	200,000	4350.00 – 6250.00
100,000	3000.00 – 4450.00	300,000	6300.00 – 9200.00

275 gallons . .	1270.00 – 1590.00	1,000 gallons . .	3175.00 – 3975.00
550	2060.00 – 2675.00	1,500	4350.00 – 5450.00

CLOCK THERMOSTATS

Single setback thermostats will range in cost from $ 237 to $ 292 plus $12.25 for each additional setback. Multistage programmable thermostats, used to automatically raise or lower temperature at preselected times, cost from $467 to $ 715. Hazardous location therrmostats for controlling heating systems in explosive environments cost $ 376 to $ 580.

AUTOMATIC VENT DAMPERS

Automatic vent dampers cost $ 414 to $ 565 installed.

AUTOMATIC VENT DAMPERS

For roof ventilators and blowers, see Section UIP 4.

SPACE HEATERS – SUSPENDED

Costs are for each gas-fired unit complete with propeller type fans, including installation. High efficiency, power vented units are at the high end of the range. Add $ 498 for blower-type operation. For poly tube adapter, add $ 253. Add $ 306 to $ 505 for electric ignition. Add $142 for propane-powered units. For oil fired units, add 150%. For steam heat costs, use Section SEG 3 or 4.

RATED CAPACITY	COST RANGE	RATED CAPACITY	COST RANGE
35,000 BTU .	1620.00 – 2500.00	150,000 BTU . .	2700.00 – 3675.00
50,000	1870.00 – 2675.00	175,000	2775.00 – 3975.00
60,000	1920.00 – 2700.00	200,000	3000.00 – 4275.00
75,000	1990.00 – 2950.00	250,000	3675.00 – 4975.00
100,000	2150.00 – 3175.00	300,000	4200.00 – 5800.00
125,000	2360.00 – 3375.00	400,000	5800.00 – 8300.00

SPACE HEATERS – FLOOR

Costs are for oil fired industrial heaters complete with fans, filters, controls, limited ductwork, storage and piping.

RATED CAPACITY	COST RANGE	RATED CAPACITY	COST RANGE
100,000 BTU	$ 6300.00	400,000 BTU	$ 16600.00
150,000	8050.00	500,000	20000.00
200,000	9700.00	750,000	28400.00
300,000	13300.00	1,000,000	37600.00

RADIANT HEATERS – SUSPENDED

Costs are for gas fired units complete with piping and controls. Continuous pipe loop systems with reflectors and vacuum pump are at the high end of the range.

RATED CAPACITY	COST RANGE	RATED CAPACITY	COST RANGE
15,000 BTU . .	2150.00 – 2500.00	75,000 BTU . .	3825.00 – 4500.00
30,000	2500.00 – 2950.00	100,000	4275.00 – 5150.00
45,000	2775.00 – 3375.00	125,000	5450.00 – 6500.00
50,000	2950.00 – 3575.00	150,000	6300.00 – 7350.00

FANJET DISTRIBUTION

Costs are for suspended fanjets including housing. Add $445 for motorized shutters.

FANJETS SIZE	FANJETS COST (EACH)	POLYTUBE DUCTING COST RANGE (PER LINEAR FOOT)
12"	$ 1160.00	0.88 – 1.10
18"	1390.00	1.00 – 1.19
24"	1750.00	1.13 – 1.46
30"	1990.00	1.24 – 1.59

VENTILATION FANS – WALL

Costs are for each fan unit complete with square or slant-wall housing. Automatic wall shutters are included at the high end of the range.

SIZE	COST RANGE	SIZE	COST RANGE
24"	1750.00 – 2360.00	42"	2500.00 – 3350.00
30"	1870.00 – 2550.00	48"	2925.00 – 3875.00
36"	2270.00 – 3025.00	54"	3425.00 – 4600.00

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HEATING
SECTION UIP 3

SOLAR HEATING SYSTEMS

The following costs are averages, including all ducting and ancillary equipment necessary for space heating either by use of liquid transfer-type or direct air-type collector systems. The costs for individual installations can vary greatly and every application must be examined for its own special design costs, locational considerations, varying capacity, type of absorption, medium and storage facilities employed. Any conventional backup system must be priced separately. For large commercial applications use the complete system costs only, where 5,000 to 10,000 square feet of collector area will fall within the Good cost range and 50,000 square feet and over will normally fit the Low cost range for pricing purposes.

LIQUID SYSTEM	LOW	AVG.	GOOD
Complete system based on square feet of collector area	$ 87.50	$ 139.00	$ 230.00
Cost of individual component items:			
Collectors, per sq. ft.	44.50	69.50	110.00
Storage tank, per gal./capacity	5.80	7.25	8.57
Pipe loops, heat exchangers, ducts and controls, complete	11300.00	18000.00	27800.00
Insulation, tank and pipes	2060.00	3600.00	6000.00

AIR SYSTEM	LOW	AVG.	GOOD
Complete system based on square foot of collector area	$ 84.00	$ 135.00	$ 215.00
Cost of individual component items:			
Collectors, per sq. ft.	42.25	67.00	104.00
Pebble bed storage container with gravel, 200 – 400 sq. ft., each	2750.00	4375.00	7200.00
Air handler, ducts, blowers and controls, complete	13100.00	20500.00	31800.00
Insulation, tank and pipes	1750.00	3100.00	5250.00

HEAT-RECOVERY SYSTEMS

The following costs are rough averages for complete air to air heat recovery systems. The costs for individual design systems can vary greatly due to the many variables involved.

SIZE	COST – MCFM
2,500 – 5,000 CFM	$9650.00 – $ 12100.00
5,000 – 15,000	5250.00 – 9850.00
15,000 – 20,000	3725.00 – 6750.00
20,000 – 30,000	2150.00 – 3800.00

HOT WATER BOILERS

Cost are for small natural gas or propane fired cast iron boilers or generators with insulated jackets and standard controls and include pumps and gauges. For oil-fired, add 10% to 20%; electric hydronic boilers, add 25% to 40%. Deduct $276 to $383 for standing pilot ignition systems. For thru-wall power venting, add $680 . Costs do not include piping or electric wiring. For large commercial and industrial boilers, see Section UIP 9. Add for expansion tank below.

RATED CAPACITY	COST RANGE
33,000 BTU	$ 3675.00 – $ 4725.00
60,000	3875.00 – 5150.00
75,000	3975.00 – 5450.00
100,000	4275.00 – 5800.00
125,000	4450.00 – 6150.00
150,000	4725.00 – 6650.00
175,000	4900.00 – 7000.00

FANJET DISTRIBUTION

Costs are for high-temperature steel expansion tanks installed with a maximum working pressure of 100 to 125 PSI

2 gallon	$ 230.00	– $ 314.00	10 gallon	$ 498.00	– $ 690.00
5	314.00	– 460.00	15	635.00	– 920.00
7	383.00	– 560.00	20	795.00	– 1160.00

HYDRONIC BASEBOARD HEATERS

Costs are for installed 7” high heaters used in hot-water systems with boilers and include tubing, panels, end caps and pivot-mounted dampers to regulate heat flow. Add $15.90 to $25.50 ($63.50 to $104.00 for 1-1/4” tube) for each corner application. For two-tier 1-1/4” tube assembly, add 50%; for three-tier, add 75%. For electric-heated hydronic, use electric baseboard costs from Page 3.

Length	1/2" Tube Cost Range	3/4" Tube Cost Range	1-1/4" Tube Cost Range
2’	$ 81.00 – $ 99.50	$ 99.50 – $ 126.00	$ 139.00 – $ 168.00
3’	93.50 – 119.00	119.00 – 152.00	177.00 – 215.00
4’	109.00 – 139.00	136.00 – 168.00	215.00 – 261.00
6’	139.00 – 177.00	177.00 – 222.00	283.00 – 352.00
7’	161.00 – 206.00	206.00 – 261.00	314.00 – 398.00
8’	206.00 – 261.00	261.00 – 314.00	359.00 – 436.00
9’	222.00 – 276.00	276.00 – 345.00	390.00 – 491.00
10’	230.00 – 292.00	292.00 – 368.00	421.00 – 530.00

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HEATING
SECTION UIP 3

ELECTRIC WALL FURNACES

Costs are for wall or recess mounted counterflow electric wall furnace and includes thermostat switches and installation. For rear package that directs some of the heated air to an adjoining room, add $153.00 to $276.00

RATED CAPACITY	COST RANGE	RATED CAPACITY	COST RANGE
2,000 watts	760.00 – 1060.00	6,000 watts	1190.00 – 1890.00
4,000	995.00 – 1430.00	8,000	1380.00 – 2310.00

RADIANT CABLE IN WALLS, FLOOR OR CEILING

Costs are for installed system and include cables, switches, controls and thermostats.

1,000 watts	535.00 – 735.00	4,000 watts	1430.00 – 2060.00
2,000	820.00 – 1190.00	5,000	1750.00 – 2430.00
3,000	1130.00 – 1490.00	6,000	1990.00 – 2900.00

RADIANT CEILING PANELS

Costs are for suspended or surface-mounted panel including controls.

200 watts	292.00 – 359.00	800 watts	635.00 – 690.00
400	414.00 – 467.00	1,000	760.00 – 795.00
600	535.00 – 590.00	1,200	870.00 – 915.00

INFRARED CEILING OR WALL HEATERS

Costs are for indoor heaters and include installation and prorated share of electrical circuits. For outdoor infrared installations add 10% to 20%.

500 watts	383.00 – 560.00	3,000 watts	900.00 – 1250.00
1,000	535.00 – 735.00	5,000	1170.00 – 1650.00
2,000	690.00 – 970.00	7,000	1800.00 – 2575.00

ELECTRIC BASEBOARD HEATERS

Costs are for baseboard heaters mounted directly on finished floor and include installation. Add $ 43.75 to $184.00 for wall thermostat. Add $63.50 to $314.00 for each built-in thermostat.

WATTS	LENGTH	COST RANGE	WATTS	LENGTH	COST RANGE
500	2’	230.00 – 620.00	2,000	8’	414.00 – 1170.00
750	3’	253.00 – 690.00	3,000	10’	515.00 – 1390.00
1,000	4’	283.00 – 840.00	5,000	11’	590.00 – 1610.00
1,500	6’	352.00 – 970.00	7,000	12’	645.00 – 1800.00

BATHROOM HEATERS

Costs are for ceiling or wall heaters including installation. Small wattage is bulb type. For ceiling light unit, add $63.50 to $104.00. For floor or kickspace units, add $ 206.00 to $276.00.

WATTS	WITHOUT FAN	WITH FAN
250	$ 135.00 – $ 276.00	$ 206.00 – $ 352.00
500	177.00 – 314.00	245.00 – 383.00
750	206.00 – 352.00	276.00 – 429.00
1,000	253.00 – 414.00	345.00 – 515.00
1,500	352.00 – 498.00	452.00 – 665.00
2,000	414.00 – 560.00	535.00 – 840.00

ELECTRIC INDUSTRIAL HEATERS WITH FAN

Costs are for 60-Hz fan-forced, ceiling or wall-mounted heaters used in industrial, commercial and farm applications, and include installation, summer fan switches and thermostats.

KW	COST RANGE	KW	COST RANGE
3	$ 1360.00 – $ 1870.00	15	$ 3025.00 – $ 4600.00
5	1470.00 – 2270.00	20	3800.00 – 5650.00
7.5	1990.00 – 3000.00	30	5100.00 – 7500.00
10	2270.00 – 3350.00	50	6950.00 – 10400.00

NOTE: For electric duct heaters with thermostat and relay, use above table deducting 25% to 40% from the cost range.

ELECTRIC CABINET UNIT HEATERS

Costs are for electric cabinet heaters with built-in thermostat and relays, and include installation, miscellaneous materials, connections and prorated share of electrical circuits. For recessed units, add 5% to 10%.

SURFACE MOUNTED

	LENGTH				
KW	3'	4'	5'	6'	7'
3	$ 3725.00	-----	-----	-----	-----
4	-----	$ 3800.00	-----	-----	-----
6	-----	3950.00	-----	-----	-----
8	-----	4100.00	$ 4725.00	-----	-----
10	-----	4275.00	4975.00	$ 5200.00	-----
12	-----	4375.00	5150.00	5300.00	-----
16	-----	-----	5250.00	5500.00	-----
20	-----	-----	-----	5650.00	$ 5950.00
24	-----	-----	-----	-----	6300.00

DUCTWORK

Costs are per linear foot of insulated, flexible, round polyester ducts and include supports, accessories and installation.

DIAM.	GRAY	METALIZED	DIAM.	GRAY	METALIZED
3”	$ 12.80	$ 13.45	12"	$ 25.50	$ 27.00
4”	13.20	14.05	13"	27.50	29.50
5”	14.05	14.90	14"	28.50	31.00
6”	14.70	15.70	15"	34.00	36.50
7”	17.70	18.65	16"	35.75	38.25
8”	18.55	19.65	17"	38.00	41.00
9”	19.25	20.70	18"	39.50	42.25
10”	19.65	21.15	19"	41.00	44.50
11”	24.90	26.50	20"	42.25	46.50

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HEATING AND COOLING

SECTION UIP 3

REFRIGERATION

The rated horsepower of the motor is approximately equivalent to the number of tons of refrigeration. One ton of refrigeration equals 12,000 B.T.U.

WINDOW UNITS

CAPACITY	COST EACH	CAPACITY	COST EACH
1/2-Ton	$1350.00 – $1610.00	1 Ton	$2120.00 – $2525.00
3/4	1610.00 – 2200.00	2	3175.00 – 3950.00

For units installed in wall sleeves, the cost on new work is approx. the same. For remodeling, add cost of making hole and repairing. Add $460.00 for 220-volt units. Add 25% to 50% for reverse cycle (heat pump) window or wall units.

PACKAGE UNITS

Average cost includes single duct and outlet, or use of heating ducts. Add $ 292.00 per additional ducted outlet or intake.

CAPACITY	COST EACH	CAPACITY	COST EACH
2 Ton	6100.00 – 6800.00	7½ Ton	21400.00 – 23900.00
3	9000.00 – 10000.00	10	27800.00 – 31500.00
5	14600.00 – 16200.00	15	41100.00 – 46300.00

COMPLETE COOLING SYSTEM

Costs are averages of engineered systems, including complete ductwork, zone controls, power, and electrical connections. Open buildings such as auditoriums, industrials, and markets will usually have a lower cost per ton than buildings requiring a larger number of temperature zones and intricate piping or ductwork such as good offices, hospitals, etc. The costs shown are medians in three cost ranges.

	COST PER TON OF CAPACITY		
CAPACITY	LOW	AVERAGE	GOOD
5 Ton	$ 4600.00	$ 5650.00	$ 7000.00
10	4375.00	5450.00	6700.00
15	4275.00	5200.00	6500.00
25	4100.00	5100.00	6250.00
50	3875.00	4825.00	5800.00
75	3800.00	4600.00	5650.00
100	3725.00	4450.00	5600.00
150	3575.00	4350.00	5300.00
250	3450.00	4200.00	5200.00
400	3350.00	3975.00	4975.00

EVAPORATIVE COOLERS

WINDOW UNITS

Cubic Feet per Minute	Cost per Unit	Cubic Feet per Minute	Cost per Unit
1500	775.00 – 995.00	4500	1520.00 – 1990.00
2500	995.00 – 1300.00	5500	1780.00 – 2280.00
3500	1230.00 – 1610.00	6500	1990.00 – 2575.00

ROOF OR WALL UNITS

Costs include one outlet, add $292 per additional outlet. For farm application, deduct 15% to 25%.

Cubic Feet per Minute	Cost per Unit	Cubic Feet per Minute	Cost per Unit
3,000	2150.00 – 2700.00	8,000	3950.00 – 5150.00
4,000	2500.00 – 3025.00	9,000	4275.00 – 5600.00
5,000	2925.00 – 3575.00	12,000	5450.00 – 6950.00
6,000	3350.00 – 3975.00	14,000	6400.00 – 8100.00
7,000	3800.00 – 4825.00	16,000	7600.00 – 9600.00

AIR CURTAINS

Air curtains in place, including necessary connections, cost from $ 55.50 to $ 93.50 per square foot of entrance opening. Add 70% to 80% for particulate filtered units and 80% to 100% for heated units.

COMBINED HEATING AND COOLING

PACKAGE UNITS

Costs are averages per installation, with single duct system and necessary vents, plumbing, power, and electrical connections for each unit. Costs are for commercial split systems of gas fired, forced air heating with gas or electric refrigerated cooling, and for heat pumps (reverse cycle refrigeration). Residential-type systems will cost 20% less for a 2-ton unit to 40% less for 5 tons.

Add $ 359 to $ 445 for each additional ducted outlet or intake, and $565 for each control to commercial systems only.

In cooler climates, heat pumps need supplemental heat. If suppl mental electric coils are installed in ducts, add $845 for each coil. If a complete extra heating system is used, price separately.

RATED COOLING CAPACITY	SPLIT SYSTEM	HEAT PUMP
2 Ton	$ 9000.00 – $ 9650.00	$ 8050.00 – $ 8750.00
3	13100.00 – 14000.00	11800.00 – 12800.00
5	20800.00 – 22700.00	19200.00 – 20800.00
7½	30500.00 – 33300.00	27800.00 – 30500.00
10	39600.00 – 43000.00	36800.00 – 40400.00
15	57500.00 – 63000.00	54000.00 – 59000.00
20	74750.00 – 82250.00	71000.00 – 77750.00
25	92000.00 – 101000.00	87750.00 – 96250.00
30	108000.00 – 119000.00	104000.00 – 115000.00

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HEATING AND COOLING

SECTION UIP 3

COMBINED HEATING AND COOLING

ENGINEERED SYSTEMS

Costs of these systems vary greatly with climate and degree of temperature control required. Buildings with large open areas and few controls need much less ductwork and fewer control items than buildings with many subdivided rooms with individual controls. In cooler climates larger heating units are involved, while costs are listed per ton of cooling capacity. Costs are averaged medians of three cost ranges and include the complete unit, installation, power, connections, and all ancillary items.

The lowest priced installation would normally be in a sheltered area in an industrial plant or market with few separate temperature zones. The highest priced installation per ton would normally be found in good and excellent buildings with occupancies such as offices, hospitals, hotels, and others which require many temperature zones, in an open area experiencing severe wind chill.

The determining cost factors are the year round ambient temperatures and the number of temperature controls, plus the quality of equipment and design of the complete system.

RATED CAPACITY	COST PER TON		
(Tons).	LOW	AVERAGE	GOOD
10	$ 5800.00	$ 8450.00	$ 12500.00
15	5550.00	8200.00	12100.00
20	5450.00	8050.00	11900.00
30	5250.00	7750.00	11600.00
40	5100.00	7600.00	11200.00
50	4975.00	7450.00	11000.00
75	4825.00	7200.00	10800.00
100	4675.00	7000.00	10500.00
150	4575.00	6800.00	10200.00
200	4450.00	6650.00	9950.00
300	4225.00	6400.00	9650.00
500	4050.00	6100.00	9350.00
750	3900.00	5950.00	9050.00
1,000	3800.00	5800.00	8900.00
1,500	3675.00	5600.00	8550.00

MISCELLANEOUS RESIDENTIAL ITEMS

Costs installed in place with necessary vents and/or connections.

	COST RANGE
Air purifier, electronic	$ 1640.00 – $ 2650.00
Air purifier, filtered air	765.00 – 1330.00
Humidifier	795.00 – 1160.00
Ceiling fan	206.00 – 590.00
add for lighting	79.50 – 735.00
Sauna heater, door unit	1610.00 – 3050.00
Rock unit	2750.00 – 4675.00

Attic exhaust fans and ventilators, see Section UIP 4.

AIR TERMINAL UNITS

VARIABLE AIR VOLUME

Costs are each and include reheat coils, controls and installation.

CAPACITY (CFM)	COST RANGE
100 – 200	$1190.00 – $ 1650.00
200 – 350	1230.00 – 1740.00
350 – 500	1350.00 – 1900.00
500 – 750	1390.00 – 2000.00
750 – 1,000	1520.00 – 2170.00
1,000 – 1,250	1620.00 – 2280.00
1,250 – 1,500	1740.00 – 2420.00
1,500 – 2,500	2220.00 – 3150.00
2,500 – 5,000	3175.00 – 4575.00

AIR TERMINAL UNITS

CONSTANT VOLUME

Costs are for single duct units and include reheat coils, controls and installation. Double row reheat coils cost $ 321 to $ 760 per square foot.

CAPACITY (CFM)	COST RANGE
100	$ 995.00 – $ 1390.00
300	1100.00 – 1520.00
500	1160.00 – 1650.00
750	1220.00 – 1740.00
1,000	1420.00 – 2000.00
1,250	1490.00 – 2080.00
1,500	1560.00 – 2220.00
2,000	2050.00 – 2850.00
2,500	2575.00 – 3600.00
3,000	2850.00 – 4025.00
4,000	3325.00 – 4725.00

DEHUMIDIFIERS

Costs are for residential dehumidifiers with built in frost and overflow controls and automatic humidistat to maintain selected humidity. Add $50 to $ 99 dollars for separate hose connection and drip tray

CAPACITY PINTS/24 HRS.	COST RANGE
15	$ 715.00 – $ 1090.00
25	820.00 – 1260.00
30	920.00 – 1350.00
40	1090.00 – 1520.00
50	1320.00 – 1810.00

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HEATING, COOLING AND VENTILATING

SECTION UIP 3

AIR-CONDITIONING REQUIREMENTS

Air conditioning requirements are greatly dependent on the occupancy of the structure. The following figures give typical quantities by occupancy in square feet per ton of cooling capacity, except as otherwise stated. Figures do not include cooling for ice making, cold storage, etc. The range of areas includes approximately 80% of all cases.

OCCUPANCY	SQUARE FEET/ TON
Fast food restaurants, small bars and taverns	100 – 225
Lounges and restaurants	125 – 275
Radio & TV stations, laboratories, barbershops, drug stores	150 – 300
Luxury high-value residential occupancies	150 – 450
Hospitals	180 – 280
Specialty retail shops, dental offices	200 – 300
Churches, auditoriums, theaters	200 – 375
Fraternal buildings, governmental, clubhouses, country clubs	225 – 400
Colleges, banks, department stores, libraries, museums	250 – 450
Schools, light industrials, offices, medical offices, telephone	275 – 500
Mortuaries, veterinary hospitals	300 – 475
Bowling alleys, retail stores, shopping centers	300 – 500
Hotels, motels, post offices	280 – 450
Discount stores, loft buildings, nursing homes	350 – 500
Dormitories, food markets	280 – 550
Vocational schools, jails	400 – 600
Residential occupancies	400 – 750
Fire stations and service garages	550 – 750
Food markets with energy recovery systems	600 – 750

Auditoriums	1 ton per each	15 **to** 25 seats
Theaters	1 ton per each	10 **to** 20 seats
Bars and taverns	1 ton per each	7 **to** 12 seats

HOME AUTOMATION SYSTEMS

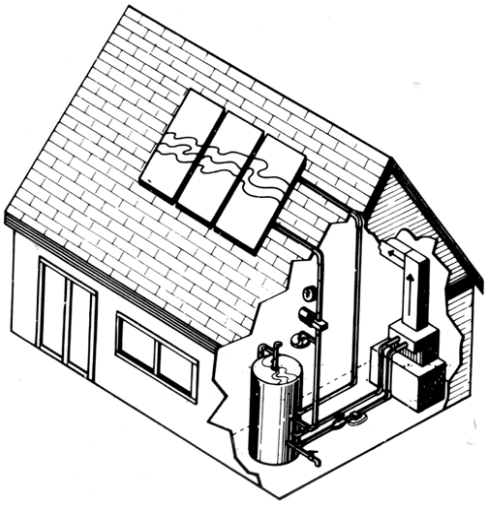
	COST RANGE
Home automation base system, one monitor and control panel for temperature, lighting, appliances and water heater control	$ 3575.00 – $ 8300.00
Custom system, including security (2040 zones), no camera base, up to 5 additional temperature zones	15500.00 – 22200.00
Two monitors, up to 20 temperature zones	27400.00 – 41000.00

BUILDING AUTOMATION SYSTEMS

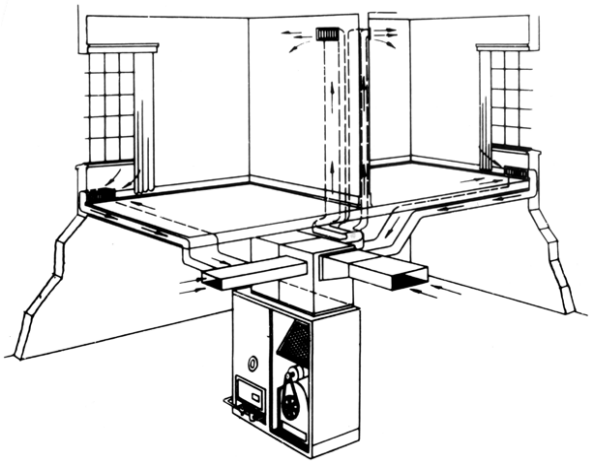
Commercial energy management or building automation system costs can vary greatly depending on the amount and sophistication of the monitoring and control equipment involved for HVAC, lighting, automation and life safety control, and energy management capabilities.

The following cost ranges are in some cases based on only a few projects and should be considered as very rough guides at best. We would suggest that whenever possible, survey, bid or contract costs should be obtained.

	COST RANGE
Small buildings, under 10,000 sq. ft., total cost	
Single function, no computer*	$ 1110.00 – $ 12200.00
Medium buildings, under 50,000 sq. ft., total cost	
multifunction, stand alone (microprocessor)	5550.00 – 22200.00
*Note - add for computer monitoring	12200.00 – 44200.00
Large buildings, over 50,000 sq. ft.,	
Fully distributed multifunction and central station	
cost per point, HVAC only	359.00 – 1800.00
Lighting only	168.00 – 875.00
Integrated, energy, fire, security	2220.00 – 4975.00



SOLAR HEATING SYSTEM

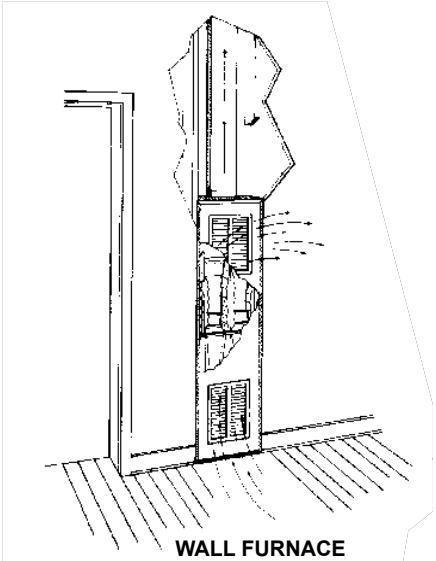


FORCED-AIR FURNACE

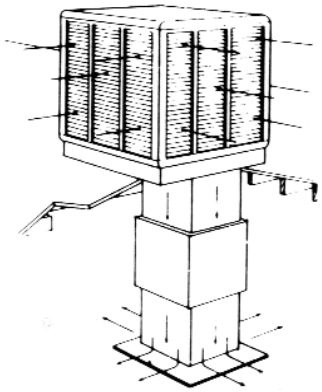
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HEATING, COOLING AND VENTILATING

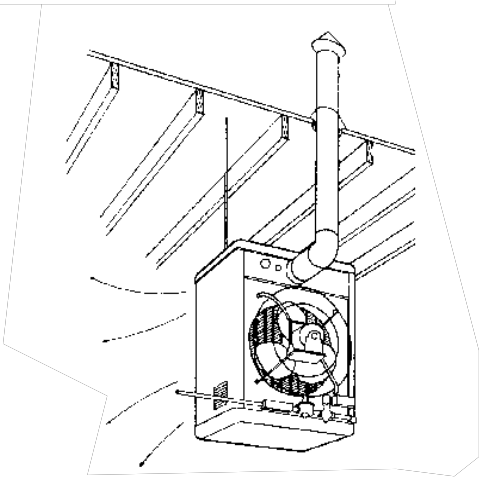
SECTION UIP 3



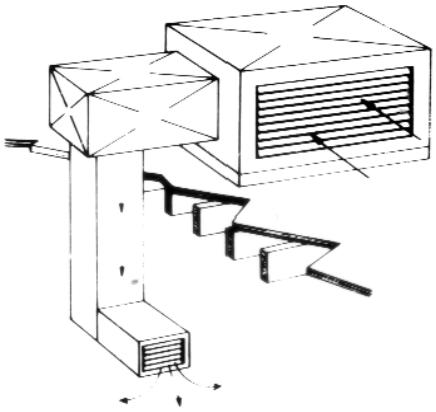
WALL FURNACE



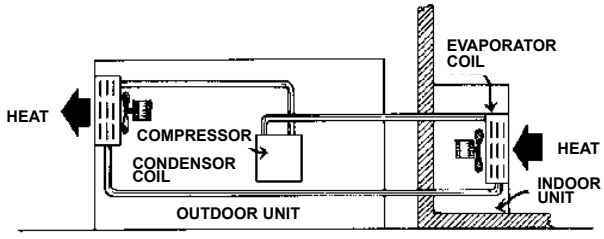
EVAPORATIVE COOLER



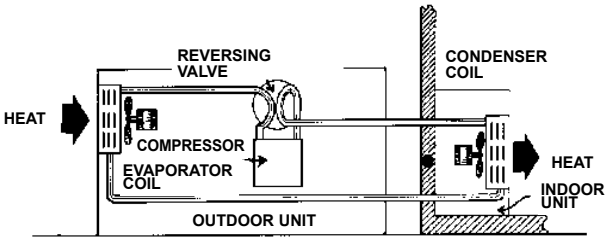
SPACE HEATERS (SUSPENDED)



PACKAGE AIR CONDITIONING

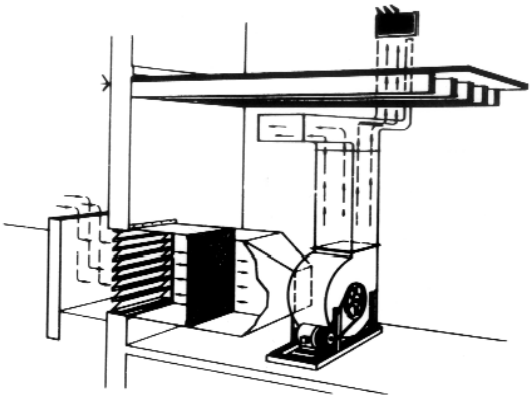


COOLING CYCLE

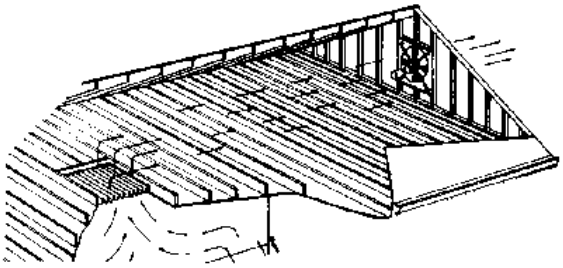


HEATING CYCLE

HEAT PUMP



VENTILATION, BLOWER



VENTILATION, FAN ONLY

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MECHANICAL COSTS

SECTION UIP 3

PERCENTAGE OF TOTAL COST

The following table records the results of studies of many recently completed buildings, by occupancy, giving the percentage of total contract cost spent on the mechanical items, exclusive of elevators and sprinklers. The average used is the median, and the high and low percentages which are given, do not include extremes, but are computed to include approximately 90 percent of all cases within the given range (45% each side of the median). For electrical percentages, see Page 18.

OCCUPANCY	HEATING ONLY			HEATING & COOLING			PLUMBING		
	LOW	MED.	HIGH	LOW	MED.	HIGH	LOW	MED.	HIGH
Apartments, Classes A and B	3.7	5.9	8.9	6.1	8.5	11.5	7.2	8.5	10.5
Apartments, Classes C and D	2.6	4.0	7.7	6.4	8.6	12.3	7.5	10.9	15.3
Auditoriums and theaters	-----	-----	-----	7.6	11.5	18.1	3.0	5.5	10.6
Banks	-----	-----	-----	4.5	7.4	12.6	2.3	4.3	7.0
Bowling alleys	-----	-----	-----	8.6	12.3	17.5	2.5	4.6	7.6
Churches	-----	-----	-----	6.6	9.8	13.9	1.8	3.6	8.3
City halls, courthouses, etc.	-----	-----	-----	7.3	10.8	15.4	5.0	7.6	9.9
Clubhouses and parish halls	3.5	6.0	10.2	8.2	10.4	14.0	5.0	7.2	11.9
Convalescent hospitals and nursing homes	4.1	6.1	8.7	7.8	11.4	17.3	8.1	10.6	13.6
Department stores	-----	-----	-----	9.1	11.8	14.9	2.8	4.0	5.8
Discount stores	3.4	5.4	9.3	8.3	10.7	13.4	2.3	4.3	7.3
Dormitories	4.7	7.1	10.6	7.0	10.1	13.8	6.6	8.4	10.9
Fraternal buildings and community centers	-----	-----	-----	8.4	12.1	17.6	4.7	7.0	10.5
Homes for the elderly	2.8	4.8	8.6	5.9	8.4	12.5	8.1	10.2	12.5
Hospitals	-----	-----	-----	11.3	16.8	22.9	7.9	10.8	15.1
Industrials	3.1*	6.2*	12.3*	11.5	16.0	23.0	3.8	6.4	11.5
Libraries and museums	4.4	6.1	8.5	8.3	12.1	17.4	3.1	5.3	7.8
Markets	-----	-----	-----	7.1	10.0	14.2	4.2	6.1	9.2
Medical office buildings and clinics	-----	-----	-----	6.2	9.7	16.4	4.6	8.5	13.0
Motels and hotels	-----	-----	-----	4.6	8.7	14.8	6.6	10.2	13.5
Office buildings	4.3	6.5	9.5	6.6	10.2	17.5	2.6	4.7	8.5
Restaurants	-----	-----	-----	8.0	12.2	15.6	5.3	9.8	13.7
Retail stores and shops	3.7	4.8	6.1	6.1	8.9	12.6	2.2	4.6	8.8
Schools, elementary	5.6**	9.0**	14.1**	10.3	13.2	16.8	4.8	7.3	11.1
Schools, secondary	-----	-----	-----	7.6	11.7	17.5	5.9	7.6	9.6
Schools, colleges	-----	-----	-----	7.4	13.4	19.8	3.8	6.7	12.1
Single-family residences	2.4	3.8	6.2	6.8	8.3	10.3	6.9	8.6	10.8
Telephone buildings	-----	-----	-----	9.5	14.0	19.2	3.7	5.5	8.0
Warehouses	1.9*	4.3*	9.1*	5.9	9.4	14.9	2.0	4.0	7.2

*With office cooling **With administrative office cooling

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PLUMBING
SECTION UIP 3

EXPLANATION

In the Segregated Cost Sections, plumbing is priced on the basis of floor area or average cost per fixture. Many times, it is desirable to give a more detailed breakdown, especially in industrial occupancies. These tables will enable the assessor to detail plumbing costs by type of fixture.

The costs listed include cost of the fixture, labor, miscellaneous materials for rough and finish, and contractor's profit and overhead. Average amounts of water, gas, and waste lines within the building are included but not exterior sewer or utility lines or meters.

Industrial plumbing may be priced from this page, but long pipe and house costs range from $ 3875 to $11000 per fixture; it is recommended that if priced from the following table, add for pipe and sewer runs of over 25' per fixture. Store and office fixtures have a normal range from $ 3125 to $ 5800 per fixture and may be priced from the table in a like manner. Toilet partitions and restroom accessory costs are found in Section UIP 2.

WATER HEATERS

Residential type:	20 Gal.	30 Gal.	40 Gal.	50 Gal.	70 Gal.
Minimum, competitive	1080.00	1110.00	1170.00	1250.00	-----
Average, 5 year guarantee	1250.00	1360.00	1440.00	1580.00	-----
Good, 10 to 15 year guarantee	-----	1590.00	1740.00	1970.00	2380.00

For hot-water recovery expansion chamber, add $620.00.
Insulation jacket, add $69.00.
Point-of-use water heaters cost $328.00- $ 550.00.

Tankless:

Gas	1910.00 – 4575.00
Electric	915.00 – 2290.00

Commercial heater:

100 GPH recovery	4050.00 – 5600.00	150 GPH recovery	4700.00 – 6200.00
Storage tank, lined		Recirculating pumps	
120 gallons	3400.00 – 4700.00	1 inch	610.00 – 1100.00
240	4700.00 – 6200.00	2	1040.00 – 1470.00
360	6250.00 – 7600.00	3	1470.00 – 2525.00

SOLAR HOT WATER HEATERS

Liquid heat collectors (two to three panels), 80 to 120 gallon water tank with heat exchanger and pump. Where system is integrated with space heat, reduce cost by 50%.

Sunbelt climates	6850.00 –10400.00
Northern climates	9000.00 –12400.00
Hot water collector, tank combination, single unit:	3125.00 – 4575.00

WATER SOFTENERS

Average cost of automatic units completely installed with necessary fittings. Rated capacity is in grains of hardness converted per week. Commercial, per regeneration, add 100%.

RATED CAPACITY	COST RANGE	RATED CAPACITY	COST RANGE
20,000	2320.00 – 2675.00	60,000	2675.00 – 3075.00
30,000	2400.00 – 2700.00	100,000	2750.00 – 3250.00
40,000	2440.00 – 2775.00	200,000	4275.00 – 5250.00

WATER FILTRATION Residential type:

complete treatment system (filter, clarify, soften, refine)	5600.00 –12900.00

FIXTURES

	LOW	AVG.	GOOD	EXCL.
Average for residential fixtures	1070.00	1640.00	2500.00	3875.00
high value	3950.00	4875.00	6200.00	7900.00
Bathtub	1090.00	1740.00	2825.00	4575.00
Handicapped, walk–in door	7600.00	8800.00	10400.00	12100.00
Fiberglass tub–shower	2030.00	2700.00	3550.00	4800.00
Handicapped unit	2675.00	4425.00	7300.00	12300.00
Deluxe or whirlpool	4775.00	7050.00	10400.00	15300.00
Environmental enclosure	42900.00	48900.00	55750.00	63000.00
Bidet	1320.00	1830.00	2525.00	3550.00
Deluxe	3875.00	4725.00	5900.00	7500.00
Drinking fountain	1040.00	1240.00	1500.00	1830.00
Refrigerated water coolers	1910.00	2250.00	2625.00	3075.00
Water coolers, hot and cold water	-----	2310.00	2825.00	3475.00
Floor drain	550.00	725.00	960.00	1290.00
Large sinks	1290.00	1460.00	1710.00	1970.00
Gas piping, residential, per unit	411.00	535.00	655.00	840.00
Hose bib	153.00	214.00	290.00	396.00
Hydrant, commercial wall	442.00	670.00	1080.00	1680.00
Laundry tray, single	620.00	820.00	1060.00	1390.00
Double	775.00	1040.00	1380.00	1830.00
Lavatory/Sink	670.00	1030.00	1520.00	2310.00
Deluxe (hand painted, vessel, etc.)	2775.00	3825.00	4800.00	6250.00
Polished metals (brass, Copper, etc.)	2775.00	3900.00	5200.00	6950.00
Marble or granite	3250.00	4575.00	6200.00	8250.00
Pedestal mount	2100.00	2350.00	2625.00	2950.00
Deluxe (hand painted, etc.)	3600.00	4575.00	5550.00	6800.00
Carved marble or granite	8550.00	12500.00	18800.00	27700.00
Roof drain	540.00	795.00	1120.00	1650.00
Shower, stall*	1090.00	1430.00	1830.00	2350.00
Prefabricated unit	1500.00	2030.00	2700.00	3550.00
Metal cabinet	481.00	620.00	795.00	970.00
Deluxe (add steam below)	3600.00	5250.00	7500.00	10800.00
Environmental enclosure	28300.00	38600.00	53250.00	72750.00
Handicapped unit	2575.00	4100.00	6700.00	10700.00
Shower base only	481.00	670.00	925.00	1290.00
Deluxe	1460.00	1770.00	2220.00	2700.00
Shower, over tub or extra hardware	304.00	396.00	540.00	700.00
Deluxe hardware sets, each	780.00	1090.00	1560.00	2160.00
Sink, kitchen, single	840.00	1050.00	1320.00	1640.00
Multibowl	1030.00	1410.00	1920.00	2675.00
Deluxe bowls	3825.00	5200.00	7150.00	9850.00
Undersink, hot water dispenser	365.00	457.00	540.00	640.00
Water purification, faucet	570.00	795.00	1080.00	1440.00
Sink, service (janitor)	1110.00	1430.00	1850.00	2380.00
Steam generators, residential baths	2525.00	2900.00	3300.00	3800.00
Sump pump, 1/3 to 1/2 hp, 1–1/4 " outlet	695.00	850.00	1060.00	1300.00
Urinal	1430.00	2030.00	2825.00	4050.00
Water closet	1120.00	1640.00	2340.00	3375.00
Deluxe	3600.00	4800.00	6450.00	8550.00
Handicapped	1290.00	1590.00	1970.00	2440.00
Air compressor, add	1710.00	2030.00	2380.00	2825.00
Electric, incinerator type	3400.00	3875.00	4250.00	4725.00
Composting	3675.00	4275.00	5050.00	5900.00
Wet bar	915.00	1080.00	1240.00	1470.00
Deluxe	1830.00	2625.00	3800.00	5400.00
Rough–in only, for fixture	570.00	685.00	840.00	1030.00
Clean outs	183.00	261.00	343.00	481.00
Vent, dryer or appliance	214.00	268.00	365.00	489.00
Add for oil-rubbed bronze, nickel, or stainless hardware, per set	76.50	130.00	222.00	381.00

*Tile, glass doors, and enclosures are included under Interior Construction in the Segregated Cost Sections. If separate costs for these items are desired, tile showers cost $1180 to $2100; tile floor only, $268 to $ 396 shower doors, $290 to $1050; tile tub surround, $665 to $1110; and tub enclosure, $381 to $1340. Custom shower or tub enclosures will cost from $2400 to $21400. Grab bars, see Section UIP 2.

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PLUMBING AND WELL DRILLING

SECTION UIP 3

INDUSTRIAL WASH SINKS

	COST RANGE
Enameled cast iron, rectangular, 30'' wide:	
4 feet long, four faucets	$3075.00 – $4025.00
8 feet long, eight faucets	5200.00 – 6800.00
Add 20% for stainless steel.	

	36"	54"
Circular wash fountains:		
Polished cement (granite chips)	$4575.00 – $5850.00	$5200.00 – $6750.00
Terrazzo (marble chips)	4725.00 – 6050.00	5450.00 – 6950.00
Enameled steel	5200.00 – 6750.00	5850.00 – 7500.00
Stainless steel	5600.00 – 7150.00	6450.00 – 8250.00
Semicircular wash fountains:		
Polished cement (granite chips)	3975.00 – 5200.00	4575.00 – 5900.00
Terrazzo (marble chips)	4100.00 – 5450.00	4800.00 – 6200.00
Enameled steel	4700.00 – 5900.00	5250.00 – 6800.00
Stainless steel	5200.00 – 6650.00	5900.00 – 7350.00
Two-person wash fountains:		
Enameled steel	2000.00 – 2775.00	
Stainless steel	2440.00 – 3250.00	
For infrared control, add	1340.00 – 2700.00	

INSTALLED PIPE

In large buildings with few fixtures, some consideration must be given to the length of pipe runs from the fixture to the point where the pipe is stubbed out of the building. Cost of runs longer than an average of approximately 25' per fixture should be added from Section UIP 9, or the following abbreviated table. Costs are per linear foot of run including fittings and valves.

	GALVANIZED	COPPER	CAST IRON	PLASTIC
1/2"	$ 16.00	$ 15.05	-----	-----
3/4"	19.45	18.10	-----	-----
1"	26.50	22.70	-----	-----
1-1/2"	34.75	34.00	$ 27.50	$ 15.65
2"	42.75	44.00	35.50	17.60
3"	61.00	76.50	42.00	23.75
4"	84.00	125.00	50.50	30.75
6"	161.00	229.00	69.00	46.50
8"	-----	-----	107.00	-----
10"	-----	-----	153.00	-----

WATER-SUPPLY METERS

Installed costs do not include piping.

TYPE	SIZE	CAPACITY	COST RANGE
Bronze, screwed	3/4"	30 gpm	$ 335.00 – $ 465.00
	1"	50	465.00 – 570.00
	1-1/2"	100	805.00 – 1040.00
	2"	160	1410.00 – 1740.00
Bronze, flanged	3"	360	4400.00 – 5250.00
	4"	500	6600.00 – 7600.00

INDUSTRIAL SHOWERS

Average cost-in-place including rough and finish plumbing.

	ENAMELED STEEL	STAINLESS STEEL	ADD FOR RECEPTORS
Column showers:			
Circular, 5 person	$ 4250.00	$ 5900.00	$ 3250.00
Semicircular, 3 person	3400.00	4625.00	2650.00
Corner, 2 person	3200.00	4100.00	2700.00
Multi-stall showers:			
Circular, 5 person	10000.00	13000.00	3600.00
Semicircular, 3 person	7800.00	10600.00	3125.00
Corner, 2 person	6800.00	9150.00	2775.00
Emergency shower	1880.00	– 2650.00	
Multi-nozzle, up to 12 spray	3175.00	– 4700.00	
decontamination	8150.00	– 9850.00	
Add for eye wash	955.00	– 1350.00	

GREASE INTERCEPTORS

As found in restaurants and meat packing houses. Installed costs do not include piping.

SIZE, GPM	CAPACITY, LB	COST RANGE
CAST IRON		
7	14	$ 1530.00 – $ 2470.00
15	30	2440.00 – 4100.00
50	100	5250.00 – 8400.00
FABRICATED STEEL		
100	200	$ 9850.00 – $13000.00
250	500	18900.00 – 23700.00
500	1,000	30200.00 – 36900.00

SEWAGE DISPOSAL

SEPTIC TANK: Average costs installed and connected in normal soil.

750 gallon	$1430.00 – $2290.00	2,000 gallon	$ 3975.00 – $5550.00
1,000	1970.00 – 2850.00	4,000	8650.00 – 11100.00
1,250	2650.00 – 3500.00	6,000	13200.00 – 16500.00
1,500	3125.00 – 4250.00	10,000	22200.00 – 27700.00

Drainfields will typically cost 1 to 1.5 times the tank cost. Add 10% to 15% for elevated fields (for fill, see Section UIP 1) plus $ 7950 to $ 18300 for a grinder pump system.

Leaching lines, tile, per linear foot	$ 14.20 – $ 22.65
Plastic pipe, per linear foot	8.57 – 14.20
add for gravel or stone backfill, per cubic foot	0.69 – 2.18

Cesspools, $1110 plus $ 43.25 per linear foot of depth.

WATER WELLS

Average costs of water wells, 100 to 1,000 feet deep. Costs include drilling, casings, gravel pack, setup, testing, and miscellaneous costs up to point of actual operation excluding pumps. Costs of vertical turbine pumps are listed in Volume I, page 267. The low and high costs do not represent the minimum or maximum possible, but the centers of low and high cost ranges.

	COST PER FOOT OF DEPTH		
SIZE	LOW	AVERAGE	HIGH
5"	$ 33.50	$ 50.50	$ 64.00
9"	59.50	76.50	97.50
13"	84.00	109.00	135.00
17"	110.00	135.00	168.00
21"	132.00	161.00	207.00
25"	161.00	191.00	237.00
29"	183.00	222.00	268.00

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FIRE PROTECTION
SECTION UIP 3

FIRE PUMP EQUIPMENT

HORIZONTAL SHAFT (CENTRIFUGAL, 100 PSI)

Prices include installation costs, coupling and motor or engine on a steel base, plus relief valve and waste cone. Controller must be added for electric units. Diesel engine costs include battery, trickle charger, coupling and automatic controller. Electric motors are 230/460 volt, 3 phase, 60 Hz.

GPM	HP	RPM	ELECTRIC	DIESEL
500	60	1800	$ 39600.00	$ 123000.00
750	75	1800	48400.00	133000.00
1000	100	1800	58250.00	141000.00
1500	125	1800	72500.00	153000.00
2000	150	1800	86000.00	162000.00
2500	200	1800	98250.00	168000.00

VERTICAL SHAFT (TURBINE, MULTISTAGE)

The following prices include a vertical electric motor with thrust bearing. Costs for diesel engines include a right angle drive, coupler, metal skid, fuel tank, battery, trickle charger and automatic controller.

GPM	HP	RPM	PUMP ONLY	ADD FOR ELECTRIC	ADD FOR DIESEL
500	50	1800	$ 48700.00	$ 6500.00	$ 79750.00
750	75	1800	55500.00	8050.00	89000.00
1000	100	1800	62000.00	10600.00	97250.00
1500	125	1800	73500.00	14700.00	109000.00
2000	150	1800	86500.00	19100.00	119000.00
2500	200	1800	98750.00	24000.00	127000.00

PUMP DRIVERS

Diesel engines with coupling, skid, battery, fuel tank and battery charger. For horizontal drive.

For 1000-gal., 100-psi pump . . . . . $ 51000.00
For 2000-gal., 100-psi pump 62750.00

Electric motors (squirrel-cage induction, 3-phase, 60-cycle, driproof).

VOLTAGE	RPM	HORSEPOWER					
		30	50	75	100	150	200
230 – 440	1800	2470.00	4200.00	6450.00	8550.00	12700.00	17100.00

ELECTRIC DRIVE CONTROLLERS

Combined Manual and Automatic, Across-the-Line

VOLTS	AMP*	HORSEPOWER					
		30	50	75	100	150	200
220	30000	14700.00	18200.00	24400.00	32000.00	-----	-----
	75000	27600.00	30100.00	31600.00	34300.00	40200.00	47800.00
440	25000	13800.00	15300.00	17000.00	18200.00	19700.00	22000.00
	60000	27100.00	28600.00	31300.00	32000.00	34300.00	35000.00

Combined Manual and Automatic, Reduced Voltage

VOLTS	AMP*	HORSEPOWER					
		30	50	75	100	150	200
220	30000	18200.00	22200.00	27100.00	-----	-----	-----
	75000	31300.00	34000.00	38400.00	41900.00	52250.00	66000.00
440	25000	14700.00	18600.00	24000.00	27600.00	34000.00	39600.00
	60000	27600.00	32600.00	38400.00	40200.00	45700.00	51500.00

DIESEL DRIVE CONTROLLERS 13500.00–14100.00

*Circuit breaker interrupting capacity.

FLOW METERS

PUMP SIZE	COST RANGE	PUMP SIZE	COST RANGE
500 gpm	2220.00 – 3825.00	2000 gpm	3075.00– 7600.00
750	2320.00 – 4400.00	3000	3825.00– 9000.00
1000	2440.00 – 5200.00	4000	4700.00–10600.00
1500	2750.00 – 6450.00		

AUXILIARY LIGHT PACKS

Average cost in place for emergency lighting, including ancillary connections. Low end of range applies to lead acid batteries and the high end to nickel cadmium batteries.

Single . . . . . $ 805.00 – $1240.00
Double . . . . . 810.00 – 1330.00
Multipacks . . . . . 1040.00 – 1650.00

SPRINKLER SYSTEMS

Costs per square foot may be found in the Segregated Cost Sections. Refer to the section applicable to the type of building under consideration (see discussion in SEG INTRO).

COST PER HEAD			
LOW	AVERAGE	GOOD	HIGH COST
$ 320.00	$ 411.00	$ 555.00	$ 725.00

SMOKE-AND HEAT-ACTUATED ALARM SYSTEMS

Most commercial installations are leased. The costs below represent the installation costs charged to the user. For in place costs see if dependent detectors under Fire Alarm Systems costs on next page. range.

Commercial base cost $ 2200.00 plus $0.69 per square foot of protected areas.

Residential ionization smoke detectors cost $ 153.00 to $ 283.00 installed. Residential gas detectors, carbon monoxide, radon, etc., cost $76.50 to $161.00 installed.

Residential battery operated smoke detectors cost $ 50.50 to $ 145.00 installed.

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FIRE PROTECTION
SECTION UIP 3

FIRE ALARM SYSTEMS

Buildings under 75' height.	COST RANGE
Control panel at lobby	$1680.00 – $2110.00
Add for each zone	343.00 – 481.00
Pull station	472.00 – 810.00
Smoke detector	635.00 – 955.00
Water leak detector	670.00 – 805.00
*Pull station with horn or bell alarm	222.00 – 268.00
*Pull stations	136.00 – 161.00
*Rate of rise, heat detectors	176.00 – 290.00
*Smoke–actuated door controls	374.00 – 550.00
*Smoke detectors	183.00 – 320.00
*Water leak detectors	365.00 – 489.00

*Use same costs for independent stations and detectors in buildings over 75' high.

Buildings 75' and above.	
Control panel at fire control room (with street access for fire dept.) 75 zones	$19100.00–$28700.00
Speakers connected with microphone at control panel	222.00– 261.00
Emergency telephone, 5 jacks	465.00– 775.00
Sprinkler water flow detector	261.00– 605.00
Time, date, location printer	13000.00– 18500.00
Battery standby system	13500.00– 19700.00

STANDPIPE

Costs per story, installed with necessary fittings. Add for hydrants below.

	2"	4"	6"	8"
Outside installation	$1040.00	$2250.00	$3450.00	$4575.00
Add for 1st-story Siamese connection	1290.00	1830.00	2350.00	4925.00
Inside wet standpipe	1360.00	2775.00	4100.00	5450.00

HYDRANTS

1-1/2" hose connection	$389.00	2-1/2" hose connection	$640.00
1-1/2" gate valve	635.00	2-1/2" gate valve	765.00
2-way Siamese 4" connection	1740.00	2-way Siamese 6" connection	2310.00

Inside fire hydrants including 75' of 1-1/2" hose, valve, rack nozzle installed without cabinet, cost $1040.00 to $1460.00.

HOSE CABINETS

Steel	$ 472.00–$ 670.00	Aluminum	$550.00 –$ 810.00
Stainless Steel	870.00– 1290.00		

PLAYPIPES (Each)

Short, 15"	$ 183.00	Long, 30"	$ 335.00

HOSES (Per linear foot)

1-1/2"	$ 3.63 –$ 4.65	2 -1/2"	$ 5.42 –$ 7.24

HOSEHOUSES

Steel	$1970.00–$2350.00	Aluminum	$2625.00–$3225.00

EXTINGUISHER CABINETS

Steel	$335.00 –$ 535.00	Aluminum	$389.00 –$ 620.00
Stainless Steel	640.00 – 1050.00		

EXTINGUISHERS

	COST EACH		COST EACH
Portable, 2-1/2 gal. (pressurized) antifreeze	$ 365.00	water	$ 268.00
Carbon dioxide, with hose and horn, 2-1/2#	268.00	20#	670.00
5#	320.00	*50#	2440.00
10#	396.00	*100#	4700.00
Dry chemical, regular types 2-1/2#	130.00	30#	625.00
5#	183.00	*45#	2440.00
10#	283.00	*150#	4250.00
20#	411.00	*350#	6450.00

*Costs include wheeled carts.

For all-purpose dry chemical types, add 5% to 10%. For halon type, add 100%.

CARBON DIOXIDE SYSTEMS

Flooding Systems, cost per cubic foot.

TANK SIZE	SMALL (500 cu. ft.)	MEDIUM (3,000 cu. ft.)	LARGE (30,000 cu. ft.)
Standard hazards	$ 5.70	$ 4.41	$ 2.89
Electric hazards	7.70	5.50	3.04
Fur vaults	10.30	6.93	3.34

Local Application, cost per square foot.

	SMALL (25 sq. ft.)	MEDIUM (250 sq. ft.)	LARGE (1,000 sq. ft.)
Coated surfaces	$ 222.00	$ 135.00	$ 94.50
Liquid surfaces	283.00	191.00	138.00

AIR FOAM SYSTEMS

High Expansion, cubic feet per minute.

5,000 cfm	$ 8250.00	10,000 cfm.	$ 11200.00
15,000 cfm	15500.00		

Low Expansion, (protein), cost per tank.

TANK SIZE	FUEL OIL	GASOLINE
500 sq. ft.	$ 15300.00	$ 16500.00
1,000	17300.00	19100.00
1,500	19500.00	22200.00
2,000	22300.00	25600.00
2,500	25000.00	29500.00
3,000	28400.00	34300.00

HALON 1301 SYSTEMS

Cost in place per cubic foot including ionization detection with approximately 6% total flooding of electrical hazards. Recharging of system may or may not be viable, since costs may double. Acceptable substitutes, such as FM200, will currently cost about 10% more, while Inergen systems will cost 10% to 20% less than costs listed below.

1,000 cu. ft.	$13.35 – $16.70	3,000 cu. ft.	$ 8.00 – $10.65
30,000 cu. ft.	3.43 – 5.32		

DRY CHEMICAL SYSTEMS

Restaurant Hood/Duct	LOW	AVG.	HIGH COST
Cost per nozzle	$ 740.00	$ 1040.00	$ 1430.00

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ELECTRICAL
SECTION UIP 3

SERVICE ENTRANCE EQUIPMENT

SINGLE PHASE, 120/240 V

Includes combination meter socket and circuit breaker panel, circuit breakers, riser conduit cables, weatherhead, ground rod, clamp, cable and fittings. For group meters, add $290 to $428 per meter. Add $151 for each ground fault interrupter breaker. For underground service, deduct 5% to 10%.

CAPACITY	LOW	AVG.	GOOD
30 amperes	$ 585.00	$ 710.00	$ 865.00
60	865.00	1030.00	1230.00
100	1310.00	1540.00	1800.00
200	2310.00	2675.00	3100.00

3-PHASE, 120/208 V

Cost includes meter socket, main breaker or switch, riser conduit, cable, entrance cap, ground rod, clamp, cable and fittings.

60 amperes	$2110.00	$2310.00	$ 2525.00
100	2925.00	3175.00	3450.00
200	5050.00	5450.00	5950.00
400	9200.00	9900.00	10700.00
600	13400.00	14500.00	15600.00
800	17800.00	19100.00	20500.00
1000	21700.00	23400.00	25300.00
1200	25900.00	28100.00	30100.00

GROUNDING SYSTEMS

Grounding rod, complete, up to 10' conductor cable	$ 265.00	$ 347.00	$ 459.00
Water pipe clamp system	142.00	186.00	250.00
Foundation connector system	372.00	685.00	1290.00
add for lightning terminal points	159.00	194.00	226.00
add for each additional foot of cable	1.67	3.95	9.31
add for arrestor, 175 V to 650 V	226.00	314.00	428.00
Electrolytic tube system, complete	1890.00	3000.00	4800.00
add for ground resistance tester	7200.00	8450.00	10000.00

SAFETY SWITCHES

Fused, single throw, 3-pole, 600-V, NEMA #1, indoor type. For weatherproof boxes, add 20% to 40%; dustproof, add 100% to 200%; explosionproof, add 300% to 400%.

SIZE	COST RANGE	SIZE	COST RANGE
30 amperes	$411.00 – $ 476.00	400 amperes	$3450.00 – $ 4100.00
60	585.00 – 685.00	600	5450.00 – 6600.00
100	960.00 – 1170.00	800	9350.00 – 10900.00
200	1590.00 – 1920.00	1,200	12300.00 – 14400.00

SWITCHGEAR

Listed costs are typical cost ranges of standard types and styles of switch and panelboard equipment and vary with voltage, number of circuits, number of wires and phases of current, and safety features. Add costs of circuit breakers.

SERVICE SWITCHGEAR COST RANGE

AMPS	RESIDENTIAL Light Commercial	COMMERCIAL Light Industrial	INDUSTRIAL Institutional
100	525.00 – 835.00	----- – -----	----- – -----
200	1010.00 – 1460.00	----- – -----	----- – -----
225	3725.00 – 4575.00	6100.00 – 6800.00	----- – -----
400	6300.00 – 7450.00	9600.00 – 10600.00	----- – -----
600	9200.00 – 10600.00	12900.00 – 14400.00	33900.00 – 46200.00
800	11900.00 – 13600.00	16100.00 – 18100.00	41400.00 – 56500.00
1000	14700.00 – 16600.00	19100.00 – 21400.00	48900.00 – 65750.00
1200	17300.00 – 19400.00	21900.00 – 24500.00	55500.00 – 74500.00
1600	----- – -----	27200.00 – 30600.00	68750.00 – 91000.00
2000	----- – -----	32200.00 – 36200.00	81250.00 – 107000.00
2500	----- – -----	----- – -----	95750.00 – 124000.00
3000	----- – -----	----- – -----	109000.00 – 141000.00
4000	----- – -----	----- – -----	134000.00 – 170000.00

DISTRIBUTION SWITCHGEAR (Light, Heat or Power Centers)

AMPS			
100	1300.00 – 1700.00	----- – -----	----- – -----
150	1670.00 – 2110.00	2140.00 – 2925.00	----- – -----
225	2050.00 – 2850.00	3075.00 – 3900.00	5800.00 – 7550.00
400	2925.00 – 3950.00	4375.00 – 5850.00	8450.00 – 11300.00
600	3800.00 – 5100.00	5950.00 – 7550.00	11300.00 – 14900.00
800	4500.00 – 6050.00	7200.00 – 9200.00	13900.00 – 18400.00
1200	5800.00 – 7800.00	9600.00 – 12300.00	18400.00 – 24400.00
1600	----- – -----	11700.00 – 14800.00	22500.00 – 30100.00
2000	----- – -----	----- – -----	26100.00 – 34700.00

CIRCUIT BREAKERS FOR SWITCHGEAR

AMPS	120 VOLTS	240 – 480 VOLTS	600 VOLTS
20 – 60	$100.00 – $323.00	$339.00 – $ 635.00	$428.00 – $945.00
70 – 100	306.00 – 670.00	620.00 – 1070.00	835.00 – 1590.00
125 – 225	915.00 – 1450.00	1170.00 – 1860.00	1640.00 – 2650.00
400 – 600	----- – -----	3400.00 – 4175.00	3800.00 – 5250.00
700 – 800	----- – -----	4675.00 – 6300.00	5400.00 – 7450.00

CIRCUIT BREAKERS FOR METALCLAD SWITCHGEAR
600 VOLTS

AMPS		AMPS	
800	$ 15100.00 – $24500.00	3,000	$ 71750.00 – $109000.00
1,600	28600.00 – 47900.00	4,000	115000.00 – 164000.00
2,000	40400.00 – 63500.00		

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ELECTRICAL
SECTION UIP 3

TRANSFORMERS

DRY TYPE Single phase, 240/480 V primary, 120/240 secondary		OIL FILLED Three phase or Y, 5 KV or 15 KV with taps 277/480 V secondary	
SIZE	COST RANGE	SIZE	COST RANGE
3 KVA	1190.00– 1300.00	150 KVA	32000.00 – 37900.00
5	1590.00– 1840.00	300	46100.00 – 54500.00
7.5	2010.00– 2310.00	500	60250.00 – 71750.00
10	2450.00– 2825.00	750	76250.00 – 89500.00
15	3100.00– 3650.00	1000	88500.00 –105000.00
25	4325.00– 5050.00	1500	111000.00 –130000.00
37.5	5650.00– 6450.00	2000	130000.00 –153000.00
50	6650.00– 7800.00	2500	146000.00 –173000.00
75	8500.00–10000.00	3000	161000.00 –191000.00

SUBSTATIONS

High-voltage unit substations, complete with transformers, breakers and grounding, cost from $151 to $570 per KVA.

EXAMPLES

RATING	COST	COST/ KVA	RATING	COST	COST/ KVA
150 KVA	86250.00	570.00	1,000 KVA	206000.00	217.00
500	150000.00	306.00	2,000	286000.00	142.00

POWER WIRING

The following tables may be used in lieu of actual costs.

The tables indicate an estimate of the average costs installed in place. They include (where applicable) wire, fittings, hangers, bends, termination, and contractor's overhead and profit. Costs are based on a maximum height above the floor of 12'. Power wiring for motors and motor costs are found in Section UIP 9, Page 6.

RIGID CONDUIT AND WIRING (EXPOSED)

Costs include three conductors of the maximum wire size in each size conduit. They also include: tees, ells, junction boxes, bends, hangers and fittings.

SIZE	COST/ FOOT	SIZE	COST/ FOOT
1/2"	16.15	2"	39.00
3/4"	19.30	2-1/2"	56.00
1"	22.50	3"	76.50
1-1/4"	26.25	3-1/2"	106.00
1-1/2"	31.00	4"	150.00

WIREWAY GUTTERS
(Per linear foot, without wires)

4" x 4" 41.00 6" x 6" 56.00 8" x 8" 85.00

BUS DUCT, 3-PHASE, 600-VOLT

Cost per linear foot for indoor plug in type, including typical fittings, but not disconnect or circuit breaker type switches to machines. For weatherproof duct, add 15% – 25%.

3-POLE COST RANGE			4-POLE COST RANGE		
AMPS	ALUMINUM	COPPER	AMPS	ALUMINUM	COPPER
225	124.00 – 161.00	178.00 – 250.00	225	151.00 – 201.00	258.00 – 354.00
400	178.00 – 217.00	275.00 – 347.00	400	209.00 – 265.00	362.00 – 476.00
600	234.00 – 283.00	372.00 – 468.00	600	290.00 – 339.00	476.00 – 585.00
800	306.00 – 347.00	476.00 – 585.00	800	339.00 – 420.00	580.00 – 725.00
1000	372.00 – 411.00	585.00 – 700.00	1000	428.00 – 484.00	685.00 – 835.00
1350	459.00 – 515.00	780.00 – 915.00	1350	530.00 – 610.00	890.00 – 1070.00
1600	525.00 – 585.00	915.00 –1070.00	1600	620.00 – 690.00	1020.00 – 1220.00
2000	650.00 – 730.00	1130.00 –1290.00	2000	775.00 – 835.00	1260.00 – 1460.00
2500	805.00 – 875.00	1380.00 –1590.00	2500	935.00 – 1030.00	1530.00 – 1800.00
3000	945.00 – 1030.00	1660.00 –1920.00	3000	1120.00 – 1220.00	1800.00 – 2090.00

UNDERGROUND WIRING

Average costs per linear foot in place, in trench. Includes three con ductors of the maximum wire size and terminations for each size of conduit, trenching not included.

Excavation, backfill and compaction for the trench, assuming average soil conditions, use $ 1.17 per cubic foot.

PLASTIC DUCT		TRANSITE DUCT		GALVANIZED RIGID CONDUIT	
SIZE	COST	SIZE	COST	SIZE	COST
2"	25.25	2"	25.25	2"	34.75
3"	52.50	3"	51.50	3"	70.00
3-1/2"	75.00	3-1/2"	74.00	3-1/2"	98.50
4"	106.00	4"	105.00	4"	139.00

POWER POLE

40' high 1890.00

5' wood cross arm with insulators and hardware 500.00 each

TRANSFER SWITCHES
(Automatic, single phase, 3 pole, 600 V, or indoor type.)

AMPS	COST	AMPS	COST
30	5850.00	600	34500.00
60	7550.00	800	40400.00
100	9900.00	1000	49500.00
150	12900.00	1200	57500.00
225	18900.00	1600	68000.00
400	26700.00	2000	76750.00

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ELECTRICAL
SECTION UIP 3

GENERATORS

Costs of home generator sets including connection. Lower end of range is with rope starter. Upper end is with battery and automatic starter.

	COST RANGE
1,000 watts	1500.00 – 1960.00
1,500	2010.00 – 2550.00
2,000	2460.00 – 3300.00
3,000	3325.00 – 5650.00
4,000	4050.00 – 6800.00
5,000	4825.00 – 8050.00
7,000	6600.00 –10800.00

STANDBY POWER

Emergency generators for institutional, commercial and other buildings, complete with controls for immediate operation in the event of loss of the primary power source, cost from $570 to $2360 per KW. The costs vary with the size and type of driver. 120-volt battery systems cost $3275 to $ 4850 per kilowatt.

EXAMPLES

RATING	GAS/ GASOLINE DRIVE	COST/ KW	RATING	DIESEL DRIVE	COST KW
10 KW	23600.00	2360.00	30 KW	51000.00	1690.00
15	30000.00	2000.00	100	109000.00	1090.00
30	45800.00	1530.00	150	139000.00	935.00
100	96000.00	960.00	300	219000.00	730.00
150	122000.00	810.00	500	294000.00	585.00
			750	428000.00	570.00
			1,000	572000.00	570.00

ELECTRICAL OUTLETS

The following costs may be used to arrive at a more detailed estimate of electrical costs than is obtained by using the costs per square foot of floor area that are given in the Segregated Cost Sections. The following costs apply to convenience and lighting outlets only, not to power wiring for equipment or heating. Cost per outlet includes allowance for service, but not for fixtures, panelboards, safety switches or circuit breakers.

Normal residential wiring will usually be in the low and average ranges, and commercial and public buildings in the average to high ranges. Large industrial buildings and warehouses with relatively few outlets and long, heavy wiring runs may run much higher on a cost-per-outlet basis. Explosion proof receptacles can run 300% to 400% more.

COST PER OUTLET

TYPE	LOW	AVG.	GOOD	EXCL.
Nonmetallic sheathed cable (Romex)	83.50	100.00	123.00	150.00
Armored cable (BX)	95.50	119.00	145.00	178.00
Flexible conduit	124.00	155.00	194.00	234.00
Thinwall conduit (EMT)	139.00	178.00	217.00	283.00
Rigid conduit	159.00	194.00	250.00	306.00
Low–voltage, telephone, TV, sound	54.50	59.00	63.00	68.50
Coaxial cable or fiber optic	61.00	70.50	80.50	92.00

For ground fault interrupter outlets, add . . . 12.20 – 24.35 each

OTHER CIRCUITS (NONMETALLIC)

	LOW	AVG.	GOOD
Garbage disposer (includes switch and receptacle)	209.00	265.00	323.00
Dishwasher	194.00	250.00	314.00
Dryer	323.00	362.00	420.00
Range and oven	372.00	451.00	540.00
Room air conditioner	379.00	428.00	468.00
Water heater	298.00	354.00	444.00

For flex conduit – add 20% to 40% to above circuits.
For EMT – add 30% to 50%.

LIGHTING FIXTURES

TYPE			
Incandescent, surface	97.00	201.00	595.00
Open commercial, standard	127.00	161.00	217.00
Recessed or adjustable	169.00	306.00	540.00
Pendant	74.00	130.00	250.00
Vaportight	234.00	347.00	500.00
Explosion proof	570.00	990.00	1700.00
Illuminated exit	178.00	323.00	525.00
add for battery backup	372.00	428.00	500.00
Chandeliers	515.00	2060.00	7250.00
High value*	15500.00	29400.00	57500.00
Fluorescent, surface or pendant			
Strip, one lamp	151.00	226.00	347.00
Two lamps	161.00	250.00	372.00
Four lamps	226.00	339.00	500.00
Six lamps	500.00	610.00	730.00
Eight lamps	730.00	810.00	930.00
Recessed, troffer or diffused			
Two to four lamps	226.00	372.00	570.00
Six to eight lamps	790.00	945.00	1170.00
add for air–handling type	108.00	142.00	178.00
add for emergency lighting ballast	515.00	570.00	620.00
High intensity discharge			
Mercury vapor, recessed	730.00	1070.00	1590.00
Pendant, vaportight	865.00	990.00	1170.00
Explosionproof	1590.00	1920.00	2310.00
High bay	1290.00	1700.00	2370.00
High pressure sodium, low bay	1440.00	1610.00	1800.00
High bay	1870.00	2450.00	3175.00
Pendant, vaportight	1220.00	1350.00	1510.00
Explosion proof	1950.00	2090.00	2230.00
Metal halide, low bay	1170.00	1380.00	1660.00
High bay	1590.00	2170.00	2875.00
add for twin bay	226.00	290.00	379.00
add for bay wire guards	75.50	134.00	226.00
Occupancy sensors,			
Photoelectric cell	476.00	570.00	700.00
Daylighting, automatic			
Dimming system	2800.00	3525.00	4600.00
each extra sensor, add	620.00	685.00	790.00
Auto switch sensors,			
Manual dimmer	155.00	283.00	515.00
Track lighting, four foot track	372.00	459.00	555.00
Eight foot track	700.00	845.00	1070.00

***NOTE:** Chandelier costs will vary greatly due to the materials, finish and intricacy of design. Fixtures classified by age or beauty as having antique or historical value, or designed by name artists, must be valued as art objects by the fine arts specialists, where the costs can easily run seven to ten times the listed costs.

For emergency lighting fixtures, see Page 11. For outdoor lighting, see Sections UIP 13 and 14.

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SECURITY PROTECTION
SECTION UIP 3

SECURITY ALARM SYSTEMS

SENSORS:	RESIDENTIAL COST RANGE	COMMERCIAL COST RANGE
Point area detectors (per opening)		
Alarm glass (wired)	159.00 – 323.00	323.00 – 935.00
Screen (burglarproof, wired)	159.00 – 620.00	620.00 – 1230.00
Capacitance proximity detectors	----- – -----	935.00 – 4675.00
Photoelectric, beam interrupt	685.00 – 1550.00	1550.00 – 4675.00
Field of view changes	18.60 – 159.00	18.60 – 159.00
Stress detectors, floor	217.00 – 476.00	620.00 – 1550.00
pressure mat	159.00 – 620.00	----- – -----
Trip or pull switches (wire trap)	18.60 – 75.50	----- – -----
Switch sensors,		
Magnetic, (per switch)	169.00 – 339.00	169.00 – 935.00
Mercury	169.00 – 339.00	178.00 – 339.00
Self contained (battery)	83.50 – 159.00	----- – -----
Vibration detectors	92.00 – 234.00	169.00 – 620.00
Volume detectors		
Motion detectors,		
Infrared	108.00 – 685.00	234.00 – 1230.00
Microwave	178.00 – 685.00	323.00 – 1230.00
Ultrasonic	178.00 – 685.00	323.00 – 1230.00
Self contained (battery)	178.00 – 620.00	----- – -----
Sound detectors,	–	–
Sensing units	178.00 – 685.00	323.00 – 1230.00
Remote monitoring	1460.00 – 7750.00	1460.00 – 7750.00
Keypad locks, stand alone,	–	–
per door	585.00 – 1890.00	1420.00 – 2850.00
ANNUNCIATORS:		
Alarms, local		
Bells	159.00 – 323.00	476.00 – 935.00
Buzzers	18.60 – 37.00	18.60 – 37.00
Horns	75.50 – 347.00	476.00 – 935.00
Sirens	75.50 – 347.00	476.00 – 3100.00
Lights	159.00 – 323.00	476.00 – 935.00
CONTROL UNITS (TRANSMISSION–DISPLAY):		
Control panel	585.00 – 1010.00	1530.00 – 3125.00
Remote annunciator	323.00 – 935.00	610.00 – 1870.00
add for: event recorders	----- – -----	459.00 – 1590.00
Transponders	----- – -----	775.00 – 3100.00
Electrical switch locks	234.00 – 935.00	775.00 – 1550.00
Remote transmitters, ea.	83.50 – 118.00	----- – -----
Standby power, battery	37.00 – 314.00	234.00 – 6200.00
Wireless key fob	100.00 – 347.00	----- – -----
PERSONNEL/ID VERIFICATION SYSTEMS:		
Coded badge/card access	----- – -----	890.00 – 17500.00

CCTV SYSTEM

	COST RANGE
Camera, fixed, standard resolution	161.00 – 1610.00
High resolution	1610.00 – 20100.00
Each simulated or dummy camera	16.15 – 121.00
Each miniature covert camera	242.00 – 3225.00
Mounts	16.15 – 242.00
Monitor	161.00 – 1800.00
Recorder, video	79.50 – 1610.00
Digital recording system	970.00 – 11300.00
Web broadcast system	323.00 – 4025.00
Complete systems	3225.00 – 10500.00

COMPLETE RESIDENTIAL UNITS

(Includes sensors, annunciators and controls)

Hardwired, basic economy system	347.00 – 1780.00
Extensive, deluxe unit	2700.00 – 29500.00
Wireless	476.00 – 4725.00

ACCESS SYSTEMS

Metal detector, wand	323.00 – 530.00
Walk-thru	5600.00 – 12400.00
Portable	14100.00 – 16100.00
X-ray, table top	42500.00 – 49800.00
Conveyerized	62750.00 – 92500.00
Fluoroscopic letter/parcel	7700.00 – 18900.00
Biometrics, standalone, per station, palm	4725.00 – 7100.00
Finger print	2850.00 – 6150.00
Hand geometry	4225.00 – 6850.00
Retinal/iris	3800.00 – 6500.00
Facial recognition	11800.00 – 23600.00
Voice	2390.00 – 6150.00
System control monitor	23600.00 – 119000.00
Access portals, bulletproof glass,interlocking door system	
with volumetric sensing	51000.00 – 93750.00
add for facial/print recognition	7700.00 – 11800.00
add for visitor recognition system	7100.00 – 8300.00
ID verification, coded badge/card base cost	1190.00 – 3550.00
add for each reader	347.00 – 700.00
add per card/ badge	8.06 – 22.55
add for computer system (up to 8 readers)	4725.00 – 7100.00
Generic, state driver license reader	2850.00 – 5150.00

DURESS PANIC BUTTON SYSTEMS

Push button, wall or desk, visual/audio console	1190.00 – 3525.00
add per button station	306.00 – 347.00
add per station with PA speaker	347.00 – 420.00
Wireless, location monitor/alarm console	47300.00 – 119000.00
add per ID/tracking pager	585.00 – 1190.00

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PHOTOVOLTAIC
SECTION UIP 3

PHOTOVOLTAIC POWER SUPPLY

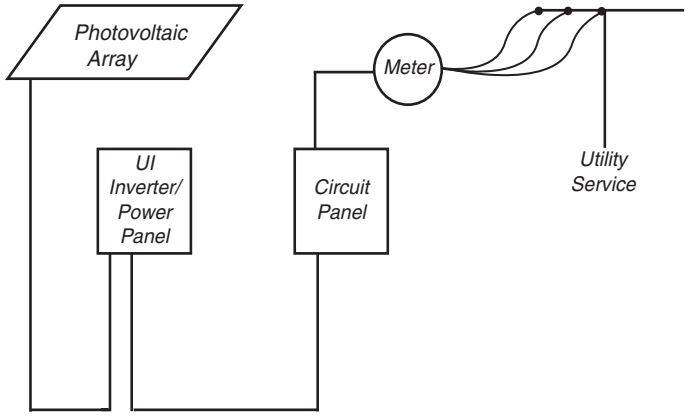
These solar power systems run from basic direct current systems, which consist of a simple solar module directly running DC equipment to Utility Inter-tie (UI) systems incorporating inverters, control panels and meters. The UI system allows the user to use the utility rather than a battery bank for storage. The excess power is sold back to the utility at a rate determined by the utility or credited to the user through the use of additional utility meters.

PRE-ENGINEERED RESIDENTIAL UTILITY INTER-TIE SYSTEM

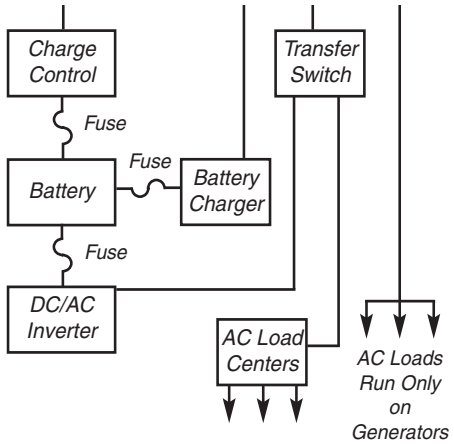
Includes solar modules, inverter, panel and meter. The number of panels required is determined by several factors including power demand, percentage of solar supplementation, and available square footage. For battery backup, additional components to be added from below.

	COST RANGE	
12 - 24 module system, installed, per module	2450.00 –	3275.00
36 - 48	2110.00 –	2675.00
72 - 96	1890.00 –	2310.00
Unit Costs:		
Solar Modules, 10 watt - 30 watt	323.00 –	700.00
40 watt - 60 watt	580.00 –	945.00
75 watt - 120 watt	1050.00 –	1750.00
285 watt - 300 watt	4050.00 –	4525.00
Module mounts, top-of-pole, 1 - 6 panels, per panel	108.00 –	372.00
8 - 16 panels	100.00 –	226.00
side-of-pole, 1 - 4 panels	226.00 –	428.00
ground/roof mounts, 1 - 6 panels	186.00 –	331.00
low - profile	159.00 –	314.00
two-tier, 6 - 12 panels	92.00 –	178.00
Automatic tracker, 4 - 12 panels, per panel	379.00 –	610.00
Power panel, includes inverter, DC disconnect, bypass switch, 1500 watt - 3000 watt	5850.00 –	7800.00
3600 watt - 4800 watt	7450.00 –	10900.00
7200 watt - 8000 watt	14100.00 –	19500.00

Unit Costs continued:	COST RANGE	
Inverters, true sine wave, 125 watt	484.00 –	635.00
250 watt	875.00 –	1010.00
600 watt	1280.00 –	1400.00
1100 watt	1590.00 –	2030.00
modified sine wave, 500 watt	850.00 –	1070.00
1500 watt	2090.00 –	2260.00
2400 watt	2900.00 –	3125.00
3600 watt	3325.00 –	3525.00
grid inter-tie only, 1000 watt	3850.00 –	4050.00
1500 watt	4725.00 –	4925.00
2000 watt	4775.00 –	5000.00
2500 watt	5800.00 –	5950.00
true sine wave grid-tie, 4000 watt - 5500 watt	7500.00 –	8500.00
Industrial inverter, three-phase, grid-tie, 10000 watt - 20000 watt	19800.00 –	41200.00
30000 watt - 50000 watt	50750.00 –	88250.00
100000 watt - 150000 watt	169000.00 –	228000.00
225000 watt - 300000 watt	262000.00 –	303000.00
Standard deep-cycle battery, 6 volt	226.00 –	484.00
12 volt	428.00 –	580.00
industrial deep-cycle battery packs, 12 volts	4375.00 –	6150.00
24 volt	8100.00 –	11700.00
48 volt	15500.00 –	22300.00
Charge controllers, 12, 24, or 48 volt, 12 - 20 amps	169.00 –	484.00
30 amp - 35 amp	290.00 –	635.00
40 amp - 60 amp	428.00 –	1110.00
Battery charger, 30 amp - 40 amp	372.00 –	610.00
75 amp	790.00 –	970.00
Transfer switch, (DC) 120 amp	186.00 –	250.00
240 amp	314.00 –	610.00
Transfer switch, (AC) 120 amp	234.00 –	585.00
240 amp	835.00 –	1230.00
Meters	570.00 –	730.00



UTILITY INTER-TIE BASIC COMBINATION



PHOTOVOLTAIC ARRAY / GENERATOR COMBINATION

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ELECTRICAL
SECTION UIP 3

PERCENTAGE OF TOTAL COST

The following table records the results of studies of many recently completed buildings, by occupancy, giving the percentage of total contract cost spent on the electrical items. The average used is the median and the high and low percentages which are given do not include extremes, but are computed to include approximately 90 percent of all cases within the given range (45% each side of the median).

OCCUPANCY	TOTAL ELECTRICAL			SERVICE & DISTRIBUTION			LIGHTING & CONTROL			SPECIAL SYSTEMS		
	LOW	MED.	HIGH	LOW	MED.	HIGH	LOW	MED.	HIGH	FIRE ALARM & DETEC-TION MEDIAN	STANDBY POWER MEDIAN	OTHER* MEDIAN
Apartments, Classes A and B	6.6	8.9	11.9	4.1	5.2	6.8	1.5	1.9	2.7	0.5	0.4	0.9
Apartments, Classes C and D	5.3	7.5	10.6	3.1	4.3	6.3	1.7	2.1	2.9	0.5		0.6
Auditoriums and theaters	6.1	9.1	13.5	2.6	4.2	7.1	2.2	2.9	4.1	0.4	0.7	0.9
Banks	8.0	11.2	16.2	3.0	4.0	6.1	2.2	3.3	5.6	0.8	0.4	2.7
Bowling alleys	7.9	10.6	13.9	4.7	6.2	8.3	2.5	3.2	4.2	0.5		0.7
Churches	6.4	9.1	13.0	2.5	3.5	5.3	3.1	4.2	6.0	0.7		0.7
City halls, courthouses, etc.	7.6	10.3	13.6	4.2	5.5	7.5	2.8	3.4	4.4	0.5	0.6	0.3
Clubhouses and parish halls	6.3	9.3	13.5	3.1	4.2	6.1	2.6	3.9	5.9	0.6		0.6
Convalescent hospitals and nursing homes	8.4	11.5	15.7	3.3	4.8	7.3	2.9	3.9	5.4	0.9	0.6	1.3
Department stores	9.4	12.3	16.6	5.1	6.9	9.7	3.4	4.1	5.3	0.6	0.5	0.2
Discount stores	6.0	8.8	13.2	3.1	4.5	7.0	2.5	3.4	4.8	0.4		0.5
Dormitories	7.0	8.6	10.8	3.8	4.0	4.6	2.0	2.8	3.9	0.6		1.2
Fraternal buildings and community centers	6.2	8.9	12.7	3.0	4.1	5.9	2.6	3.5	5.0	0.6		0.7
Homes for the Elderly	7.4	9.4	13.8	4.2	5.6	8.0	1.5	1.7	2.5	0.7		1.4
Hospitals	9.8	13.5	17.9	5.6	7.4	10.0	2.6	3.5	5.0	0.3	0.8	1.5
Industrials†	7.3	10.9	16.9	3.7	5.4	7.8	3.2	4.6	6.2	0.4		0.5
Libraries and museums	8.4	12.7	18.0	4.0	5.7	8.3	3.5	4.8	6.8	0.6		1.6
Markets	9.9	13.3	16.9	6.2	7.8	10.0	2.9	3.6	4.9	0.4	0.7	0.8
Medical office buildings and clinics	7.4	10.4	13.9	4.0	5.2	7.0	2.4	3.2	4.6	0.5	0.6	0.9
Motels and hotels	4.9	8.4	11.0	2.8	4.2	6.4	1.8	2.6	3.8	0.3	0.4	0.9
Office buildings	6.7	10.0	15.6	3.1	4.6	7.2	2.6	4.0	6.4	0.4	0.5	0.5
Restaurants	8.0	11.8	15.7	5.4	7.3	9.0	2.3	3.9	5.8	0.3		0.3
Retail stores and shops	7.1	10.0	15.5	4.0	6.0	9.4	3.0	3.8	5.1			0.2
Schools, elementary	7.2	9.6	13.9	3.2	4.4	6.5	2.6	3.5	5.2	0.5	0.5	0.7
Schools, secondary	7.0	9.3	12.7	3.0	4.2	6.1	2.4	3.0	4.2	0.5	0.5	1.1
Schools, colleges	6.2	9.5	13.5	3.1	4.3	6.3	2.5	3.4	5.0	0.5	0.6	0.7
Single-family residences	4.5	5.3	7.8	3.6	3.9	4.5	0.7	1.2	2.4	0.2		
Telephone buildings	8.2	12.3	16.6	3.4	5.3	8.6	0.9	1.2	1.9	0.9	4.7	0.2
Warehouses	4.5	7.5	13.0	2.2	4.3	8.6	2.0	2.7	3.8	0.5		

†Lighting and utility outlet only, without power wiring.

*Other special systems include intercom, sound, TV cabling, security, etc.

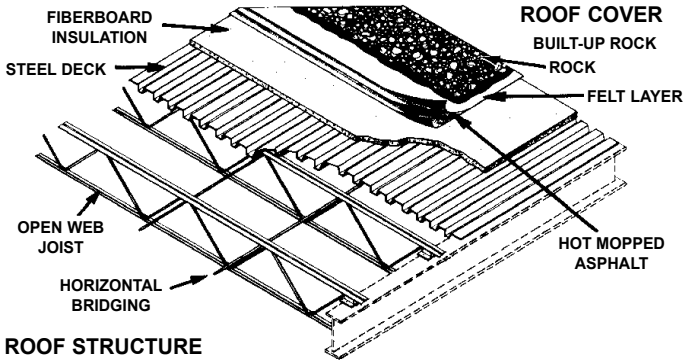
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ROOFS
SECTION UIP 4

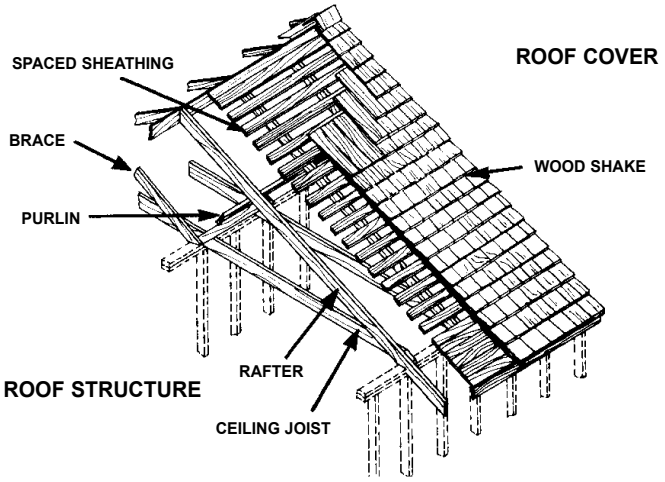
EXPLANATION

The general headings for roof structures in the Segregated Cost sections include the complete roof structure: rafters, purlins, sheathing or deck, bracing and ties. If a more detailed cost is needed, use the tables in this section.

TYPICAL ROOFS



ROOF STRUCTURE



ROOF STRUCTURE

WOOD RAFTERS

Cost per square foot includes necessary ties, ridges and bracing. Ceiling joists should be added. For pitched roofs, use the multipliers in the Roof Slope table. For prefabricated trussed rafters, see Page 3, in this section.

SPACING

DIMENSIONS (inches)	12" o.c.	16" o.c.	20" o.c.	24" o.c.	32" o.c.	48" o.c.
2 x 4	3.45	2.78	2.34	2.06	1.66	1.24
2 x 6	4.10	3.31	2.81	2.50	2.03	1.46
2 x 8	4.76	3.87	3.27	2.87	2.34	1.75
2 x 10	6.01	4.89	4.15	3.67	2.99	2.24
2 x 12	7.09	5.75	4.89	4.29	3.45	2.58
3 x 4	5.57	4.51	3.86	3.39	2.77	2.06
3 x 6	7.66	6.16	5.24	4.58	3.70	2.77
3 x 8	9.65	7.84	6.64	5.79	4.73	3.51
3 x 12	13.55	11.00	9.25	8.13	6.60	4.89
4 x 6	9.18	7.38	6.26	5.48	4.46	3.27
4 x 8	11.10	8.94	7.54	6.63	5.32	3.93
4 x 12	15.55	12.50	10.60	9.25	7.50	5.57

ROOF SLOPE MULTIPLIERS

For roof components of sloping roofs, multiply by the figures given below to obtain the cost per square foot of horizontal projection.

RISE	MULTIPLIER	RISE	MULTIPLIER
1½ to 12	1.01	6 to 12	1.12
2 to 12	1.01	8 to 12	1.20
3 to 12	1.03	12 to 12	1.42
4 to 12	1.06	15 to 12	1.60
5 to 12	1.08	18 to 12	1.80

These multipliers may also be used to convert the horizontal projected area to the actual roof area for sloping roofs.

WOOD CEILING JOISTS

DIMENSIONS (inches)	SPACING 12" o.c.	16" o.c.	20" o.c.	24" o.c.
2 x 4	2.78	2.17	1.77	1.46
2 x 6	3.31	2.58	2.15	1.78
2 x 8	3.87	3.06	2.67	2.33
2 x 10	4.89	3.85	3.15	2.67
2 x 12	5.75	4.49	3.70	3.15

EXPOSED WOOD BEAMS

Costs per linear foot including sanding and finish. For rough sawn ornamental beams, add 40%. Simulated plastic beams cost 10.45 to 13.10 per linear foot.

SIZE	COST	SIZE	COST	SIZE	COST
2 x 6	6.04	4 x 8	15.55	6 x 12	42.00
3 x 6	11.20	4 x 12	21.60	6 x 14	47.25
3 x 8	13.40	6 x 8	26.75	8 x 12	57.50
4 x 6	12.85	6 x 10	34.00	8 x 14	63.50

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ROOFS
SECTION UIP 4

STEEL JOISTS

Costs of solid rolled or open web joists, per square foot of roof area, including connections, ties, and braces. Depth is for open web joists.

DEPTH	SPAN	LIGHT	MEDIUM	HEAVY
8”	10’ to 14’	1.78	2.67	-----
10”	14’ to 16’	1.84	3.03	4.26
10”	18’ to 20’	2.99	4.79	6.76
12”	16’ to 20’	2.36	3.70	4.96
12”	20’ to 22’	3.33	5.07	6.95
14”	18’ to 22’	2.86	4.17	5.52
14”	22’ to 26’	3.73	5.32	7.06
16”	20’ to 24’	3.01	4.29	5.79
16”	24’ to 28’	3.59	5.10	6.75
16”	28’ to 32’	4.60	6.67	8.77
18”	22’ to 26’	3.33	4.82	6.48
18”	26’ to 30’	3.92	5.69	7.57
18”	30’ to 34’	5.01	7.29	9.65
20”	24’ to 28’	3.74	5.42	7.16
20”	28’ to 32’	4.29	6.23	8.38
20”	32’ to 36’	5.45	7.88	10.35
24”	28’ to 34’	4.42	6.47	8.62
24”	34’ to 40’	5.10	7.38	9.81
24”	40’ to 46’	6.19	8.94	11.90
28”	40’ to 48’	4.74	8.38	10.85
28”	48’ to 54’	6.03	10.60	13.90
32”	50’ to 56’	5.52	9.25	12.30
32”	56’ to 62’	6.76	11.25	15.25
36”	58’ to 64’	6.03	10.05	13.50
36”	64’ to 70’	7.16	12.05	16.15
40”	66’ to 72’	6.53	10.85	14.95
40”	72’ to 78’	7.60	12.50	17.25
48”	82’ to 88’	7.43	12.50	17.10
48”	88’ to 94’	8.38	13.80	19.00

WOOD-TRUSSED JOISTS

Costs of light wood flange with solid plywood web or open tubular steel web joists, per linear foot, including ancillary connections.

TYPE	SPAN	LIGHT	MEDIUM	HEAVY
Plywood	15’ to 30’	8.84	10.70	13.15
Tubular steel	40’ to 70’	8.22	10.05	13.55

PURLINS

Costs per linear foot of purlin including connections or hangers. Light Z or C flanges for pre-engineered metal buildings may run 30% to 50% of the structural channel costs. Light wood nailing strips for wood pole frame buildings may run 65% to 85% of the purlin costs.

LIGHTWEIGHT STEEL CHANNELS		LIGHTWEIGHT I-BEAMS		SIZE	WOOD WITHOUT STRUTS	WITH STRUTS
SIZE	COST	SIZE	COST	SIZE	COST	COST
4”	29.25	4"	34.50	1 x 6	2.18	-----
6”	39.00	6”	45.25	2 x 4	2.71	7.00
8”	49.00	8”	57.00	2 x 6	3.30	7.75
10”	59.00	10”	68.00	2 x 8	4.13	8.84
12”	69.00	12”	79.50	2 x 10	4.63	10.35
15”	83.00	15”	95.00	2 x 12	6.04	11.65

STEEL SPACE FRAMES

The following square foot costs are averages of typical three dimensional roof structures for buildings requiring long two way spans. They do not apply to architectural embellishments or patented systems primarily designed for appearance and skylight areas. Costs include sheathing.

LIGHT	MEDIUM	HEAVY
25.00 – 37.50	36.50 – 47.25	45.25 – 69.00

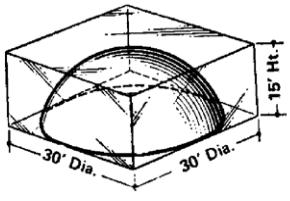
DOMES

Typical range of costs of dome structures where the dome forms both roof and walls. Cost per square foot of horizontal area which is covered, for dome structure only, excluding floor, finish, lighting, etc.

Skylight cover on metal frame, D = 5' to 40'	84.50 – 181.00
Metal geodesic, D > 70'	37.50 – 71.00
Reinforced concrete, cast-in-place	51.00 – 76.50
Thin shell, concrete	33.50 – 55.00
Wood deck and frame	26.00 – 57.00

Domes may also be estimated by adding up the unit costs of the component parts and multiplying by an area calculated by converting the dome into a rectangular structure as shown in the illustration.

30' x 30' = 900 sq. ft.
4 x 30' x 15' = +1,800 sq. ft.
Apply unit cost to 2,700 sq. ft.



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ROOFS
SECTION UIP 4

STEEL ROOF TRUSSES

Tapered steel girders will cost from 15% to 30% less for short spans to 25% to 35% less for long spans (over 80'). Costs include end connections.

	COST FOR EACH TRUSS		
SPAN	LIGHT	MEDIUM	HEAVY
30'	4375.00	5350.00	6550.00
40'	5950.00	7150.00	8700.00
60'	9050.00	10800.00	13100.00
80'	12300.00	14500.00	17500.00
100'	15200.00	18600.00	22000.00
120'	-----	22000.00	26500.00
140'	-----	26100.00	31100.00
160'	-----	-----	35400.00

WOOD-TRUSSED RAFTERS

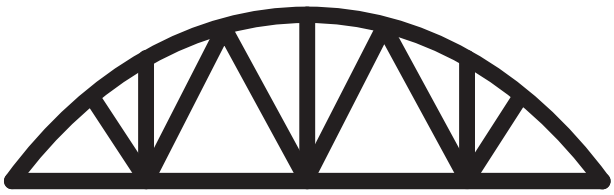
Costs are for prefabricated roof trusses common to light wood framing (12" to 48" on center) using 2" x 4" and 2" x 6" (up to 45' span) and 2" x 8", 2" x 10" and 2" x 12" (over 45' span) members, attached with steel plates or plywood gussets, including ancillary items and ties.

For pole-frame construction (8' to 14' spacing) with members 2" x 8" and larger with spans below 50', add 20%. Add 10% for site built trusses. For double trusses, count both and deduct 5% from the total.

	COST FOR EACH TRUSS		
SPAN	LIGHT	MEDIUM	HEAVY
20'	132.00	139.00	149.00
30'	203.00	227.00	250.00
40'	295.00	333.00	386.00
50'	401.00	469.00	550.00
60'	520.00	610.00	725.00

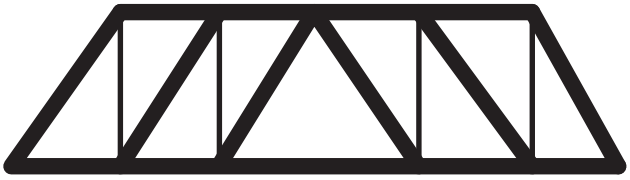
INDUSTRIAL WOOD TRUSSES

Costs are for timber or glulam industrial trusses, including end connections. For finished commercial trusses add 20% to 25%. Glued, laminated wood girders will cost approximately 25% to 50% less than bowstring trusses for short spans (up to 25'), 20% more for mid-range spans (up to 45'), to 50% to 75% more for long-span girders (up to 75'). Glued, laminated arches with ties, will cost a proximately 15% to 20% less than bowstring trusses for short spans to 10% more for long spans (over 80'). Cost ranges are for trusses bearing roof loads only, not for those carrying hoists or cranes.



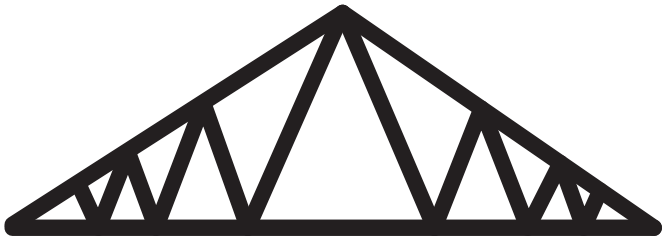
BOWSTRING TRUSS

BOWSTRING SPAN	COST FOR EACH TRUSS		
SPAN	LIGHT	MEDIUM	HEAVY
20'	1620.00	1900.00	2120.00
40'	4025.00	4650.00	5400.00
60'	6600.00	7800.00	9350.00
80'	9500.00	11400.00	13800.00
100'	12700.00	15200.00	18600.00
120'	15800.00	19400.00	23700.00
140'	19300.00	23700.00	29200.00



FLAT HOWE

BELGIAN OR HOWE SPAN	COST FOR EACH TRUSS		
	LIGHT	MEDIUM	HEAVY
20'	1030.00	1260.00	1740.00
40'	3275.00	4125.00	5250.00
60'	6450.00	8050.00	10200.00
80'	10400.00	13000.00	16200.00
100'	15200.00	19000.00	23400.00



BELGIAN TRIANGULAR

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ROOFS
SECTION UIP 4

ROOF DECKS AND SHEATHING

Per square foot of decking or sheathing

TYPE	COST RANGE
Concrete: hollow precast plank	11.65 – 18.40
Lift slab	13.90 – 21.75
Lightweight channels, 2”3”	9.49 – 14.35
Plank, 2”3” T&G	10.75 – 14.35
Reinforced slab, 4”6”	14.20 – 22.55
Topping, 1 3/4” 2 1/2” foamed concrete	2.08 – 4.35
Lighweight concrete, reinforced, 2”3”	4.60 – 6.19
Prestressed tees or double tees	12.60 – 20.90
Fiberboard, prefinished one side,	
2” T&G	4.60 – 6.53
3” T&G	6.11 – 7.63
Gypsum: 2” plank with metal edge	6.53 – 10.20
2”-3” poured over gypsum board	5.11 – 7.12
2”-3” poured over fiberglass	6.11 – 7.71
Steel:18 gauge, corrugated or crimped	4.24 – 8.22
26 gauge	2.83 – 5.67
Cellular	10.85 – 19.50
Tectum, 2”, prefinished one side	6.44 – 8.77
Wood sheathing: 5/16” plywood	1.56 – 1.93
3/8” plywood	1.71 – 2.18
1/2” plywood	1.90 – 2.49
5/8” plywood	2.24 – 3.05
3/4” plywood	2.64 – 3.36
1-1/8” plywood, T&G	4.01 – 5.44
1” spaced board	1.78 – 3.15
1” solid board	2.78 – 4.26
1” T&G	2.87 – 4.64
2” T&G	4.77 – 8.07
3” T&G	7.99 – 11.90
2” plank	4.63 – 9.33
3” plank	7.75 – 13.15
for oriented strand waferboard, deduct	0.12 – 0.27
for fire treated wood, add	0.66 – 1.21
Wood Stresskin sandwich panels	9.33 – 20.50

ROOF COVER

Cost per square foot of roof area, including typical flashing, valleys and underlayment as required by the roofing type. Extremely steep or cut-up roofs may be 20% above the high end of the range. For vertical or mansard applications, add 20%. Curved metal panels, add $1.49 to $4.18 .

Aluminum, sheet, corrugated, crimped or tile panels	
preformed, light (.0175” .024” thick)	3.34 – 6.10
heavy (.032” .050” thick)	
including prefab lock and seam	5.69 – 8.84
formed seam, flat or standing	5.86 – 14.60
batten seam	6.85 – 16.65
shingles	4.45 – 9.58
Built up, 2–ply	1.75 – 2.80
3–ply	2.06 – 3.45
4–ply	2.42 – 3.99
5–ply	2.87 – 4.98
add for gravel or small rock	0.40 – 0.63
add for large rock	0.47 – 0.80
Cement fiber, shingles	4.49 – 9.74
corrugated or sheet (Transite)	3.87 – 15.60
Composition, roll	1.10 – 2.46
Composition shingle, light, to 235#	1.96 – 3.39
heavy, over 235# or laminated	2.50 – 5.92

ROOF COVER

(Continued)

TYPE	COST RANGE
Copper, flat or standing seam	14.05 – 26.00
Batten seam	15.30 – 28.25
Shingles	14.30 – 19.95
Elastomeric, single ply	3.59 – 10.50
Reinforced sheet or spray fluid coat	6.51 – 12.60
Fiberglass, corrugated or sheet	2.36 – 4.45
Structural or FRP panels over 8 oz.	4.45 – 11.35
Modified bitumen, single ply	3.68 – 10.85
Reinforced sheet	6.10 – 12.40
Slate (red slate, add 100%)	8.77 – 18.70
Steel, corrugated, crimped or tile panels, galvanized	
Preformed, light (30 26 gauge)	2.36 – 5.94
Heavy (24 18 gauge)	
Including prefab lock and seam	5.02 – 8.69
Formed seam, flat or standing	5.35 – 13.30
Batten seam	6.51 – 15.40
Shingles	4.26 – 9.33
add for porcelain enamel finish	4.01 – 6.85
add for sandwich panels	4.60 – 15.75
add for interior metal liner	2.36 – 4.26
add for stainless steel	4.18 – 8.62
Terne, flat or standing seam	13.10 – 24.10
Batten seam	14.15 – 26.00
Tile, clay	7.12 – 21.05
Concrete	6.01 – 11.10
Plastic or rubber	5.19 – 9.33
Wood, shakes	4.35 – 8.07
Shingles	4.01 – 7.37
Fiber shingles	2.52 – 5.27
add for fire–resistant finish	0.66 – 1.34
Roof walkways	6.51 – 8.77
Bird spikes, per linear foot	4.45 – 15.60

INSULATION

(Per square foot of insulated area)

Aluminum foil, single ply	R 5 – R 9	0.49 – 0.59
add, per additional ply		0.19 – 0.31
Expanded mica, 3-1/2”	R 7 – R 9	1.75 – 2.46
Fiberboard, 1/2” wood	R 1 – R 2	1.34 – 1.56
1”	R 2 – R 3	1.58 – 2.15
2”	R 5 – R 6	2.49 – 3.21
Fiberglass or rock wool:		
For blown–in, use appropriate batt Rvalue plus 20%.		
1” board	R 3 – R 4	1.92 – 2.92
2” board	R 7 – R 8	2.86 – 4.11
batts or roll, 2-1/2”	R 7 – R 9	0.88 – 1.24
3 1/2”	R11 – R13	1.02 – 1.34
6”	R19 – R22	1.41 – 1.90
9”	R28 – R30	1.93 – 2.61
12”	R36 – R38	2.53 – 3.40
add for vinyl faced or wire support		0.28 – 0.55
add for colored band support		0.41 – 0.83
Foamglass, 1” board	R 2 – R 3	4.54 – 5.41
2” board	R 4 – R 6	5.67 – 7.15
3” board	R 7 – R 8	7.43 – 8.77
Polyurethane, 1” board	R 7 – R 8	2.36 – 2.93
2” board	R14 – R17	3.08 – 4.35
3” board	R22 – R25	4.39 – 5.27
Polystyrene, 1” board	R 5 – R 6	1.77 – 2.49
2” board	R10 – R12	2.67 – 3.49

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ROOFS
SECTION UIP 4

SKYLIGHTS

Costs are given per square foot of framed area.

TYPE	PLAIN GLASS OR PLASTIC	WIRE GLASS
Flat	54.00 – 73.00	59.00 – 81.00
Pitched, without vent	59.50 – 84.50	63.50 – 102.00
Pitched, with vent	61.00 – 105.00	66.50 – 119.00
	Small	**Large**
Plastic domes, fixed	75.00 – 130.00	56.00 – 95.00
Roof windows or		
Vented domes	136.00 – 218.00	95.00 – 158.00
add for blinds	11.25 – 17.10	6.44 – 11.25
Screens	18.85 – 23.50	15.10 – 18.85

Electric controls, cost $655 to $795 per opening.

Tubular, cost each, 10' . . . . 840.00 14" . . . 1130.00 21" . . . 1260.00

add $166 for each light kit and $446 for each ventilation kit

ROOF HATCHES AND FIRE VENTS

Cost per square foot of framed area.

	COST RANGE
Small (under 10 sq. ft.)	166.00 – 227.00
Medium	136.00 – 189.00
Large (over 25 sq. ft.)	101.00 – 158.00

VENTILATORS

Cost for each range includes galvanized steel (lowest), aluminum, and stainless steel or copper (highest). For light residential installations, deduct 50%.

DIAMETER	STATIONARY COST RANGE	ROTARY COST RANGE
6"	174.00 – 423.00	227.00 – 565.00
12"	340.00 – 735.00	461.00 – 1030.00
18"	535.00 – 1020.00	710.00 – 1490.00
24"	710.00 – 1300.00	960.00 – 1920.00
30"	900.00 – 1530.00	1240.00 – 2330.00
36"	1120.00 – 1800.00	1490.00 – 2750.00
48"	1520.00 – 2270.00	2030.00 – 3500.00

CONTINUOUS RIDGE VENTS

Cost range per linear foot including screen and damper.

THROAT SIZE (inches)	COST RANGE	THROAT SIZE (inches)	COST RANGE
4"	74.50 – 101.00	12"	115.00 – 147.00
9"	96.50 – 126.00	14"	129.00 – 158.00

EXHAUST VENTS

Costs each are for propeller type in galvanized housings, including installation. For residential attic fans, deduct 50%.

SIZE (inches)	CAPACITY (CFM)	MOTOR SIZE	FLAT ROOF	PITCHED ROOF
12"	800	1/20 HP	980.00	1210.00
15"	1,200	1/15 HP	1170.00	1410.00
18"	2,400	1/10 HP	1380.00	1660.00
24"	4,200	1/4 HP	1900.00	2270.00
30"	6,000	1/2 HP	2650.00	3100.00
36"	11,000	3/4 HP	3675.00	4300.00
42"	15,000	1 HP	5100.00	5900.00
48"	20,000	2 HP	7050.00	8050.00

GUTTERS AND DOWNSPOUTS

Costs are given per linear foot of gutter or downspout. For scupper and conductor head, add $ 227 to $ 386 each.

	COST RANGE
Aluminum	9.02 – 19.35
Copper or stainless steel	20.65 – 31.50
Fiberglass or vinyl	8.46 – 13.10
Galvanized steel	16.50 – 23.80
Wood	8.29 – 16.65
Add for mesh guard	2.12 – 2.80
Add for solid guard	6.70 – 11.10

FALSE-MANSARD FASCIA

False mansard framework is not normally a part of the roof structure. It can be priced on a cost per square foot of vertical projection, exclusive of sheathing and cover which can be added from Page 4.

TYPE	COST RANGE
Metal frame	10.95 – 20.65
Wood frame	7.50 – 13.95

For finished soffit, add $6.70 to $10.50 for plaster or stucco, $4.26 to $9.49 for wood, $6.10 (residential) to $11.00 for metal or vinyl, per square foot of soffit area.

FASCIA BOARD

Cost per linear foot for wood, metal or vinyl fascia boards. For residential construction, use low end of range. For 1" wood material reduce costs by 50%.

	COST RANGE		COST RANGE
6" and less	4.18 – 5.35	8" and over	4.45 – 8.07

STEEPLES

Costs are given per linear foot of height except where noted.

CUPOLAS

TYPE	SMALL (up to 10') COST RANGE	MEDIUM COST RANGE	LARGE (up to 20') COST RANGE
Copper or terne	900.00 – 2120.00	1410.00 – 3375.00	2120.00 – 5250.00
Aluminum or steel	326.00 – 1350.00	469.00 – 1870.00	735.00 – 2575.00
Fiberglass	250.00 – 965.00	423.00 – 1520.00	690.00 – 2430.00
Wood	333.00 – 655.00	520.00 – 1030.00	780.00 – 1590.00

SPIRES

Copper or terne	760.00 – 1210.00	1220.00 – 1890.00	1890.00 – 3100.00
Aluminum or steel	250.00 – 640.00	326.00 – 965.00	386.00 – 1470.00
Fiberglass	236.00 – 432.00	287.00 – 725.00	386.00 – 1240.00
Wood	189.00 – 287.00	243.00 – 432.00	401.00 – 630.00

DOMES

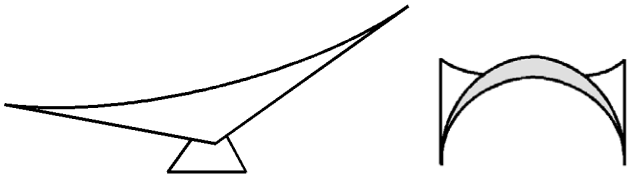
Cost per linear foot of dome diameter, up to 8'

	COST RANGE
Copper or terne	545.00 – 1020.00
Aluminum or steel	236.00 – 401.00
Fiberglass	149.00 – 236.00
Wood or stucco	149.00 – 287.00
Add for cupola clocks, per face (three to eight feet)	3000.00 – 15500.00
Crosses, per linear foot of height	287.00 – 1270.00
Spikes, per linear foot of height	227.00 – 655.00
Balls, cost each	243.00 – 1210.00
Weather vanes, cost each	295.00 – 1750.00

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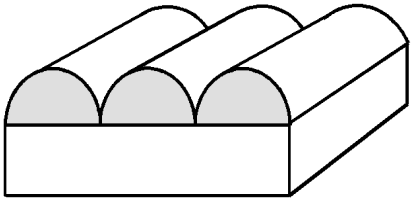
ROOFS
SECTION UIP 4

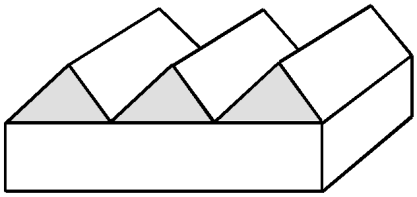
ROOF SHAPES



HYPERBOLIC PARABOLOIDS

BARREL SHELL

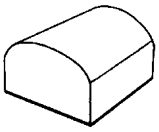




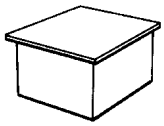
FOLDED PLATE

ROOF SHAPES

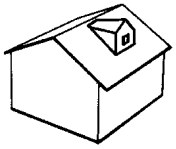
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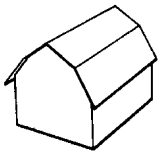
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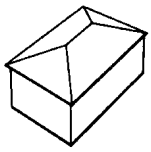
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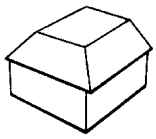
GABLE WITH DORMER



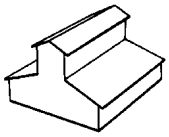
GAMBREL



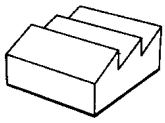
HIP



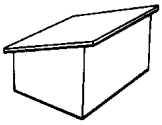
MANSARD



MONITOR



SAWTOOTH



SHED

THIN-SHELL REINFORCED CONCRETE

Cost ranges per square foot of horizontal area. For domes, see page 2.

	COST RANGE
Barrel shell	25.00 – 44.50
Folded plate	23.00 – 41.50
Hyperbolic-paraboloid, small bays	22.25 – 43.00
Large span	25.00 – 47.25

FABRIC ROOF STRUCTURES

Costs per square foot of floor area for permanent single layer fabric structures completely installed. For interior thermal liners, add $ 5.19 to $10.20 per square foot.

	COST RANGE
Air supported	54.00 – 75.00
Tension supported	61.50 – 92.00

DORMERS

Costs of exterior construction of dormer windows and necessary dormer structure vary greatly with the size of the dormer. It may be approximated by applying 1/3 to 2/3 of the dwelling square foot cost to the horizontal area of the dormer at the dormer eave line.

MONITOR AND SAWTOOTH ROOFS

Costs of monitor and sawtooth roofs may be built up as with any other roof structure, adding for trusses if they exist and then pricing the vertical window sections as follows:

	FIXED WINDOWS	VENTED WINDOWS
Fiberglass	24.25 – 30.25	32.75 – 40.50
Glass	27.25 – 34.00	36.50 – 44.50
Wire glass	33.25 – 41.25	43.00 – 54.00

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WALL COSTS
EXPLANATION

This section is not a part of the Segregated Cost Method. It is presented here to provide a detailed cost buildup where it is felt that a specific wall being valued varies significantly from the typical walls on the Segregated Cost pages or where such detail is demanded. The costs in this section differ from the Segregated Costs of Sections SEG 1 through SEG 6 in that each individual wall component is priced separately, while the segregated wall costs include, in the one cost, an allowance for each of the following components:

1. **BASIC WALL**
2. **EXTERIOR COVER**
3. **INTERIOR FACING**
4. **FINISH (exterior and interior)**
5. **OPENINGS (windows and doors)**

Some walls will be made up of many component parts, while others may have only two or three. The examples, on the next page, illustrate the method to use in building up a total wall cost. The costs are given per square foot of actual area of the various components except as otherwise stated. If any component, such as windows or doors, does not occupy the total wall area, the unit price should be multiplied by the proportion of the area of the component to the total wall area. Costs include full allowance for the general contractor's overhead and profit however, the architect's fee must be added.

Wall costs are the sum of the costs of all their parts. Basic wall costs for wood frame, brick, stone, etc., and for wall coverings such as face brick, wood shingles, and many others, are priced for the entire wall area regardless of the number and size of openings. To these costs are added the unit costs for windows and doors. These unit costs are applied in the proportion the windows and doors bear to the entire wall area.

This principle is demonstrated in the examples. In Example 1, for instance, where 10% of the total wall area consists of windows, 10% of the unit cost for windows is applied to the total wall area. There are exceptions to the above rule. For instance, in the building elevation illustrated in Figure A, the basic wall cost would be applied only to the lower half of the wall which is shown and the window costs applied to the upper half.

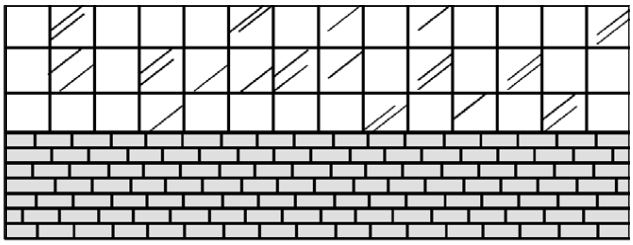


FIGURE A

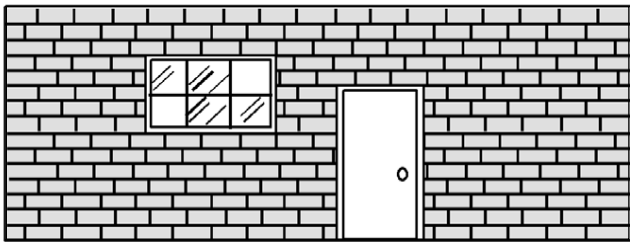


FIGURE B

Between this type of wall and the completely blank wall, there is a complete chain of gradations, which leaves much to the experience and judgment of the assessor.

Figure B shows the front elevation of a typical industrial building, where the basic wall costs are applied to the entire wall and the cost of the windows and door is applied to the actual area of the openings.

This so called doubling of areas for computation, as in Figure B, provides an allowance for the added cost of building the wall around openings.

This section can also be used to test the proper rating choice in the Segregated Cost section. For example, the assessor may encounter a wall which is not typical either in the number of openings, type of interior finish, or some other feature. They may profitably build up the cost of that particular wall in detail and not only use the cost for that particular appraisal, but also could use it as a guide in choosing the proper rating number in the Segregated Costs when a similar wall is encountered in the future.

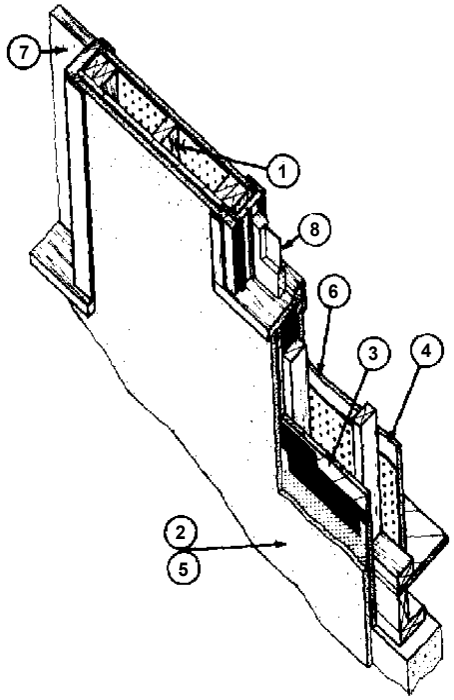
It is sometimes desirable to build up the cost of interior partitions instead of pricing them on a square foot basis. Some of the component parts for interior construction may be priced from this section; however, interior doors and built in cabinets should be priced from Section UIP 2.

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WALL COSTS

EXAMPLE 1

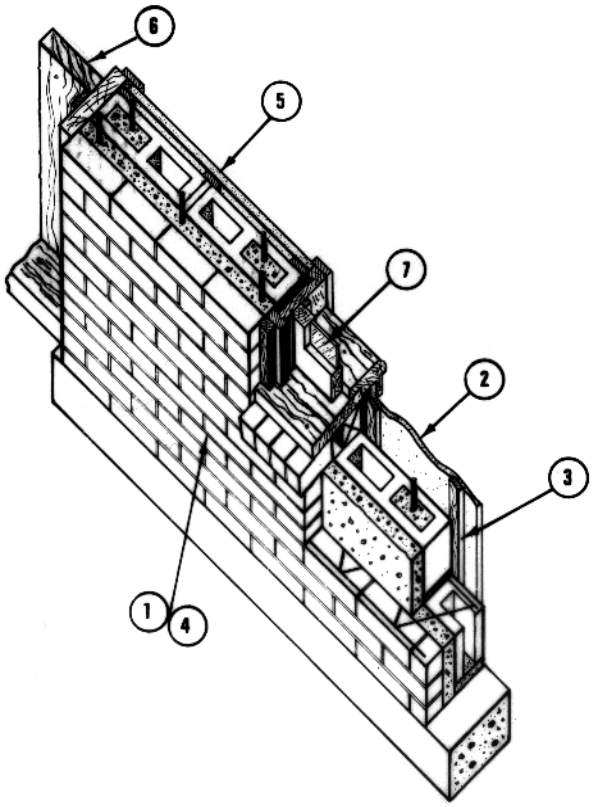
	Cost per Square Foot of Total Wall Area
BASIC WALL	
1. 2 x 4, 16 in. o.c.	2.93
EXTERIOR COVER	
2. Stucco, three coats on wire and paper	6.46
3. Sheathing, 1" polystyrene, extruded	2.06
INTERIOR FACING	
4. Plaster, two coats on gypsum lath,	5.71
Keene's cement finish	1.24
FINISH	
EXTERIOR	
5. Stucco paint, two coats	1.40
INTERIOR	
6. Paint, flat wall, two coats	1.23
OPENINGS	
DOORS	
7. Wood, exterior, raised panel (5% of total wall area) .05 x $45.38 =	2.27
WINDOWS	
8. Wood, double hung, insulated, double glazed, (15% of total wall area) .15 x $41.77 =	6.27
TOTAL COST PER SQ. FT. OF WALL AREA	29.57



NOTE: *Costs are the midpoint averages between the minimum and maximum costs listed. Insulation is an additive.*

EXAMPLE 2

	Cost per Square Foot of Total Wall Area
BASIC WALL	
1. Common brick, 10" grouted, reinforced, block backup	32.50
INTERIOR FACING	
2. Gypsum board, 1/2"	2.37
3. Furring, wood	1.96
FINISH	
EXTERIOR	
4. Waterproofing, 2 coats	1.15
INTERIOR	
5. Paint, flat wall, 2 coats	1.23
OPENINGS	
DOORS	
6. Wood, exterior, flat panel (5% of total wall area) .05 x $39.38 =	1.97
WINDOWS	
7. Wood, double hung (10% of total wall area) .10 x $30.38 =	3.38
TOTAL COST PER SQ. FT. OF WALL AREA	44.56



NOTE: *Costs are the midpoint averages between the minimum and maximum costs listed.*

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WALL COSTS

COSTS PER SQUARE FOOT OF WALL AREA

MASONRY CONSTRUCTION

	COST RANGE
Adobe, grouted, reinforced, 9"11"	27.50 – 35.75
Post and girder construction, 8"	22.70 – 29.00
Brick, common, 8"	27.25 – 39.50
12"	36.00 – 51.50
8", block backup	24.25 – 34.00
12", block backup	27.00 – 39.50
16", block backup	30.25 – 45.25
9" – 10" cavity	29.50 – 40.25
9" – 10" cavity, block backup	24.95 – 35.25
9" – 10" grouted, reinforced	30.50 – 45.75
9" – 10" grouted, reinforced, block backup	26.25 – 38.75
12" – 14" grouted, reinforced, block backup	29.00 – 44.25
8", tile backup	26.50 – 38.00
12", tile backup	28.25 – 42.00
6", SCR or hollow brick	18.95 – 26.50
8"	21.50 – 29.50
For brick pilasters, add to wall cost	2.56 – 3.29
For each 4" additional thickness of common brick, add	7.78 – 11.25
For each tier or wythe of face brick in lieu of common add	3.95 – 6.10
For each wythe of oversized brick in lieu of common, deduct	1.84 – 5.45
Concrete, formed, reinforced, 6''	20.05 – 33.50
8"	22.55 – 36.25
12"	26.50 – 42.00
add for stay–in–place forming	0.72 – 3.37
Precast panels, 2"- 3", gray	15.85 – 24.55
4"- 5", gray	17.45 – 26.50
6"- 7", gray	19.75 – 29.50
add for white	8.41 – 12.00
add for ribbed or fluted openings	12.00 – 24.25
add for exposed aggregate	5.45 – 10.85
add for insulated sandwich	2.33 – 4.66
deduct 10% for tees or double tees	
Tilt–up panels, 5"- 6", gray	14.60 – 18.25
add for each 2" of additional thickness	1.36 – 1.84
add for textured finishes	1.68 – 7.06
Concrete block, reinforced, 6"	11.70 – 16.30
8"	12.65 – 18.40
12"	16.15 – 24.25
Unreinforced, 8"	11.85 – 14.80
12"	14.80 – 20.05
add for split, slumpstone, fluted or ground face	1.84 – 6.51
add for sound block	2.17 – 4.32
For concrete or block pilasters, add to wall cost	1.84 – 2.89
For concrete or block bond beams, add to wall cost	2.00 – 3.03
Glass block, white or aqua	54.00 – 78.00
Colors and solar reflective	78.00 – 107.00
Gunite reinforced, each 1'' of thickness	5.52 – 7.97
Gypsum block, 4''	10.75 – 14.00
6"	11.55 – 14.95
Tile, clay, 4"	13.35 – 16.15
6"	15.30 – 20.35
8"	17.30 – 23.30
Add for tile or block glazing, each side	8.66 – 13.10

Stone masonry – see stonework tables at end of Section UIP 6, Page 7.

WALL INSULATION

		COST RANGE
Aluminum foil, reflective,		
Single ply	R5 – R9	0.47 – 0.57
Two ply	R13 – R14	0.68 – 0.81
Three ply	R16 – R17	0.88 – 1.10
Four ply	R18 – R20	1.08 – 1.33
Five ply	R20 – R22	1.27 – 1.59
Batts or roll, fiberglass,		
1 1/2"	R4 – R6	0.66 – 0.93
2 1/2"	R7 – R9	0.81 – 1.19
3 1/2"	R11 – R13	0.94 – 1.27
6"	R19 – R22	1.33 – 1.80
add for vinyl–faced high density or mineral fiber		0.43 – 0.56
Loose fill, expanded mica,		
in cavity wall, 1"	R2 – R3	0.59 – 0.81
2"	R4 – R5	1.10 – 1.55
3"	R6 – R8	1.56 – 2.14
4"	R8 – R10	1.99 – 2.80
In block wall, 4"	R.65 – R.75	0.56 – 0.99
6"	R1.25 – R1.35	1.00 – 1.66
8"	R2 – R3	1.52 – 2.56
12"	R4 – R5	2.67 – 4.85
Rigid board, cork, 1/2"	R1 – R2	1.99 – 3.08
1"	R3 – R4	3.08 – 5.47
2"	R7 – R8	4.91 – 8.04
Fiberboard, 1"	R2 – R3	1.53 – 2.06
2"	R5 – R6	2.40 – 3.08
Fiberglass, 1"	R3 – R4	1.84 – 2.83
2"	R7 – R8	2.78 – 3.95
Foamglass, 1"	R2 – R3	4.35 – 5.20
2"	R4 – R6	5.47 – 6.90
Perlite board, 1"	R2 – R3	1.71 – 2.31
2"	R4 – R6	2.53 – 3.52
Polystyrene, extruded, 1"	R5 – R6	1.71 – 2.40
2"	R10 – R12	2.56 – 3.37
2 1/2"	R12 – R15	3.01 – 3.95
3"	R15 – R18	3.36 – 4.51
Beadboard, 1"	R3 – R4	1.31 – 1.81
2"	R6 – R8	1.87 – 2.67
3"	R9 – R12	2.37 – 3.21
4"	R12 – R16	2.78 – 3.52
Polyurethane, 1/2"	R3 – R4	1.83 – 2.15
1"	R7 – R8	2.24 – 2.83
1 1/2"	R10 – R12	2.53 – 3.79
2"	R14 – R16	2.92 – 4.20

Sprayed or foamed-in-place (urethane, styrene, etc.) use appropriate R-value rating above and add 100% to the cost. For wet-sprayed cellulose, add 50%.

AIR AND VAPOR BARRIER

Aluminum foil, paperbacked	0.47 – 0.57
Air infiltration fiberwrap	0.46 – 0.57
Air space/drainage meshwrap	0.85 – 1.38
Building paper	0.32 – 0.50
Polyethylene	0.28 – 0.44

For below grade application see waterproofing costs in Section UIP 1.

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WALL COSTS

COSTS PER SQUARE FOOT OF WALL AREA

STUD CONSTRUCTION

WOOD

		COST RANGE				COST RANGE	
2" x 3"	12" o.c.	2.74 –	3.03	2" x 8"	12" o.c.	6.34 –	7.62
2" x 3"	16" o.c.	2.17 –	2.49	2" x 8"	16" o.c.	4.82 –	5.79
2" x 3"	24" o.c.	1.43 –	2.08	2" x 8"	24" o.c.	3.03 –	3.68
2" x 4"	12" o.c.	3.54 –	3.95	2" x 10"	16" o.c.	5.61 –	6.59
2" x 4"	16" o.c.	2.49 –	3.37	2" x 10"	24" o.c.	4.02 –	4.82
2" x 4"	24" o.c.	1.84 –	2.33	2" x 10"	36" o.c.	3.03 –	3.61
2" x 6"	12" o.c.	4.32 –	5.45	4" x 4"	24" o.c.	4.32 –	5.45
2" x 6"	16" o.c.	3.61 –	4.10	4" x 4"	36" o.c.	3.54 –	4.10
2" x 6"	24" o.c.	2.49 –	3.37	4" x 4"	48" o.c.	2.74 –	3.61
For hurricane or seismic bracing and connectors, add						0.65 –	1.43

METAL

Dimen. o.c.	Light (to 25 gauge)	Medium (to 20 gauge)	Heavy (to 16 gauge)	Extra Heavy (C-Stud)
4" –16" o.c.	2.74	3.54	4.02	4.49
4" –24" o.c.	2.24	2.81	3.29	3.95
6" –16" o.c.	3.03	4.10	4.66	5.30
6" –24" o.c.	2.74	3.54	3.95	4.49

EXTERIOR WALL COVER

	COST RANGE	
Aluminum siding, corrugated or crimped,		
Light (.0175".024" thick)	3.29 –	4.82
Heavy (.032" to .050" thick)	5.52 –	8.27
residential type, painted	5.04 –	7.06
add for sandwich panels	6.98 –	21.20
add for enameled paint	0.47 –	1.36
add for curved panels	1.43 –	4.10
add for wall girts, metal	1.36 –	3.54
Wood	0.65 –	1.84
Cement fiber shingle or plank, 5/32"	3.77 –	5.52
Sheet siding, residential type	3.68 –	5.04
Heavy corrugated or sheet (Transite)	5.45 –	7.62
Sandwich panels, insulated	16.30 –	28.75
Board and batten, 1" cedar or redwood	4.82 –	9.05
Plywood with batts	3.54 –	4.73
Composition siding	2.33 –	3.68
Fiberglass, corrugated or sheet (4 – 5 oz.)	2.24 –	3.77
Sandwich panels	4.24 –	4.73
Structural or F.R.P. panels	4.66 –	12.55
Hardboard, exterior grade, 3/8" 1/2", sheet siding	2.96 –	4.66
Lap siding or shingles	3.29 –	5.14
Masonry veneer, single wythe, select common brick	13.35 –	22.85
Face brick, standard	15.25 –	25.75
Split face brick, Roman or Norman	14.05 –	23.30
Mortarless brick	15.55 –	24.80
Used brick	14.60 –	24.55
Adobe	11.30 –	17.30
Cast stone, ornamental	32.50 –	54.00
Ornamental face concrete block, 4"	12.90 –	22.05
Simulated brick veneer or panel, 1/4"– 3/8"	10.75 –	24.40
Imitation precast stone	11.85 –	29.50

Pictorial carved artwork, see Section UIP 6, Page 5.
Natural stone veneers, see under Stonework, Section UIP 6.

WALL COVER (Continued)

	COST RANGE	
Plywood, exterior grade, 3/8" 5/8", plain	2.81 –	3.77
Textured	3.54 –	4.66
Fiber overlay	3.61 –	4.96
Redwood or select cedar	4.32 –	7.15
Exposed aggregates, fine	8.52 –	10.60
Medium	10.70 –	14.35
Large	13.10 –	19.25
Rustic or log siding, veneer	6.98 –	20.05
Half round with full corners	18.95 –	25.75
Full logs, 6" - 7" diameter	17.85 –	27.50
8"	21.05 –	32.50
9"	23.30 –	35.25
10"	25.25 –	39.00
Hand hewn or oversized logs	31.00 –	54.00
Shakes, wood	4.66 –	6.66
Shingles, wood	4.10 –	6.25
Wood panels	4.96 –	8.52
Siding, vinyl	4.73 –	6.66
Wood, 1/2" 1" boards	3.54 –	5.04
Select cedar or redwood	4.24 –	10.75
Steel, galvanized, corrugated or crimped,		
Light (30 to 26 gauge)	2.49 –	4.41
Heavy (24 to 18 gauge)	4.82 –	7.97
Residential type, painted	4.73 –	6.34
add for sandwich panels	6.90 –	20.90
add for curved panels	1.43 –	4.10
add for enameled paint	0.28 –	1.36
add for porcelain enamel	3.95 –	6.66
add for stainless steel	4.02 –	8.34
add for textured protective cover (Galbestos)	4.82 –	7.62
add for wall girts, metal	1.36 –	3.54
Wood	0.65 –	1.84
Stucco, on paper and wire mesh, three coats	5.94 –	6.98
On metal lath, three coats	6.10 –	7.97
On masonry, two coats	4.10 –	6.51
Synthetic plaster, on fabric and rigid		
Insulation board, 1"	8.90 –	13.65
For decorative texture or aggregate finish, add	2.33 –	7.71
Each inch of insulation over 1", add	0.80 –	2.00
On fiber cement board	7.15 –	8.90
Add for heavy fiberglass reinforcement	2.56 –	4.73
Pictorial artwork, see Section 56, Page 5.		
Terra cotta, 2"- 4"	34.00 –	61.00
Tile, ceramic	17.15 –	29.00
Tile, mosaic panels, minimum artwork	24.10 –	51.50
Pictorial, excluding name art	38.00 –	93.50
Hand–painted	125.00 –	427.00
Vitrolite (structural glass)	27.50 –	49.75
Sheathing, asphalt composition, 1/2"	1.68 –	2.17
Ballistic shielding fiberglass panels, 1/2" – 5/8"	30.50 –	37.50
1" – 1 1/2"	90.00 –	115.00
Fiberboard, 1/2"	1.61 –	2.08
Gypsum board, water–resistant, 1/2" 5/8"	2.17 –	2.81
Oriented strand–waferboard, 3/8" 1/2"	2.08 –	2.49
Plywood, 3/8" 1/2"	2.24 –	2.56
5/8" 3/4"	2.56 –	2.96
For structural shear walls, add	0.16 –	0.28
Wood, solid 1" boards	2.74 –	3.68
Spaced 1" boards	2.17 –	2.74
Stresskin sandwich panels	8.34 –	14.20
add for treated wood	0.65 –	1.36

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WALL COSTS

COSTS PER SQUARE FOOT OF WALL AREA OR OPENING

EXTERIOR FINISH

	COST RANGE	
Epoxy or urethane	2.31 –	6.90
Masonry paint, prime coat	0.55 –	0.93
Each additional coat	0.35 –	0.57
Silicone waterproofing for masonry, two coats	0.99 –	1.30
Stain, on shingles or wood siding, two coats	1.08 –	1.62
Structural steel paint, shop coat	0.44 –	0.74
Prime field coat	0.46 –	0.84
Each additional coat	0.46 –	0.75
Stucco paint, prime coat	0.71 –	0.99
Each additional coat	0.44 –	0.66
Textured coat, one heavy coat or acrylic mod./ Polymer–based	1.52 –	2.25
Sealer coat	0.32 –	0.55
Wood siding, prime coat	0.72 –	0.99
Each additional coat	0.44 –	0.57

INTERIOR FACING

Chalkboard, hardboard	10.75 –	17.15
Slate	24.80 –	37.25
Vinyl finish	18.40 –	23.80
Furring, wood on masonry	1.43 –	2.49
Wood on wood	1.24 –	2.08
Metal	2.17 –	3.54
Gypsum board, 1/2", taped	2.17 –	2.56
5/8"	2.24 –	2.81
add for water–resistant or embossed	0.22 –	0.44
Hardboard, plastic finish, patterned	3.61 –	6.34
Pegboard	2.89 –	5.79
Tempered, plain	2.17 –	3.68
Insulation board, prefinished	2.17 –	3.54
Plaster, two coats, on metal lath	4.82 –	6.59
On gypsum lath	4.73 –	5.94
On masonry	4.10 –	5.14
Plaster, three coats, on metal lath	5.94 –	7.06
Add for Keene's finish	0.87 –	1.61
Plywood paneling, softwood	2.89 –	5.45
Hardwood	4.82 –	9.99
Custom panels and trim	10.60 –	23.80
Polylaminated panels	3.68 –	6.90
Tile, acoustical	2.49 –	3.95
Ceramic or quarry	12.25 –	34.50
Custom mosaics	23.80 –	52.50
Hand–painted	118.00 –	413.00
Metal, enameled	11.15 –	15.10
Plastic	8.52 –	11.30
Wood paneling, 1/2"-1" boards, softwood	5.79 –	8.66
Hardwood	8.66 –	17.85
Custom plank decorative	16.85 –	34.50
3/8'' cedar closet lining	8.12 –	10.60

Masonry veneers – use exterior costs plus 5% to 10%.

INTERIOR FINISH

	COST RANGE	
Canvas, cloth, and other specialties	4.10 –	9.12
Copper and wood veneer sheets	11.55 –	15.40
Gypsum impregnated jute fabric	2.81 –	4.73
Laminates, plastic, standard grades	6.34 –	9.05
Custom grades, leather, stone, wood	11.25 –	25.00
Natural wood finish	1.36 –	2.00
Paint, prime coat or sealer	0.50 –	0.93
Each additional coat	0.35 –	0.66
add for enamel, per coat of enamel	0.13 –	0.38
Textured finish, sprayed on	0.43 –	0.68
Vinyl wall covering	2.49 –	6.17
Wallpaper, standard grades	1.83 –	3.85
Custom grades	4.24 –	9.30
Hand painted, embroidered	8.34 –	28.50
Mural and aluminum specialties	4.73 –	10.60
Removal	0.46 –	0.88
Zolotone, primer and one coat	1.61 –	3.29

DOORS

COMMERCIAL AND INDUSTRIAL

(Cost per square foot of door area or opening.)

All costs include hardware and installation.

All costs include hardware and installation.

Fire doors, hollow metal, 1 3/4''	70.50 –	107.00
Kalamein, 1 3/4''	61.50 –	84.00
Kalamein, 2 1/2''	69.50 –	91.50
Metal clad, swinging	63.50 –	98.00
Metal clad, rolling	63.50 –	84.00
Weatherstripping	6.03 –	18.40
add for mechanical operation, each	1000.00 –	1410.00
add for electric operation, each	2360.00 –	3525.00
add for underwriter's label	8.04 –	18.40
For paired doors, add 35% to 50% for second door.		
Gates, steel, folding	38.75 –	50.00
Overhead doors, sectionals, manual operation	19.75 –	29.75
Roll up	27.50 –	42.75
add for mechanical operation, each	1030.00 –	1500.00
add for electric operation, each	2330.00 –	3275.00
Strip doors, plastic	12.40 –	24.80
Traffic doors, rubber, swinging	63.50 –	91.50

For entrance doors, see Storefront costs in Section UIP 6.
For air curtains, see Section UIP 3.
For hangar doors, see Section SEG 4, page 7.

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WALL COSTS

COSTS PER SQUARE FOOT OF WALL AREA OR OPENING

DOORS

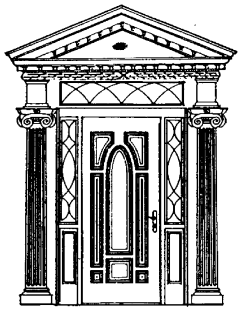
RESIDENTIAL STYLE

All costs include hardware and installation.

	COST RANGE
Metal, hollow core	26.50 – 38.00
Insulated	32.50 – 45.75
Aluminum storm and screen, stock	18.95 – 31.25
Deluxe	31.00 – 45.25
Aluminum screen	10.75 – 18.95
Plastic, laminated, solid core	67.50 – 118.00
Security, ballistic resistant	113.00 – 221.00
Raised panel	329.00 – 427.00
Wood, hardboard, slab, solid core	23.55 – 34.00
Pine or fir, flat panel	31.25 – 47.50
Raised panel	32.75 – 58.00
French	43.00 – 67.50
Dutch	41.25 – 61.00
Natural finish hardwood or plastic–faced overlay	
Flush, hollow core	21.50 – 49.25
Solid core	24.80 – 53.50
Decorative, standard pattern or carved	56.00 – 73.50
Deluxe pattern or carved	76.50 – 118.00
For sidelites, use 50% of decorative door costs	
Custom hardwood, solid core, flat panel, oak	118.00 – 153.00
Raised panel, oak	144.00 – 359.00
Teak	175.00 – 389.00
White ash	192.00 – 298.00
Philippine mahogany	259.00 – 550.00
Walnut	245.00 – 610.00
Decorative, glass lites, standard pattern	329.00 – 436.00
For plastic or metal, deduct 40%.	
Custom carved	398.00 – 670.00
For triple–insulated beveled glass, add	47.50 – 71.00
For 3/4"–thick sculptured crystal glass, add	75.00 – 99.50
Solid brass adjustable sill	24.10 – 44.50
Sidelites, sq. ft. of panel area	199.00 – 329.00
half round or ellipse toplites	276.00 – 595.00
Rectangular toplites	161.00 – 276.00
Wood storm and screen	24.10 – 31.00
Deluxe	31.00 – 41.25
Weatherstripping, compression	2.17 – 5.94
magnetic	3.77 – 6.98
Glass patio doors, sliding or prefabricated swinging	
Aluminum, single glazed, 3/16"	21.50 – 35.25
double glazed, 5/8"	35.50 – 58.00
triple glazed	39.75 – 63.50
Wood, softwood	61.00 – 79.50
Deluxe hardwood	90.00 – 130.00
For exterior metal or vinyl cladding, add	10.30 – 15.10
For diamond grilles, add	12.90 – 18.40
For curved–top grilles, add	35.25 – 42.75
For insulated glass, add	23.30 – 30.50
For insulated Low E glass with Argon gas, add	41.25 – 46.00
For solar bronze or solar gray insulating Glass, add	15.10 – 21.05

For special glass treatments, use additives from Page 7.

ORNAMENTAL DOORWAYS



Cost per square foot of ornamental opening, excluding the entry door area.

	COST RANGE
Fiberglass, stock (no lites)	56.00 – 99.50
Wood, Stock trim	72.00 – 141.00
Custom wood, hand carved, including lites	259.00 – 535.00

GARAGE DOORS

	SINGLE	DOUBLE
Overhead, single panel,		
Plastic	16.30 – 22.70	10.85 – 20.35
Metal	14.60 – 25.50	9.99 – 22.05
Wood, plywood or Hardboard	13.05 – 19.90	8.52 – 19.50
Redwood or select cedar	20.35 – 32.75	16.30 – 29.00
Custom, red oak or cherry	31.75 – 49.25	26.75 – 45.25
Sectional, plastic	17.95 – 26.50	12.55 – 22.70
Metal	16.30 – 29.50	11.70 – 24.80
Wood or hardboard	14.60 – 23.30	9.99 – 22.05
Redwood or select cedar	23.80 – 43.50	20.90 – 37.50
Custom, red oak or cherry	41.25 – 62.00	38.25 – 55.00
Raised panels	51.00 – 72.00	47.25 – 65.00
Fan top, door headers,		
Plastic, add	6.51 – 12.25	5.94 – 11.15
Custom wood	7.78 – 13.75	7.06 – 12.90

	COST RANGE
Add for electronic operator, radio control, each	458.00 – 840.00
Extra remote controls, each	76.00 – 175.00
Add for electric operator, key or pushbutton, each	625.00 – 1030.00
Extra remote stations, each	93.50 – 199.00
Weatherstripping	2.81 – 6.25
MISCELLANEOUS	
Entry gates, metal	29.00 – 58.00
Custom estate, aluminum	69.50 – 104.00
Hand–wrought iron	115.00 – 298.00
Add for spear tops	4.24 – 12.75
Scrolls	21.20 – 36.75
Gold leaves	4.10 – 18.25
Add for electronic operator, up to 1 H.P., sliding, each	4925.00 – 9950.00
Swinging	3325.00 – 8300.00
Add for control accessories, each, digital or magnetic card unit	1840.00 – 2330.00
Key or single push button	1130.00 – 1500.00
Magnetic vehicle detector	1410.00 – 2000.00
Basement entry, prefabricated steel bulkhead, each	1030.00 – 1500.00
Pet doors, each	161.00 – 473.00
Automatic, each	860.00 – 1620.00

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WALL COSTS

COSTS PER SQUARE FOOT OF WALL AREA OR OPENING

WINDOWS

All costs include single glazing, trim, hardware, and installation. Add 25% to 50% for double glazed, 45% to 75% for triple glazed, or see glass additives below.

	COST RANGE	
Aluminum sash, projected (awning/hopper)	26.50 –	43.75
Basement	20.90 –	29.75
Casement	21.90 –	38.00
Double hung	20.90 –	31.75
Fixed, acrylic glass block	43.00 –	57.00
Greenhouse	39.50 –	64.00
Jalousie	25.25 –	41.25
Single hung	17.95 –	27.50
Sliding, horizontal	14.50 –	27.00
For bronze or white finish add 15%.		
Vinyl sash, solid, add 20% to aluminum costs above.		
Steel sash, casement	20.65 –	41.00
Fixed, industrial	16.60 –	37.50
Vented, industrial	18.95 –	46.75
Wood sash, projected (awning/hopper)	34.25 –	58.00
Basement	25.25 –	33.50
Casement	29.50 –	47.50
Double hung	24.25 –	36.50
Greenhouse	51.00 –	84.00
Single hung	22.85 –	31.25
Sliding, horizontal	20.05 –	30.75
For metal or vinyl clad, add	12.25 –	18.25
Large windows, crystal glass, fixed	18.25 –	29.50
add for metal or vinyl–clad wood or solid vinyl	4.10 –	9.92
Bay or bow, prefabricated units	46.75 –	74.50
Greenhouse, window wall, 1’ to 3’ projection	73.50 –	98.00
Polygon or round windows, small	127.00 –	175.00
add for curved glass	175.00 –	230.00
add for bent (90°) glass	680.00 –	795.00
add for opaque glass	2.89 –	4.10
add for wire glass, clear	19.50 –	23.45
add for 1/4’’ plate glass, polished	4.02 –	8.34
add for 1/4’’ tempered plate glass	9.92 –	20.65
add for painted trim	2.08 –	2.49
add for aluminized reflective glass or solar film	9.92 –	12.25
add for double glazing (insulating glass)	7.71 –	11.85
add for impact (storm) resistant	27.00 –	48.00
add for transparent heat–reflective filament	9.30 –	15.30
add for Low E insulated with Argon gas	18.80 –	29.00

FIRE ESCAPES

Two-story building, with ladder to roof, each	7750.00 –	9850.00
add for counterbalance	4200.00 –	6350.00
Each additional story	4200.00 –	5600.00

GLASS

Cost per square foot of glass, including material and installation. For sloped glazing, add 10%.

PLATE GLASS	LIGHT AREA (Square Feet)				
1/4"	10 – 50	50 – 75	75 – 100	100 – 125	125 – 150
Clear	14.30	18.40	22.40	27.00	32.75
Tinted (bronze or grey)	17.15	21.90	25.75	31.00	36.50

Single strength	9.75	Obscure, 1/8”	18.40
Double strength	10.30	Obscure, wire, 1/4”	23.80
Crystal, 3/16"	12.90	Polished wire, 1/4” plate	32.75
Double-glazed, 1/2"	34.00	Tempered plate, 1/4”	22.70
Bullet resistant, 1-3/16"	192.00	Acrylic sheet, 3/16”	21.50
Impact resistant	75.00	Polycarbonate, 3/16”	42.75
Liquid crystal, 1" insulated	214.00		

MISCELLANEOUS

	COST RANGE	
Security mesh or iron bars	15.10 –	31.00
Wrought–iron grilles, decorative	23.55 –	42.75
Security shutters, rolling metal, small windows	91.50 –	138.00
large windows	69.50 –	87.00
Doors	42.00 –	81.00
add for motorized operation, each	590.00 –	1710.00
for plastic, deduct 10% – 30%.		
For security sensors, see Section UIP 3, Page 16.		
Shades, cotton	2.17 –	3.29
Vinyl coated	2.96 –	5.52
Matchstick	2.89 –	4.73
Foil faced, insulated layers	11.70 –	18.55
add for motorized operation, each	7.78 –	18.40
Shutters, exterior, plain	11.70 –	23.30
Louvered	12.25 –	31.75
Interior, movable louvers	19.50 –	40.25
For custom hardwood, add 100% – 200%.		
Storm windows	6.66 –	12.75
Venetian blinds, horizontal slats	4.73 –	27.25
Vertical steel slats	7.71 –	46.75
For custom or mirror finish, add 100%.		
Window grilles, snap–in	6.66 –	12.75
Window screens, aluminum frame and mesh	4.73 –	27.25
White frame and mesh	7.71 –	46.75
Bronze frame and mesh	4.73 –	6.25
Aluminum solar screen	6.17 –	8.52
Bronze solar screen	17.70 –	29.00
Wood frame and mesh	8.52 –	12.25
Basement window wells, corrugated metal, each	61.50 –	101.00
add for grate cover	71.50 –	84.00
4’ escape well	230.00 –	305.00

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